IBM Storage Scale System 6000

Hardware Planning and Installation Guide

SC28-3485-10

Note

Before using this guide and the product it supports, read the information in Chapter 2, “Notices,” on page 3.

This edition applies to Version 6 release 2 modification 0 of the following product and to all subsequent releases and modifications until otherwise indicated in new editions:

- IBM Storage Scale Data Management Edition for IBM Storage Scale System 6000 (product number 5765-DME)
- IBM Storage Scale Data Access Edition for IBM Storage Scale System 6000 (product number 5765-DAE)

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About this information

Who should read this information

This information is intended for administrators of IBM Storage Scale System 6000 that includes IBM Storage Scale RAID.

Conventions used in this information

Table 1 on page xv describes the typographic conventions used in this information. UNIX file name conventions are used throughout this information.

Table 1. Conventions

Convention	Usage
bold	Bold words or characters represent system elements that you must use literally, such as commands, flags, values, and selected menu options. Depending on the context, **bold** typeface sometimes represents path names, directories, or file names.
bold underlined	bold underlined keywords are defaults. These take effect if you do not specify a different keyword.
constant width	Examples and information that the system displays appear in `constant-width` typeface. Depending on the context, `constant-width` typeface sometimes represents path names, directories, or file names.
italic	*Italic* words or characters represent variable values that you must supply. *Italics* are also used for information unit titles, for the first use of a glossary term, and for general emphasis in text.
<*key*>	Angle brackets (less-than and greater-than) enclose the name of a key on the keyboard. For example, <Enter> refers to the key on your terminal or workstation that is labeled with the word *Enter*.

****	In command examples, a backslash indicates that the command or coding example continues on the next line. For example:
	`mkcondition -r IBM.FileSystem -e "PercentTotUsed > 90" \`
	`-E "PercentTotUsed < 85" -m p "FileSystem space used"`

{*item*}	Braces enclose a list from which you must choose an item in format and syntax descriptions.
[*item*]	Brackets enclose optional items in format and syntax descriptions.
<Ctrl-*x*>	The notation <Ctrl-*x*> indicates a control character sequence. For example, <Ctrl-c> means that you hold down the control key while pressing <c>.
item...	Ellipses indicate that you can repeat the preceding item one or more times.

Table 1. Conventions (continued)

Convention	Usage
|	In *synopsis* statements, vertical lines separate a list of choices. In other words, a vertical line means *Or*. In the left margin of the document, vertical lines indicate technical changes to the information.

How to submit your comments

To contact the IBM Storage Scale development organization, send your comments to the following email address: `scale@us.ibm.com`.

Chapter 1. Summary of changes

Summary of changes for IBM Storage Scale System 6000 documentation.

The updates to the *IBM Storage Scale System 6000 Hardware Planning and Installation Guide* include the changes that are detailed in the following table.

Table 2. Latest updates made in the IBM Storage Scale System 6000 Hardware Planning and Installation Guide

Month of the update	Change description	Updated page
March 2025	Add MTM 5149-S05 switch.	"The MTM 5149-S05 switch" on page 115
	Add MTM 5149-S54 switch.	"The MTM 5149-S54 switch" on page 135
June 2025	Add product numbers and feature codes for cabling components that can be used with MTM 5149-S05 switch and MTM 5149-S54 switch.	"Configuration and cabling" on page 154
	Add the configuration procedure for remote code load (RCL).	"Configuring remote code load" on page 65
	Add population rules for QLC and TLC drives in a fully populated system.	"Mixed tiering for QLC and TLC drives" on page 23
September 2025	Add installation instructions for the airflow control shelf that is now included with MTM 5149-S54 switch.	"Installing the MTM 5149-S54 switch" on page 144
November 2025	Add best practices for the management of a system with IBM FlashCore® Modules and to provision the correct amount of capacity for a system.	Appendix C, "Best practices for space management in IBM FlashCore Modules," on page 83
December 2025	Introduce support to configure IBM FlashCore Modules 4 as SEDs.	• "Self-encrypting drives" on page 23 • "Encryption for IBM FlashCore Modules" on page 89
	Introduce support for the deployment of decentralized-DAT file systems.	"Decentralized DAT deployment" on page 66
	Support for mixed configuration of 36 QLC drives and 12 TLC drives.	"Mixed tiering for QLC and TLC drives" on page 23
	Starting with IBM Storage Scale System 7.0.0.0, the fan floor speed is increased to 62%.	"Fan modules" on page 26

Month of the update	Change description	Updated page
March 2026	Update the supported protocols for the feature code AJQU.	Table 24 on page 46
April 2026	Add 03KX086 cable as a requirement for the installation of an MTM 5149-S54 switch.	“Installing the MTM 5149-S54 switch” on page 144
July 2026	Include performance configurations that are supported starting with IBM Storage Scale System 7.0.1.0.	“IBM Storage Scale System 6000 performance configurations” on page 19
	Include mixed tiering configurations that are supported starting with IBM Storage Scale System 7.0.1.0.	“Mixed tiering for QLC and TLC drives” on page 23
	Add instructions to use IBM Storage Modeller to determine how to configure the drives.	Appendix D, “Best practices for space management of industry-standard NVMe drives,” on page 95
	Add airflow capacity in CFM.	“Fan modules” on page 26

Table 2. Latest updates made in the IBM Storage Scale System 6000 Hardware Planning and Installation Guide (continued)

Chapter 2. Notices

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Caution, danger, and warning notices for IBM Storage Scale System 6000

Ensure that you read and understand all notices for IBM Storage Scale System 6000.

Attention: Only a trained person must install, remove, or replace the field replaceable units (FRU).

CAUTION: To avoid personal injury, disconnect the hot-swap, air-moving device cables before removing the fan from the device. (C024)

CAUTION:

Turn off or remove the fan's connection(s) to power before attempting to service the fan. Ensure the fan has stopped moving before servicing. (L020)

CAUTION:

Only a trained personnel must remove the covers to access the inside of the system.

Enclosure Integrity

- Access covers are intended only for occasional removal.
- Follow documented procedures when opening during live or temporary service.
- When service is complete, promptly reinstall all covers, lids, and/or doors for correct operation. (L031)

CAUTION: Only trained service personnel may replace this battery. The battery contains lithium. To avoid possible explosion, do not burn or charge the battery.

Do Not:

- Throw or immerse into water
- Heat to more than 100 degrees C (212 degrees F)
- Repair or disassemble

Exchange only with the approved part. Recycle or discard the battery as instructed by local regulations. (C002)

CAUTION: Risk of fire or explosion if the battery is replaced with an incorrect battery. Dispose of used batteries according to the instructions.

警 告

本電池如果更換不正確會有爆炸的危險

請依製造商說明書處理用過之電池

Multiple power sources

Where a unit receives power from more than one sources (for example, different voltages/frequencies or as redundant power), there shall be a prominent instructional safeguard in accordance with clause F.5 near each disconnect device giving adequate instructions for the removal of all power from the unit.

DANGER:

Shock hazard!

Disconnect all power sources.

The product might be equipped with multiple power sources. To reduce risk of injury from the electric shock, remove all power cords to completely disconnect power from the system.

Protective earthing

DANGER:

The equipment must be grounded properly to earth. Failure to use properly grounded outlets may result in electrical shocks.

and/or

CAUTION: If a product has a fiber port, use only a certified laser class 1 optical transceiver.

CAUTION: This equipment is not suitable for use in locations where children are likely to be present.

此設備不適用於兒童可能出現的區域

Environmental notices

This information contains all the required environmental notices for IBM Storage Scale System 6000.

The *IBM Systems Environmental Notices* includes statements on limitations, product information, product recycling and disposal, battery information, flat panel display, refrigeration and water-cooling systems, external power supplies, and safety data sheets.

CAUTION: To reduce the risk of fire or electric shock, install the uninterruptible power supply in a temperature- and humidity-controlled indoor environment, free of conductive contaminants. Ambient temperature must not exceed 40°C (104°F). Do not operate near water or excessive humidity (95% maximum). (13)

CAUTION:

- Do not install a unit in a rack where the internal rack ambient temperatures will exceed the manufacturer's recommended ambient temperature for all your rack-mounted devices.
- Do not install a unit in a rack where the air flow is compromised. Ensure that air flow is not blocked or reduced on any side, front, or back of a unit used for air flow through the unit.
- Consideration should be given to the connection of the equipment to the supply circuit so that overloading of the circuits does not compromise the supply wiring or overcurrent protection. To provide the correct power connection to a rack, refer to the rating labels located on the equipment in the rack to determine the total power requirement of the supply circuit.
- (For sliding drawers) Do not pull out or install any drawer or feature if the rack stabilizer brackets are not attached to the rack. Do not pull out more than one drawer at a time. The rack might become unstable if you pull out more than one drawer at a time.
- (For fixed drawers) This drawer is a fixed drawer and must not be moved for servicing unless specified by the manufacturer. Attempting to move the drawer partially or completely out of the rack might cause the rack to become unstable or cause the drawer to fall out of the rack. (R001 part 2 of 2)
- The control canisters of IBM Storage Scale System 6000 are factory-sealed, contain no serviceable parts, and must be replaced by IBM-authorized service personnel only.

Electromagnetic compatibility notices

The following Class A statements apply to IBM products and their features unless designated as electromagnetic compatibility (EMC) Class B in the feature information.

IBM Storage Scale System 6000 (MTM 5149-F48). When attaching a monitor to the equipment, you must use the designated monitor cable and any interference suppression devices that are supplied with the monitor.

meets all regulatory EMC compliance requirements as offered, including auxiliary equipment and cables that may be ordered from IBM to be used in conjunction with IBM Storage Scale System 6000 (MTM 5149-F48).

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CAN ICES-3 (A)/NMB-3(A)

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This product may cause interference if used in residential areas. Such use must be avoided unless the user takes special measures to reduce electromagnetic emissions to prevent interference to the reception of radio and television broadcasts.

Warning: This equipment is compliant with Class A of CISPR 32. In a residential environment this equipment may cause radio interference.

Germany Notice

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Verantwortlich für die Einhaltung der EMV-Vorschriften ist der Hersteller:

International Business Machines Corp.
New Orchard Road
Armonk, New York 10504
Tel: 914-499-1900

Der verantwortliche Ansprechpartner des Herstellers in der EU ist:

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Technical Relations Europe, Abteilung M456
IBM-Allee 1, 71139 Ehningen, Germany
Tel: +49 800 225 5426
e-mail: Halloibm@de.ibm.com

Generelle Informationen:

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Chapter 3. Technical overview

The technical overview sections provide information about the key concepts of the IBM Storage Scale System 6000.

Target audience of this chapter: Customers and IBM Service Support Representative (SSR).

System overview

An IBM Storage Scale System 6000 (MTM 5149-F48) enclosure contains Non-Volatile Memory Express (NVMe) attached drives and a pair of server canisters.

IBM Storage Scale System 6000 is an all-flash array platform. This storage platform uses NVMe-attached drives to provide significant performance improvements as compared to SAS-attached flash drives.

IBM Storage Scale System 6000 can contain up to 48 NVMe-attached drives. You can either use 24 drives in a half-populated configuration or 48 drives in a fully populated configuration. These drives are accessible from the front of IBM Storage Scale System 6000, as shown in the following figure.

Note: The minimum number of drives is 24. In a 24-drives configuration, drives must be plugged only into slots with even numbers (2-24) in the canister 1 (top); and only into slots with odd numbers (25-47) in the canister 2 (bottom).

Figure 1. Front view of the IBM Storage Scale System 6000

Each IBM Storage Scale System 6000 contains two identical I/O server canisters. The following figure shows the rear view of the MTM 5149-F48 module canisters.

Figure 2. Rear view of IBM Storage Scale System 6000

The IBM Storage Scale System 6000 uses two AMD EPYC 9454 SP5 per server canister. The following table provides an overview of IBM Storage Scale System 6000.

Table 3. Overview of the IBM Storage Scale System 6000 system

Product	**Number of DIMMs per server canister**	**Total memory per server canister**	**Total storage (raw capacity) per IBM Storage Scale System 6000 unit**	**Server canister features**
IBM Storage Scale System 6000	24 (32 GiB DIMM)	768 GiB	Up to 1440 TB NVMe	Dual socket AMD EPYC 9454 SP5, 48-core processors. Dual 960 GB NVMe boot drives.
	24 (64 GiB DIMM)	1536 GiB		

The major key characteristics of the IBM Storage Scale System 6000 system are as follows:

- IBM Storage Scale software with enclosure-based, all-inclusive software feature licensing.
- Support for PCIe Gen 5.
  - FC: AJQS. 1 to 3 x16 G5 CX-7 (VPI, dual port 200 Gb)
  - FC: AJQQ. 1 to 3 x16 G5 CX-7 (single port 400 Gb, no auxiliary card)
- Support for dual AMD EPYC 9454 SP5, 48 cores (in the performance configuration).
- NVMe transport protocol for high performance of 2.5 inches (SFF) NVMe-attached drives and flash drive:
  - Support for industry-standard 2.5 inches, NVMe-attached drive options with the following storage capacities: 3.84 TB, 7.68 TB, 15.36 TB, and 30.72 TB (mentioned in Table 4 on page 14).

Table 4. Storage capacities and feature codes of NVMe-attached drives

Feature code	**Description**	**FRU part number**
AK0K	3.84 TB 2.5 in. PCIe Gen 4 NVMe-attached drive	03JN944
AK11	7.68 TB 2.5 in. PCIe Gen 4 NVMe-attached drive	03JN945

Feature code	Description	FRU part number
AK0L	15.36 TB 2.5 in. PCIe Gen 4 NVMe-attached drive	03JN946
AK0M	30.72 TB 2.5 in. PCIe Gen 4 NVMe-attached drive	03JN947

Table 4. Storage capacities and feature codes of NVMe-attached drives (continued)

- 24 or 48 drives per enclosure configurations are available.
- Onboard ports per canister:
  - Two RJ45 of 10 GbE for management
  - One RJ45 of 1 GbE for BMC inter-canister communication
  - One mini HDMI port for console display
  - One USB 3.0 type A port for crash-cart keyboard
  - One USB Mini-C for service
  - Various I/Os depending on the add-in cards
- Up to eight Gen 5 x16 PCIe slots per canister. These slots support the following adapter options:
  - PN: 01LL984 - CX-7 VPI, dual port, 200 Gb.
  - PN: 01LL983 - CX-7 InfiniBand, single port, 400 Gb.

The ports and LEDs details of IBM Storage Scale System 6000 are shown in the following figure.

Item	Usage
1	Dual boot drives (mirrored)
2	2x 10 GbE management network
3-10	NVIDIA CX7 network x16 G5
11	1 GbE for BMC or host
12	Push button
13	USB mini-C
14	Mini HDMI
15	Dual CXL (not used)
16	USB A

Figure 3. IBM Storage Scale System 6000 ports and LEDs

Networking details

The networking details of IBM Storage Scale System 6000 are shown in the following figure.

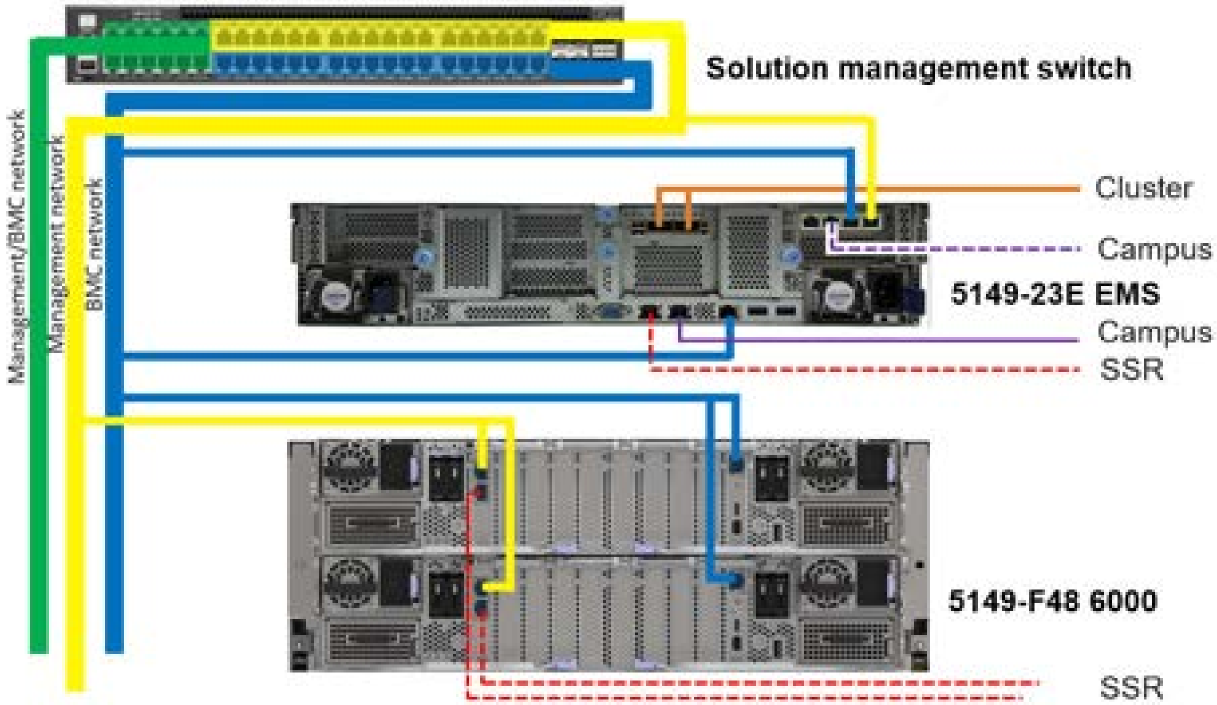


Figure 4. IBM Storage Scale System 6000 networking

- Intel Ethernet Converged Network Adapter X550-T2
  - System interface: PCIe Gen 3
  - Ethernet interface: 10 GbE / 1 GbE / 100 Mb
- FC: AJQS - CX-7 InfiniBand or VPI dual port HPC adapter
  - InfiniBand - HDR200 200 Gb / HDR100 100 Gb / EDR 100 Gb / NDR 200 Gb
  - Ethernet - 100 GbE / 200 GbE
- FC: AJQQ - CX-7 InfiniBand, single port, 400 Gb

NVMe transport protocol in IBM Storage Scale System 6000

The IBM Storage Scale System 6000 systems use the Non-Volatile Memory express (*NVMe*) drive transport protocol.

- NVMe is designed specifically for flash technologies. It is a faster and less complicated storage drive transport protocol than SAS.
- The NVMe-attached drives support multiple queues so that each CPU core can communicate directly with the drive. This feature avoids the latency and overhead of core-to-core communication to give the best performance.
- NVMe offers better performance and lower latencies exclusively for solid-state drives through multiple I/O queues and other enhancements.
- High-performance IBM Storage Scale RAID supports the following RAID code options:
  - 3WayReplication
  - 4WayReplication
  - 8+2p
  - 8+3p

  Note: For information about RAID code options, see *IBM Storage Scale RAID Administration guide*.

• The NVMe transport protocol supports industry-standard NVMe flash drives.

Enterprise-level software and support

IBM Storage Scale System 6000 consists of two server canisters that run the IBM Storage Scale software, which is a part of the IBM Storage family. For more information about the IBM Storage Scale capabilities, see IBM Storage Scale.

Enterprise-level software and support are available, based on the IBM Storage Scale software stack.

System topology

The following figure shows high-level architecture and topology.

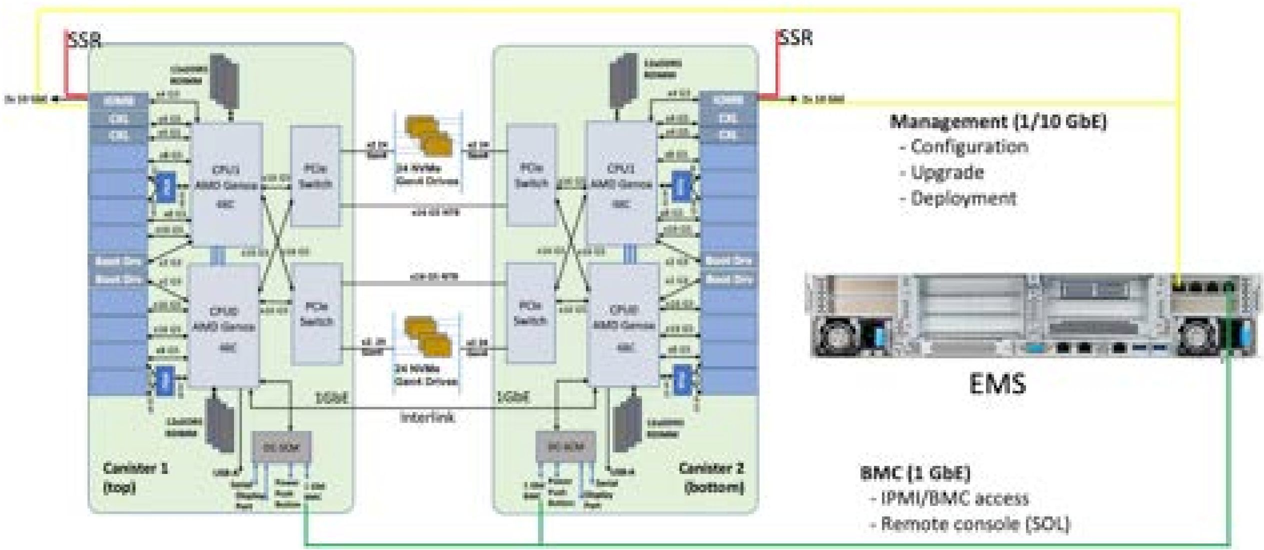


Figure 5. IBM Storage Scale System 6000 high-level architecture and topology

EMS node

The IBM Storage Scale System Utility Node is an enterprise management server (EMS). Each IBM Storage Scale System of one or more building blocks requires an EMS. You can order an IBM Storage Scale System Utility Node with IBM Storage Scale System 6000, but this configuration mostly applies to new clusters. Conversion from Power9 EMS to IBM Storage Scale System Utility Node is not yet supported. Mixing EMS types in the same environment is not supported either.

You need a minimum of one EMS node as a part of your IBM Storage Scale System cluster. The EMS node can be ordered as a part of IBM Storage Scale System 6000.

IBM Storage Scale System 6000 is supported on the IBM Storage Scale System Utility Node. The minimum release level on the IBM Storage Scale System 6000 is IBM Storage Scale System6.1.9 for IBM Storage Scale System Utility Node.

For more information about the IBM Storage Scale System Utility Node EMS, see Model 5149-23E server specifications.

The EMS node also serves as a third IBM Storage Scale quorum node in a configuration with one building block.

System management

An EMS node in an IBM Storage Scale System 6000 cluster provides system management functions. IBM Storage Scale System 6000 GUI and call home run on the EMS nodes and provide management and health monitoring capabilities. The EMS node also runs a container with Ansible playbook that can provide orchestration of complex tasks, such as cluster configuration, file system creation, and code update. Figure 5 on page 18 shows high-level view of system management topology.

IBM Storage Scale System 6000 performance configurations

MTM 5149-F48 enclosure supports the performance solution configuration.

A single MTM 5149-F48 enclosure with 12, 24, or 48 NVMe-attached drives serves as the data storage component of the performance configuration.

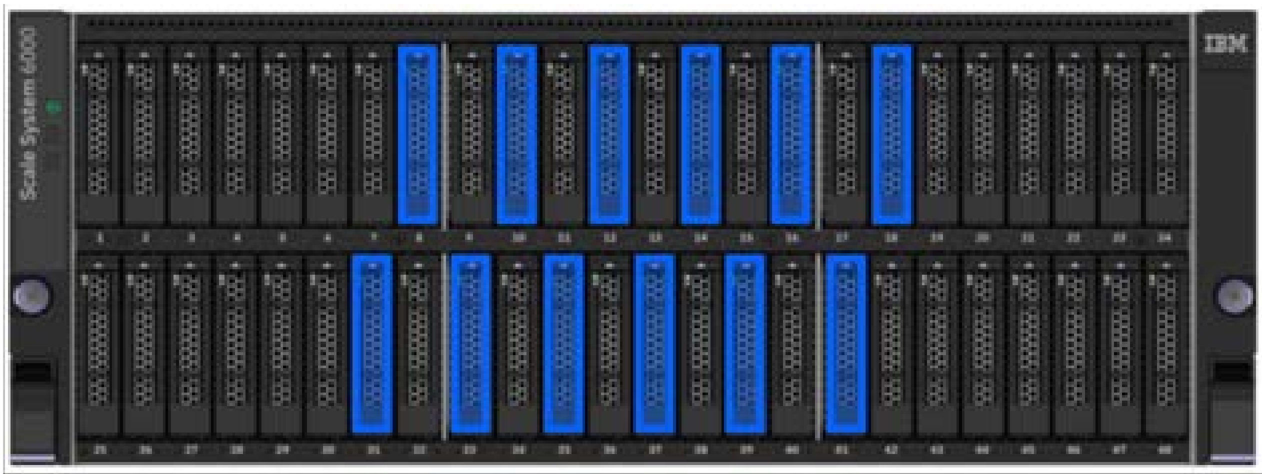


Figure 6. IBM Storage Scale System 7.0.1.0 onward: Performance configuration with 12 TLC drives and 36 fillers installed

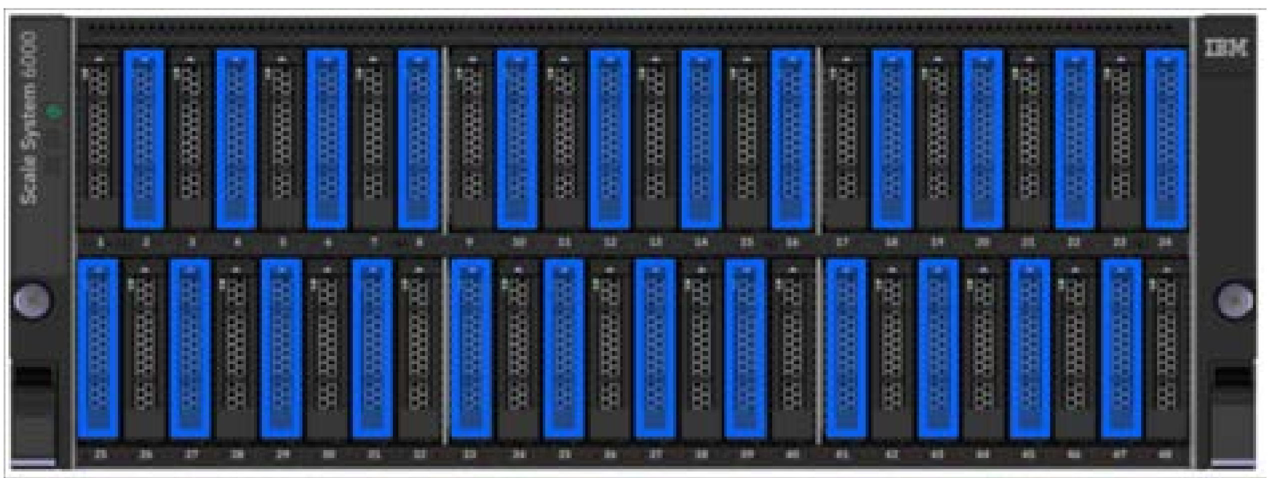


Figure 7. Performance configuration with 24 drives and 24 fillers installed

Figure 8. Performance configuration with 48 drives installed

For more information, see Table 3 on page 14 and Table 4 on page 14.

IBM Storage Scale System 6000 hybrid configurations

MTM 5149-F48 enclosure supports the hybrid solution configuration.

A single MTM 5149-F48 enclosure with 24 or 48 NVMe-attached drives serves as the data storage component of a hybrid configuration that can have 1 or more IBM Storage Scale System 5149-091 Storage Enclosures or IBM Storage Scale System 5149-F26 Storage Enclosures.

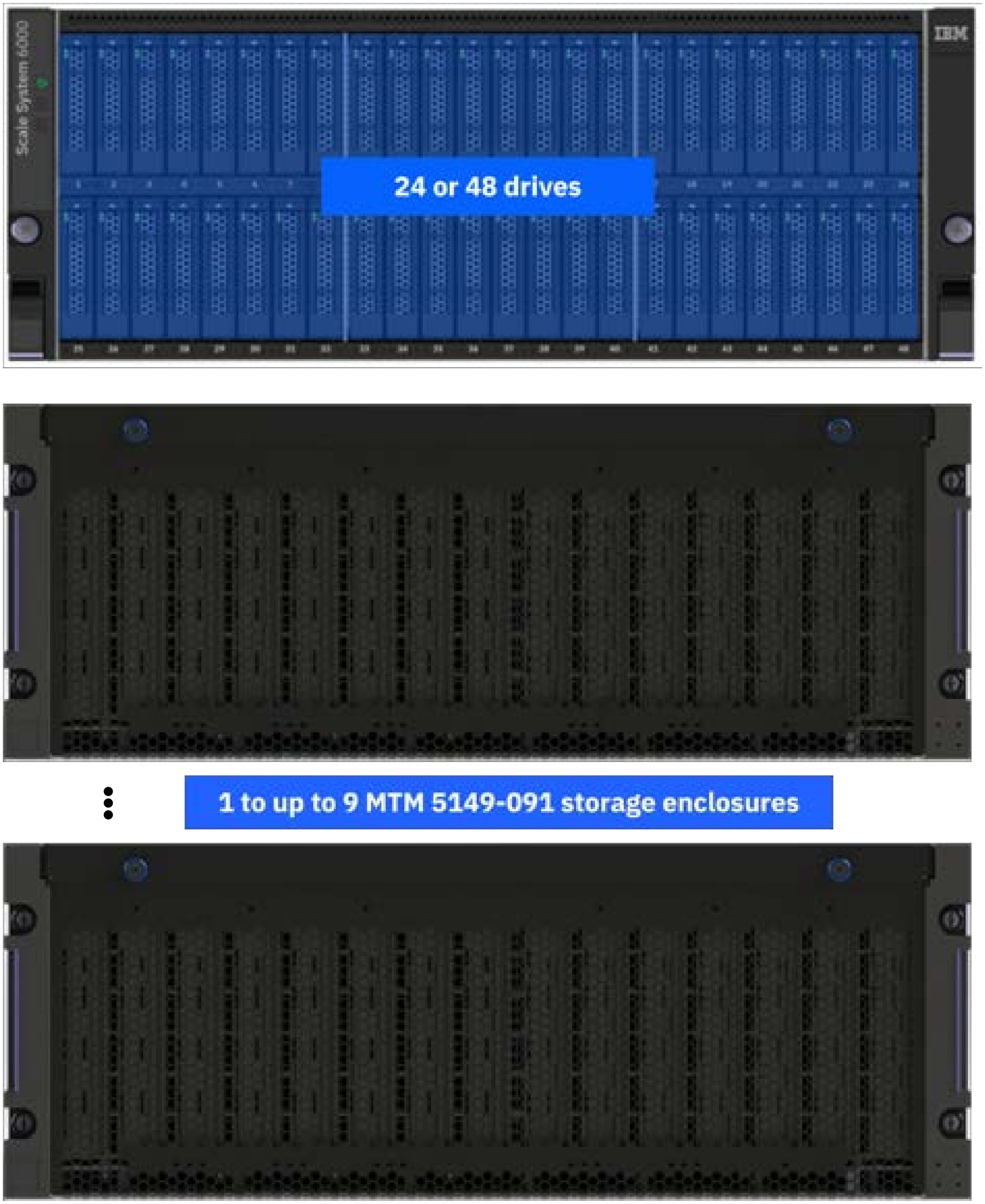


Figure 9. Hybrid configuration with 48 or 24 drives installed and one or more MTM 5149-091 storage enclosures

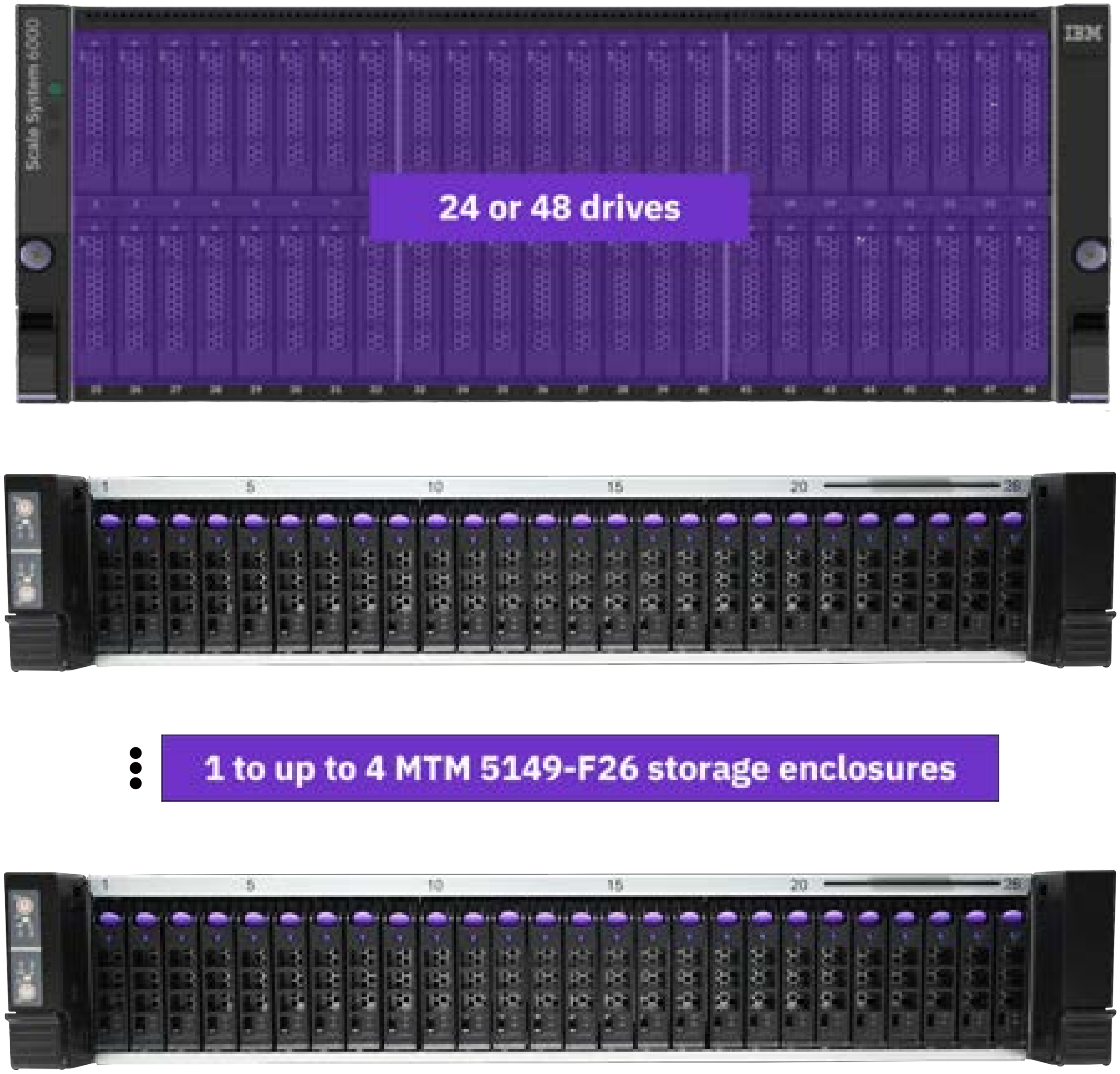


Figure 10. Hybrid configuration with 48 or 24 drives installed and one or more MTM 5149-F26 storage enclosures

For more information, see Table 3 on page 14 and Table 4 on page 14.

Drives

IBM Storage Scale System 6000 supports industry-standard 2.5-inch NVMe-attached drive options.

The following storage capacities are supported for NVMe-attached drives:

- 3.84 TB
- 7.68 TB
- 15.36 TB
- 30.72 TB

Drives are hot swappable from the front of the enclosure.

Self-encrypting drives

On IBM Storage Scale System 6000, you can configure all the supported HDDs and NVMe drives as self-encrypting drives (SEDs). Starting with IBM Storage Scale System 7.0.0.0, IBM FlashCore Modules 4 can be configured as SEDs.

Drive slots

A drive slot represents the location in an enclosure into which a drive can be inserted.

Each drive slot must contain either a drive or an empty carrier. The empty carriers do not have drive indicators. The numbering scheme for the drive slots is indicated on the enclosure.

The following figure displays an enclosure that has 48 drives, with 2.5-inch drive slots. In each canister, the drive slots are arranged in one row.

IBM Storage Scale System 6000 supports 24 and 48 drives configurations. All drives must be identical in an IBM Storage Scale System 6000 enclosure. In a 24-drives configuration, drives must be plugged only into slots with even numbers (2-24) in the top; and only into slots with odd numbers (25-47) in the bottom.

Figure 11. Front view of an IBM Storage Scale System 6000 with a 48-drive configuration

Mixed tiering for QLC and TLC drives

Follow this population rules to install quad-level cell (QLC) drives in combination with triple-level cell (TLC) drives in your IBM Storage Scale System 6000.

The default pool information for these drives is as follows.

QLC drives
Data pool.

TLC drives
Metadata or data.

Remember: An IBM Storage Scale System 6000 with a mixed drives configuration must have the same FRU for all of its QLC drives, and also all of its TLC drives must have the same FRU.

Table 5. QLC and TLC drives supported in a fully populated IBM Storage Scale System 6000		
Drive type	**Description**	**Tier**
QLC	30.72 TB, 2.5 inches, PCIe Gen 5, NVMe	2
	61.44 TB, 2.5 inches, PCIe Gen 5, NVMe	
	122 TB, 2.5 inches, PCIe Gen 5, NVMe	
TLC	3.84 TB, 2.5 inches, PCIe Gen 4, NVMe flash drive	1
	7.68 TB, 2.5 inches, PCIe Gen 4, NVMe flash drive	
	15.36 TB, 2.5 inches, PCIe Gen 4, NVMe flash drive	
	30.72 TB, 2.5 inches, PCIe Gen 4, NVMe flash drive	
	3.84 TB, 2.5 inches, PCIe Gen 5, SSD	
	7.68 TB, 2.5 inches, PCIe Gen 5, SSD	
	15.36 TB, 2.5 inches, PCIe Gen 5, SSD	
	30 TB, 2.5 inches, PCIe Gen 5, SSD	

24 QLC drives and 24 TLC drives

Restriction:

- Only a fully populated IBM Storage Scale System 6000 supports this mixed tiering configuration of QLC and TLC drives.
- Mixed tiering in an IBM Storage Scale System 6000 with IBM FlashCore Modules is not supported.
- With an IBM Storage Scale System version that is earlier than 7.0.1.0, a miscellaneous equipment specification (MES) procedure is not supported for QLC drives.
- With an IBM Storage Scale System version that is earlier than 7.0.1.0, capacity models do not support mixed tiering.

By following the next population rules, a fully populated (48 drives) IBM Storage Scale System 6000 supports 24 QLC drives in combination with 24 TLC NVMe drives.

Population rules for QLC drives

- In canister 1 (top), install QLC drives in the odd slots: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, and 23.
- In canister 2 (bottom), install QLC drives in the even slots: 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, and 48.

Population rules for TLC drives

- In canister 1 (top), install TLC drives in the even slots: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, and 24.
- In canister 2 (bottom), install TLC drives in the odd slots: 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, and 47.

The following figure shows how to apply the previous population rules.

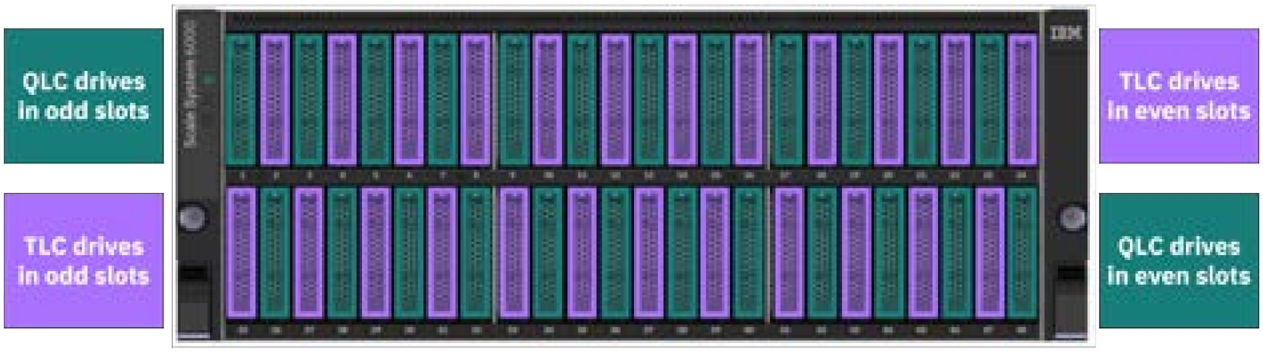


Figure 12. Fully populated IBM Storage Scale System 6000 with 24 QLC drives and 24 TLC drives

36 QLC drives and 12 TLC drives

Starting with IBM Storage Scale System 7.0.0.0, the mixed configuration of 36 QLC drives and 12 TLC drives is supported by IBM Storage Scale System 6000.

Restriction:

- Only a fully populated IBM Storage Scale System 6000 supports this mixed configuration of QLC and TLC drives.
- Tiering with IBM FlashCore Modules is not supported.
- QLC drives must *not* undergo an MES procedure (adding QLC drives to a 12-drives configuration).
- With IBM Storage Scale System versions earlier than 7.0.1.0, hybrid configurations do not support mixed tiering (the IBM Storage Scale System 6000 must not have a storage enclosure attached to it).

By following the next population rules, a fully populated (48 drives) IBM Storage Scale System 6000 supports 36 QLC drives in combination with 12 TLC NVMe drives.

Population rules for QLC drives

- In canister 1 (top), install QLC drives in these 18 slots: 1, 2, 3, 4, 5, 6, 7, 9, 11, 13, 15, 17, 19, 20, 21, 22, 23, and 24.
- In canister 2 (bottom), install QLC drives in these 18 slots: 25, 26, 27, 28, 29, 30, 32, 34, 36, 38, 40, 42, 43, 44, 45, 46, 47, 48.

Population rules for TLC drives

- In canister 1 (top), install TLC drives in these six slots: 8, 10, 12, 14, 16, and 18.
- In canister 2 (bottom), install TLC drives in these six slots: 31, 33, 35, 37, 39, and 41.

The following figure shows how to apply the previous population rules.

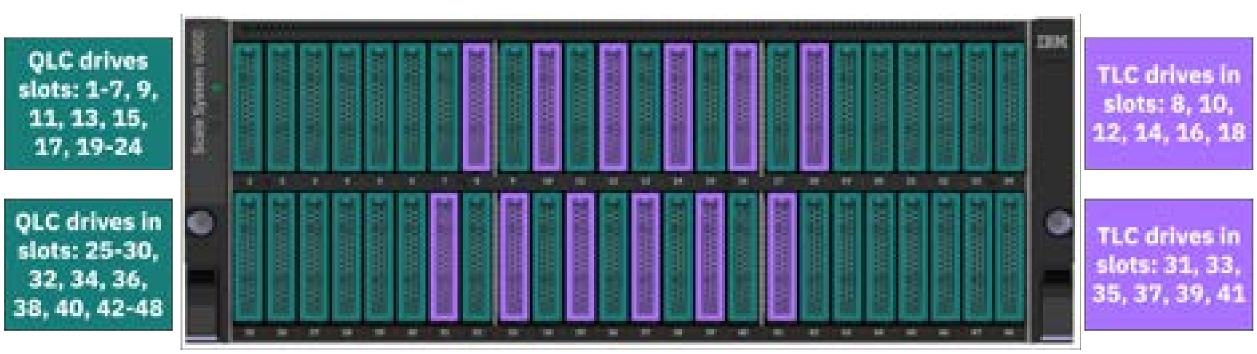


Figure 13. Fully populated IBM Storage Scale System 6000 with 36 QLC drives and 12 TLC drives

Power supply units (PSUs)

Power supply units are subcomponents of the server. A PSU takes electrical power from the power distribution units (PDUs) of the rack and distributes the power to other components in the server.

Note: The base storage system may be shipped without the power supply unit. To bare the UL mark (cULus) on a nonfunctioning enclosure, users must install the original or identical PSU model, FSM029, as evaluated in the end use (storage equipment *Regulatory Model ID: LTH-4U48* or *Alternate Supplement MT-Model: 5149-F48*). Breaching this requirement voids system safety certification.

For redundancy, four power modules are installed in the enclosure, two per canister.

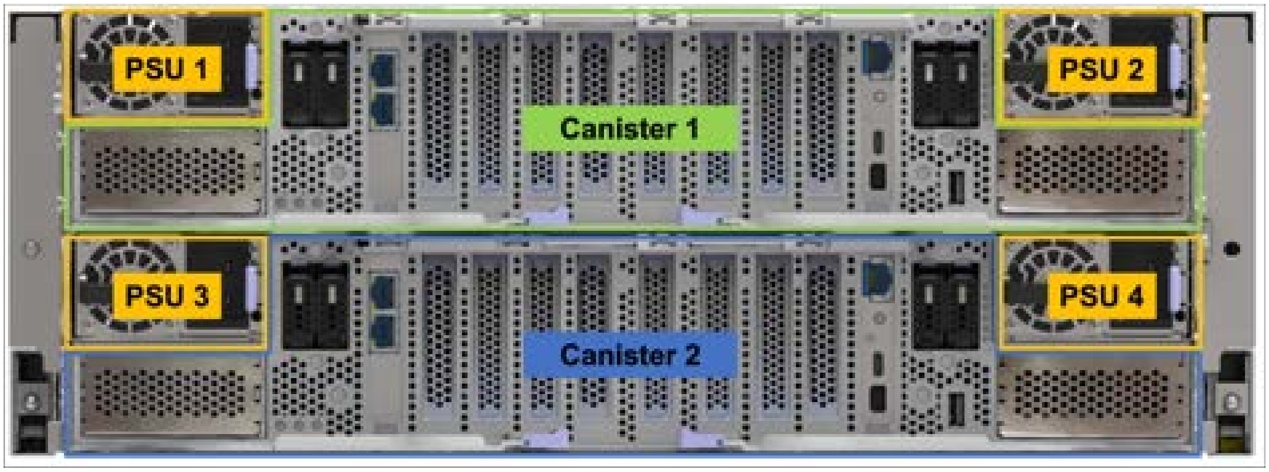


Figure 14. Power supply units

Fan modules

An IBM Storage Scale System 6000 includes six hot-swappable fan modules.

The fan modules contribute to cool the system by circulating air through the enclosure. These components are field-replaceable units that must be serviced only by IBM service personnel.

Figure 15. IBM Storage Scale System 6000 fan modules

Fan floor speed

Starting with IBM Storage Scale System 7.0.0.0 software, the fan floor speed is increased to 62%. This is a measure to accommodate the thermal requirements that are introduced by adding NVIDIA ConnectX-7 cards to all the HBA slots, which is required to connect IBM Storage Scale System 5149-F26 Storage Enclosures to an IBM Storage Scale System 6000. This change contributes to ensure adequate cooling for all the hardware components (system, HBA slots, optics).

After a code upgrade to 7.0.0.0, the increase to fan floor speed is globally applied across all MTM 5149-F48 systems. Therefore, systems without MTM 5149-F26 configurations or additional NVIDIA ConnectX-7 cards also operate at this higher fan floor.

Before 7.0.0.0, the fan floor speed was approximately 32%, with values dynamically adjusted based on ambient temperature. For example, 32% at 25°C, and higher floor speeds at 27°C, 32°C, and 35°C.

Warning: This algorithm change increases system acoustic levels due to the higher baseline fan speed.

Airflow in CFM per system

The airflow of an IBM Storage Scale System 6000 is as follows:

- 270 CFM nominal
- 450 CFM maximum

Cubic feet per minute (CFM)

CFM is an airflow characteristic that quantifies the volume of air that moves through a system. In storage enclosure specifications, the CFM value represents the airflow capacity of its fans.

SSR access port

On the back panel of the canister, the bottom RJ45 10 GbE port is the SSR access port that a service personnel can use to initialize a system.

You can use the SSR access port to set the IP address of the management and BMC interfaces. This port is also used by the SSR to update the canister software or firmware and check the health of the system before using the **Install Complete** app. The 101 ports (10.111.222.101/30) are assigned to this port; the SSR is given the 10.111.222.102 IP address. For more information, see *IBM Storage Scale Install Complete Guide*.

Note: An IBM intranet connection is required to access the guide.

In each of the IBM Storage Scale System 6000 server canisters, the RJ45 10 GbE port is designated as the SSR access port, as shown in the following figure.

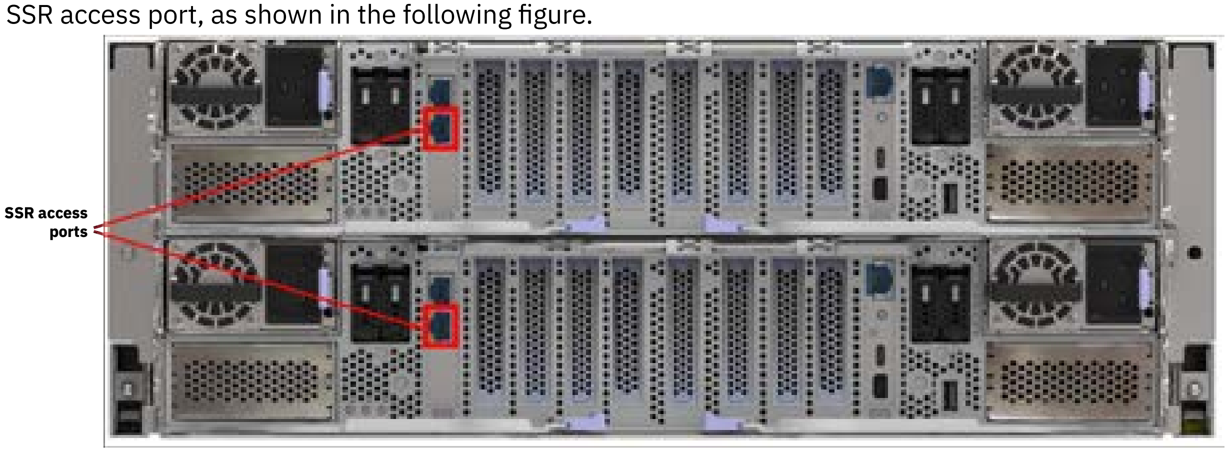


Figure 16. SSR access port

Hardware specification sheets

EMS node specifications

IBM Storage Scale System 6000 uses one EMS node in the basic configuration. The EMS node acts as the management server and provides infrastructure for hardware monitoring and hosts GUI.

The EMS server has the following roles:

- Quorum node
- GUI server
- Call home server

For more information, see the Model 5149-23E server specifications of the IBM Storage Scale System Utility Node.

Chapter 4. Planning for IBM Storage Scale System 6000 hardware

Before the IBM Service Support Representative (SSR) installs the system hardware, the customer must provide a plan that explains where and how the hardware will be installed, configured, and connected in the customer's network. Ensure that the system's physical configuration records are easily accessible to be used as a reference as needed.

The customers must work with the IBM technical team to plan for installation and deployment of an IBM Storage Scale System 6000.

IBM Storage Scale System 6000 requirements

Before you install a system, your physical environment must meet certain requirements. This includes verifying that adequate space is available and that requirements for power and environmental conditions are met.

Power requirements for each power supply (four per enclosure)

Ensure that your environment meets the following power requirements.

Table 6. System maximum measured power specifications

Product	kVA	Amps	Power Supplies	Inlet	Watts
4U IBM Storage Scale System 6000 (MTM 5149-F48), 48-drives configuration	4.50	22.50	4	C20	4800*
***Note:** This value represents the absolute maximum power draw of the power supplies. Actual power consumption will vary relative to workload/system utilization. For example, for a fully populated IBM Storage Scale System 6000 (48 NVMe drives) at a theoretical 100% system utilization, the estimated power consumption is ~3.2kW.					

To aid in power and cooling requirements planning, the following table lists the rating of each power module by enclosure.

The power that is used by the system depends on various factors, including the number of enclosures and drives in the system and the ambient temperature.

Table 7. Power rating specifications per power module

Model and type	PSU	Input power requirements	Maximum input current	Maximum power output
4U IBM Storage Scale System 6000 (MTM 5149-F48), 48-drives configuration	2400 W (4)	200 V to 240 V single phase AC At a frequency of 50 Hz or 60 Hz IEC C20 standardized	13A (x4)	2400 W

Note: The data in Table 6 on page 29 and Table 7 on page 29 measurements are presented as an example. Measurements that are obtained in other operating environments might vary. Conduct your own testing to determine specific measurements for your environment.

Each IBM Storage Scale System 6000 enclosure contains two power modules for redundancy. The total power consumption values represent the total power that is drawn by both power modules when operating together or a single power module when operating in a maintenance or failure mode.

Environmental requirements

System airflow is from the front to the rear of each enclosure:

- Airflow passes between drive carriers and through each enclosure.
- The combined power and cooling module exhausts air from the rear of each canister.

Ensure that your environment falls within the ranges that are listed in the following table.

Table 8. Temperature requirements

<table>
<tr><th>Environment</th><th>Ambient temperature</th><th>Altitude</th><th>Relative humidity</th></tr>
<tr><td>Operating</td><td>5°C to 35°C[1]
(41°F to 95°F)</td><td rowspan="2">-61 to 3048 m[2, 3]
(-200 to 10000 ft)</td><td>20% to 80% noncondensing</td></tr>
<tr><td>Nonoperating</td><td>5°C to 45°C
(41°F to 113°F)</td><td>10% to 90% noncondensing</td></tr>
<tr><td>Transit</td><td>-40°C to 60°C
(-40°F to 140°F)</td><td>-61 to 12192 m
(-200 to 40000 ft)</td><td>10% to 90% noncondensing</td></tr>
</table>

Important: Regardless of the system specifications about absolute maximum temperature, the nonoperating environmental conditions of the system solid-state devices must adhere to the *JEDEC JESD218 Enterprise-class SSD* requirements for data retention endurance. These requirements must be met particularly for the system storage, transit, or shipping maximum temperatures. To best maintain the data retention of the system solid-state devices, keep an *optimal* condition for nonoperating temperatures by making sure that the system does not remain for longer than 3 months at temperatures that exceed 40°C.

Note:

- Max ambient temperature environment = 32C / 950 meters.
- Max altitude environment = 25 degrees Celsius / 3050 meters.
- Decrease the maximum air temperature by 1 degree Celsius per 300 meters above 950 meters.
- The maximum ambient operating temperature when using an optical cable or discrete optical transceiver is 32 degrees Celsius, which includes all Ethernet (100 Gb) cables and InfiniBand (100 Gb EDR) that are greater than or equal to 3 meter in length.

Dimensions and weight requirements for rack installation

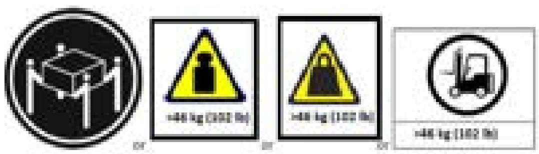
Ensure that space is available in a standard 19 in. rack (EIA Standard EIA-310-D) that is capable of supporting the enclosure. The rack rail kit supports racks with threaded round and square rail mounting holes. The following table lists the dimensions and weights of the enclosures.

Table 9. Physical characteristics of the enclosures				
Enclosure	**Height**	**Width**	**Depth**	**Maximum weight**
4U IBM Storage Scale System 6000, 48-drives configuration	176.0 mm (6.9 in.)	438.0 mm (17.25 in.)	1 m (39.4 in.) from front EIA surface to rear of CMA assembly. 823.1 mm (32.4 in.) from front EIA surface to rear surface of the enclosure arm.	62.2 kg (135 lb) per system, without rail or rack

CAUTION:

The weight of this part or unit is between 32 and 55 kg (70.5 and 121.2 lb). It takes three persons to safely lift this part or unit. (C010)

CAUTION:



The weight of this part or unit is more than 46 kg (102 lb). It takes specially trained persons, a lifting device, or both to safely lift this part or unit. (C011)

The following table shows the rack space requirements for IBM Storage Scale System 6000 in tabular form.

Table 10. Rack space requirements for the IBM Storage Scale System 6000	
Minimum rail length	**Maximum rail depth (including CMA)**
676.4 mm (266.3 in.) (+/- 5 mm or 0.20 in.)	789.8 mm (310.94 in.) (+/- 5 mm or 0.20 in.)

Service envelope illustration

The following figure provides a close-up view of the IBM Storage Scale System 6000 mounting envelope.

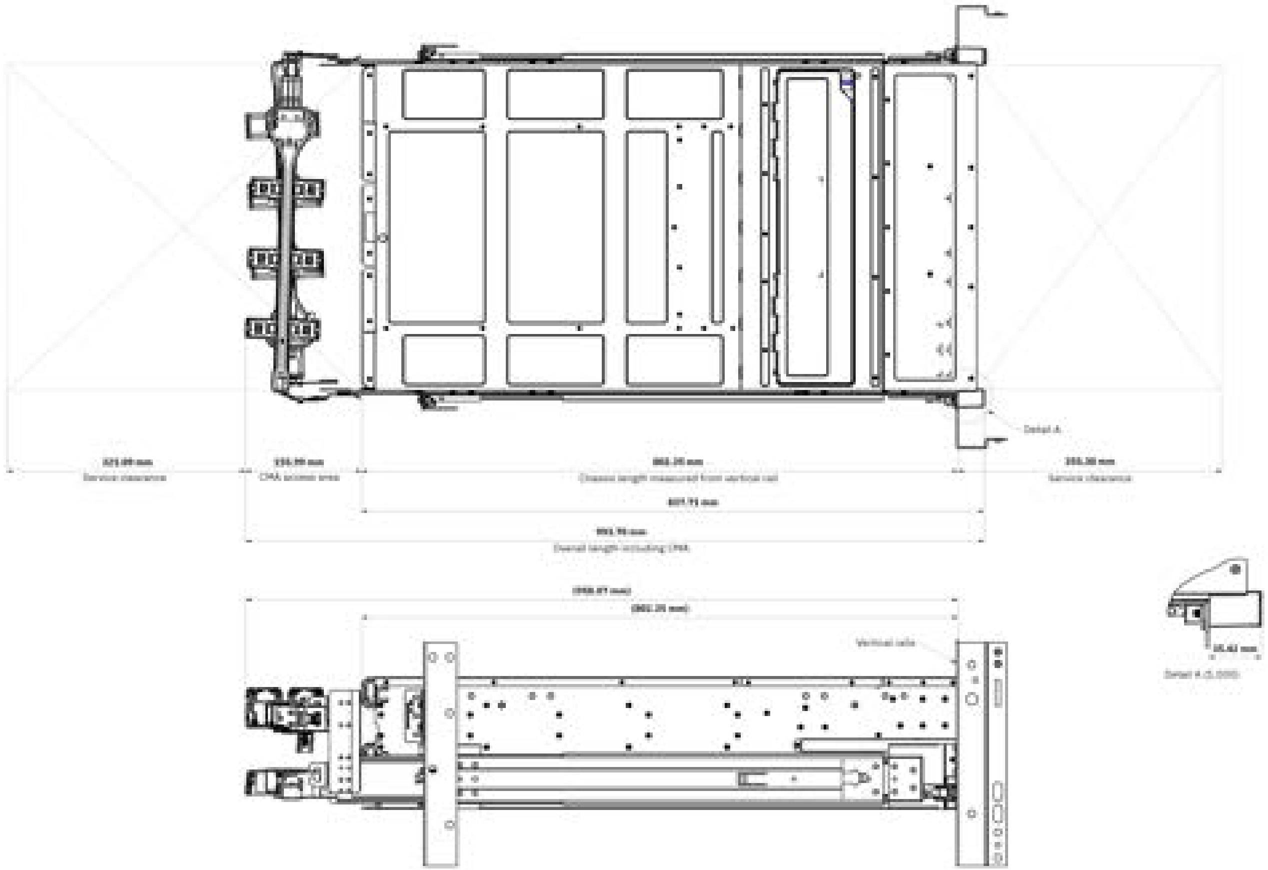


Figure 17. Service envelope illustration (dimensions in millimeters)

Service clearance for IBM Storage Scale System 6000

The service clearance area is the area around the rack that is needed for the authorized service representatives to service the enclosure as shown in the sample illustration in the following figure.

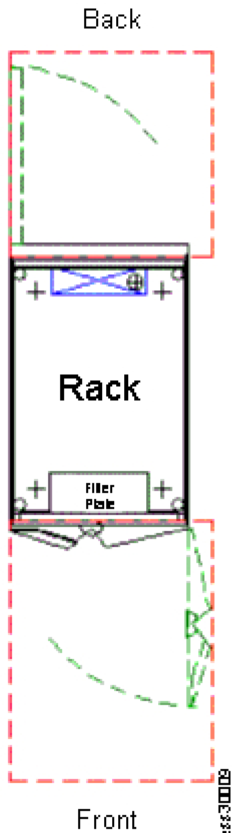


Figure 18. Sample illustration to show space around the rack

For IBM Storage Scale System 6000, use the recommended measurements that are given in the following table.

Table 11. Service clearance requirements

Front[1]	**Back**
915 mm (36 in.)	915 mm (36 in.)
[1]Storage racks require larger service clearances in the front of the rack.	

For more information about the layout of the room, see *Computer room layout*.

See the Site preparation and physical planning section to help you prepare your physical site for the installation of IBM Storage Scale System 6000.

Additional space requirements

Ensure that these additional space requirements, as shown in the following table, are available around the enclosures.

Table 12. Clearances

Location	**Additional space requirements**	**Reason**
Left and right sides	50 mm (2 in.)	Cooling air flow
Back	Minimum: 100 mm (4 in.)	Cable exit

Supported drives

The following table provides drive specifications for your IBM Storage Scale System 6000 system.

Table 13. Drive specifications

Model and type	**2.5 in. drives**
4U IBM Storage Scale System 6000, with 48 drive slots of 2.5 in.	12 or 24 dual-port NVMe drives

For more information about the feature codes of the NVMe drives, see Table 4 on page 14.

Acoustical declaration with noise hazard notice

The following figure indicates the declared noise emissions values in accordance with ISO 9296.

SSRs must be enrolled in a hearing conservation program and use hearing protection when they are servicing the system.

Declared noise emission values in accordance with ISO 9296 [1-8]						
Product description: RMID: LTH-4U48	Declared mean A-weighted sound power level, $L_{WA,m}$ (B)		Declared mean A-weighted emission sound pressure level, $L_{pA,m}$ (dB) Bystander[10] / Front[11] / Rear[11]		Statistical adder for verification, K_v (B)	
	Operating	Idling	Operating	Idling	Operating	Idling
• Configuration A per Flex Test Report[9] • Fan Duty set to 60% - ~14500 rpm inlet Rotor setpoint • Estimated Operating Point at 25 C ambient temperature and sea level altitude	8.3	8.3	68.2 / 74.3 / 78.3	68.2 / 74.3 / 78.3	0.3	0.3
• Configuration A per Flex Test Report[9] • Fan Duty set to 100% - ~23800 rpm • Estimated Operating Point at: ○ 28 C ambient and 3050m altitude ○ 35 C ambient and 950m altitude	9.2	9.2	77.5 / 86.5 / 84.7	77.5 / 86.5 / 84.7	0.3	0.3

Notes:

1. Declared level $L_{WA,m}$ is the mean A-weighted sound power level.
2. Declared level $L_{pA,m}$ is the mean A-weighted sound pressure level computed as the arithmetic average of the measurements at the 1-meter bystander positions, or it is measured as the maximum 0.5-meter operator position at the front or rear face with the doors opened.
3. The statistical adder for verification, K_v, is a quantity to be added to the declared mean A-weighted sound power level, $L_{WA,m}$ such that there will be a 95% probability of acceptance, when using the verification procedures of ISO 9296, if no more than 6.5% of the batch of new equipment has A-weighted sound power levels greater than ($L_{WA,m}$ + K_v).
4. The quantity $L_{WA,c}$ (formerly called L_{WAd}), can be computed from the sum of $L_{WA,m}$ and K_v.
5. Measurements are made in conformance with ISO 7779 and declared in conformance with ISO 9296.
6. **B**, **dB**, abbreviations for bels and decibels, respectively. 1 B = 10 dB.
7. Under certain environments, configurations, system settings and/or workloads, fan speeds are increased resulting in higher noise levels.
8. **Notice:** Government regulations (such as those prescribed by OSHA or European Community Directives) may govern noise level exposure in the workplace and may apply to you and your server installation. The actual sound pressure levels in your installation depend upon a variety of factors, including the number of racks in the installation; the size, materials, and configuration of the room where you designate the racks to be installed; the noise levels from other equipment, the room ambient temperature, and employees' location in relation to the equipment. Further, compliance with such government regulations also depends upon a variety of additional factors, including the duration of employees' exposure and whether employees wear hearing protection. IBM recommends that you consult with qualified experts in this field to determine whether you are in compliance with the applicable regulations.
9. Configuration A:
   a. 48 DDR5 DIMM
   b. 6 IBM FCM4 SSD
   c. 42 IBM Bender pass2 Loading Card SSD
   d. 10 SAS 24G HBA
   e. 2 NVIDIA ConnectX-7 400G
   f. 4 NVIDIA ConnectX-7 200G
   g. 4 FMS029 PSU
   h. 12 60x56mm Sunon Fans
10. Estimated from Average Sound Pressure at 0.5m microphone positions.
11. Operator Position; 0.5m microphone position.

Warning: IBM Storage Scale System 7.0.0.0 software introduces changes that increase the fan floor speed for IBM Storage Scale System 6000, which consequently causes higher acoustical levels.

Figure 19. Acoustical declaration with noise hazard notice

Shock and vibration specifications for IBM Storage Scale System 6000 enclosures

Table 14 on page 35 and Table 15 on page 36 provide the shock and vibration testing results for your IBM Storage Scale System 6000 system.

Table 14. Shock testing results		
Shock categories	**Test level**	**Sweep rate of shocks**
Operational	3.5 g 3 ms 1/2 Sine	5 shocks to machine base
Nonoperational	35 g trapezoidal pulse with Δ V 136 in/s	6 shocks with 1 shock to each face

Table 15. Vibration testing results			
Vibration categories	**Test level**	**Frequency range**	**Test length**
Operational[1]	0.10 g swept sine	5 - 500 Hz	30 minutes (performed in all 3 axes)
Nonoperational	1.04 g swept sine	2 - 200 Hz	15 minutes (performed in all 6 faces)
[1] A sine-on-random profile test was performed, which combined this vibration profile with 0.06 g peak sine, dwells at 50 Hz and 60 Hz. These sine tests can be performed separately.			

Table 16. Random vibration PSD profile breakpoints (operational)									
Class	**5 Hz**	**17 Hz**	**45 Hz**	**48 Hz**	**62 Hz**	**65 Hz**	**150 Hz**	**200 Hz**	**500 Hz**
V1L/V2	2.0×10^{-7}	2.2×10^{-5}	2.2×10^{-5}	2.2×10^{-5}	2.2×10^{-5}	2.2×10^{-5}	2.2×10^{-5}	2.2×10^{-5}	2.2×10^{-5}
• All values in this table are g^2/Hz. • For reference only, no test is required.									

Table 17. Random vibration PSD profile breakpoints (nonoperational)							
Class	**2 Hz**	**4 Hz**	**8 Hz**	**40 Hz**	**55 Hz**	**70 Hz**	**200 Hz**
V1L/V2	0.0010	0.0300	0.0300	0.0030	0.0100	0.0100	0.0010
• All values in this table are g^2/Hz. • For reference only, no test is required.							

Powering on IBM Storage Scale System 6000

Powering on an IBM Storage Scale System 6000 by connecting cables

The following devices are needed to power on an IBM Storage Scale System 6000:

- IBM Storage Scale System 6000
- 4 power cables with C19 connectors
- 4 AC power sockets (for more information, see Table 7 on page 29)
- A monitor with a mini display-port cable
- A USB keyboard

Complete the following steps to power on an IBM Storage Scale System 6000.

1. Keep the IBM Storage Scale System 6000 on a bench.

Figure 20. IBM Storage Scale System 6000

2. Identify all the ports that are needed to power on the system.

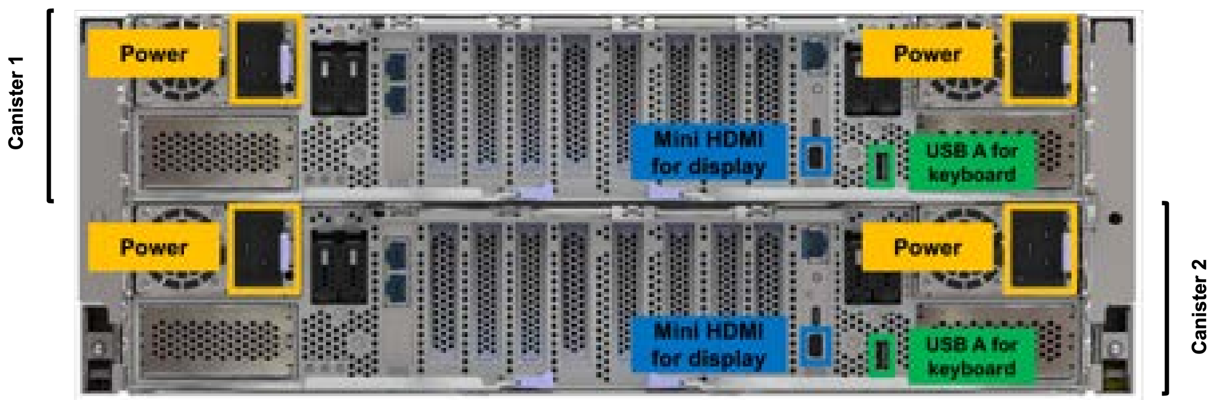


Figure 21. IBM Storage Scale System 6000 ports

3. Connect a monitor to the mini HDMI display port.

Figure 22. Monitor to display port connection

4. Connect a keyboard to the USB A port.

Figure 23. Keyboard to the USB A port connection

5. Connect the four power cables to the PSUs.

Figure 24. Power cables

6. Connect the four power cables to the power distribution unit (PDU).

   The system powers on automatically and boots. After approximately 5 minutes, the system displays a command prompt.

Planning for site preparation

This information is intended to help you prepare your physical site for the installation of an IBM Storage Scale System 6000. Marketing and installation planning representatives are also available to help you plan your installation. Proper planning for your new system facilitates a smooth installation and a fast system start.

The use of the terms, “server”, “processor”, “system”, and “all models” in the following information refer to the IBM Storage Scale System 6000.

Site preparation and physical planning

For detailed guidelines about site preparation and physical planning, see the Site preparation and physical planning document.

Planning for hardware installation

Before the IBM SSR installs the system, you must plan the physical configuration and the initial data configuration.

Certain physical site specifications must be met before you can set up your system. This activity includes verifying that adequate space is available, and that requirements for power, network and environmental conditions are met.

Important: The IBM SSR refers to these system configuration details when they perform the system installation, so it is important that these records are complete and accurate.

1. Review all the guidelines in the Planning topics to understand where the system can be installed and identify all prerequisites, such as building structure, equipment rack, environmental controls, power supply, and accessibility.

   If there are dependencies identified, you need to resolve them before the IBM SSR installs your system hardware.
2. Use the hardware locations of enclosures and other devices to identify the rack locations where the IBM SSR will install each enclosure.

Planning for racks

Plan for IBM or non-IBM racks according to the IBM Storage Scale System 6000 requirements.

For detailed rack specification information about the 5149-R42 rack and the 7965-S42 rack, see Planning for the 7965-S42 rack.

If you do not have a 5149-R42 rack or 7965-S42 rack and want to install the system into a non-IBM rack, see "IBM Storage Scale System 6000 requirements" on page 29. Ensure that the physical environment meets the specified requirements, such as rack space, power, and environmental conditions.

Reviewing IBM Storage Scale System 6000 location guidelines

Consult these guidelines when you plan the location of an IBM Storage Scale System 6000 and any existing IBM Storage Scale System in your environment, including any IBM Storage Scale client or protocol node.

IBM Storage Scale System 6000 contains two server canisters. Each IBM Storage Scale System 6000 cluster consists of the following components:

- One or more IBM Storage Scale System 6000 systems – in a rack, each system requires a space of 4U (standard rack units).
- One EMS node (IBM Storage Scale System Utility Node) – in rack, requires a space of 2U (standard rack units).
- 1 GbE network switch for management network – in rack, requires a space of 1U (standard rack units).
- High-speed InfiniBand or Ethernet network for internode communication – in rack, requires a space of 1U (standard rack units).

IBM Storage Scale System 6000 model

The model contains two hot swappable server canisters that have two CPUs each.

Server canister orientation for inserting or removing DIMMs

The following figure shows the CPUs and the location of the DIMM slots. It shows one of the two server canisters for an IBM Storage Scale System 6000 (MTM 5149-F48), which is open and oriented for DIMM removal and replacement.

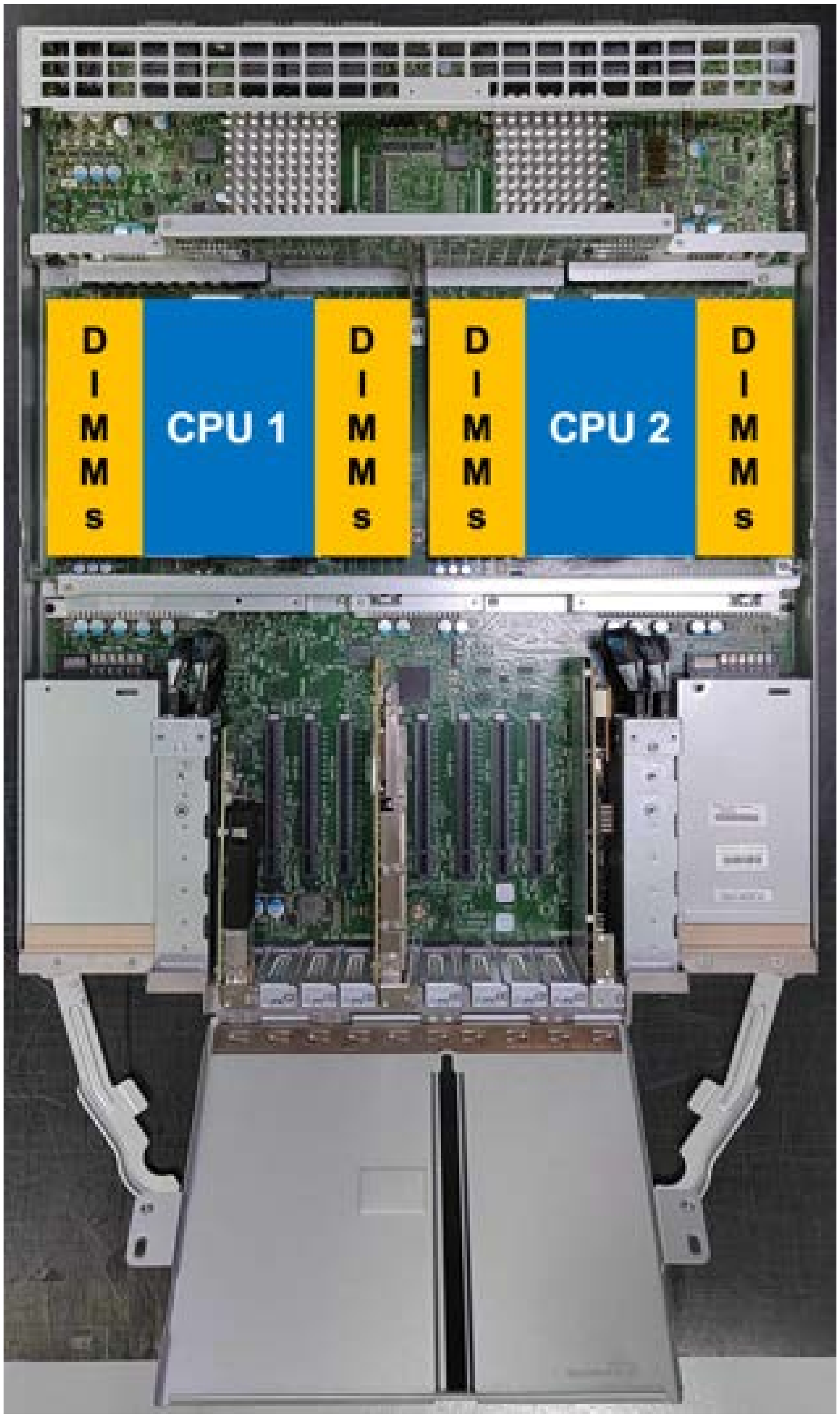


Figure 25. Location of CPU and DIMMs within the server canister

Planning for power for server canister

Each enclosure is provided power through four power supplies. Either of the power modules can power the enclosure independently if there is a loss of input power to the other power supply in the enclosure.

Plan to connect the power cords of the power supplies on the left side of the enclosures (when viewed from the rear) to one power source, and connect the power cords of the power supplies on the right side of the enclosures to another power source.

Attention: The power cords are the main power disconnect. Ensure that the socket outlets are located near the equipment and are easily accessible.

The following figure shows the rear view of an IBM Storage Scale System 6000. The power modules are located on the sides of the IBM Storage Scale System 6000.

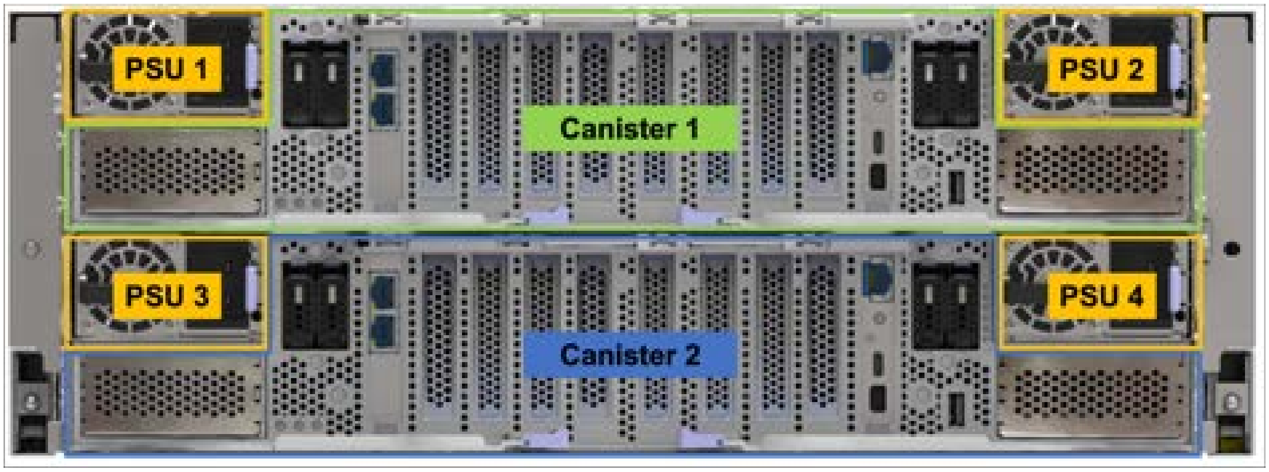


Figure 26. Rear view of an IBM Storage Scale System 6000

Planning for cable connections to PDUs

Each enclosure must be connected to a pair of power outlets by selecting appropriate feature codes while ordering the system. The following table lists the feature codes of the power cords.

Table 18. Power cord feature codes

Feature codes	Description
ELC5	Power cable - Drawer to IBM PDU (250V/10A) C13/C20
END7	Power cord - Drawer to IBM PDU (250V/10A) C13/C20
4558	Power cord - 4.3 m (14.1 ft), drawer to IBM PDU (250 V/10 A) C13/C14
END8	Power cord - 3.0 m (9.8 ft), drawer to IBM PDU (250 V/10 A) C13/C14

Attention: Ensure that sufficient power supply circuits are available to provide the total power requirements of the equipment that is connected to each power supply circuit.

EMS node power planning

The following table lists the documentation that can be used as a reference while planning for power requirements.

Table 19. Power planning reference information for IBM Storage Scale System Utility Node

For this information...	Go to...
PSU	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=overview-power-supply-units-psus
IBM Storage Scale System Utility Node specifications	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=sheets-storage-scalesystem-utility-node-specifications

Table 19. Power planning reference information for IBM Storage Scale System Utility Node (continued)	
For this information...	**Go to...**
Model 7965-S42 rack specifications	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=rack-model-7965-s42-specifications
Rack installation specifications for racks that are not purchased from IBM	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=pr-rack-installation-specifications-racks-that-are-not-purchased-from
IBM Storage Scale System Utility Node requirements	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=planning-storage-scalesystem-utility-node-requirements
Connections for the IBM Storage Scale System Utility Node	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=cables-connections-utility-node
Installing and connecting the management server for an IBM Storage Scale System	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=task-installing-connecting-ess-management-server
PSU LEDs	https://www.ibm.com/docs/en/storage-scale-system/5149-23E?topic=leds-power-supply-unit-psu

Physical installation planning

Before you set up your system environment, you must verify that the prerequisite conditions for the system are met.

This information applies to the supported hardware components. Answer the following questions before you start the installation process:

- Does your physical site meet the environment requirements for your system?

  The system requires the physical site to meet the environment requirements. For more information, see "Environmental requirements" on page 30.

- Do you have adequate rack space for your hardware?

  The system requires two Electronic Industries Alliance (EIA) units for each IBM Storage Scale System 6000. For more information, see "Dimensions and weight requirements for rack installation" on page 30.

- Do the power circuits that you are planning to use have sufficient capacity and the correct sockets for your installation?

  – A clearly visible and accessible emergency power off switch is required.

  – For redundancy, at least two independent power circuits are required.

  The appropriate power requirements for each power supply unit must be provided. For more information, see "Power requirements for each power supply (four per enclosure)" on page 29.

- Have you provided appropriate connectivity by preparing your environment?

  Consult the following sections to learn about IP addresses requirements and restrictions, also about supported hosts and user interfaces.

Operating environment

To use the system, you must meet the minimum hardware and software requirements and ensure that the other operating environment criteria are met.

Supported hosts

To access the high-performance scale-out clustering capabilities of the IBM Storage Scale System 6000, users must first install a supported IBM Storage Scale license on one or more client nodes (preferably in a remote cluster). These nodes allow clients to mount the IBM Storage Scale filesystem (or NFS or SMB protocol export), and run applications to read or write data. For a list of supported client host types and operating systems, see IBM Storage Scale Frequently Asked Questions.

User interfaces

The system provides the following user interfaces:

- The management GUI, which is a web-accessible graphical user interface (GUI) that supports flexible and rapid access to storage management information. The IBM Storage Scale System 6000 GUI also provides a directed maintenance procedure (DMP) for drive replacement.
- A command-line interface (CLI) that uses Secure Shell (SSH).

IP address allocation and usage

As you plan your installation, you must consider IP address requirements and service access for the system.

IBM Storage Scale System 6000 uses 100 GbE / 200 GbE / 100 Gb EDR / 200 Gb HDR network for cluster communication and data transport. For IP port usage requirements, see IBM Storage Scale Documentation.

Attention: The address for a management IP cannot be the same as the service IP. Using the same IP address causes communication problems.

Planning your network and storage network

You need to plan to provide the network infrastructure and the storage network infrastructure that your system requires.

Planning for high-speed network adapter

Review the prerequisite system requirements and installation considerations before deployment.

It is helpful to be familiar with the following term when you are considering the features of the high-speed network adapter for these connections.

Remote Direct Memory Access (RDMA)
RDMA is a networking standard that allows the adapters to transfer data directly to or from the endpoints in a connection without using CPU resources on either of the endpoints. These transfers occur simultaneously with other system operations and do not impact overall system performance. When the system performs an I/O operation over an RDMA-based connection, data is sent directly to the network, which reduces latency and increases the speed of data transfers.

Review the following elements when you are deciding to install the high-speed network adapters:

- The following adapter is available:
  - (FC AJZL) CX-6 VPI in PCIe form factor
    - InfiniBand: HDR200 200 Gb / HDR100 100 Gb / EDR 100 Gb

- Ethernet: 100 GbE / 200 GbE

Support for high-speed switches

The following list includes all the high-speed switches that are compatible with an IBM Storage Scale System 6000.

- IBM MTM 5149-N64 – NDR IB high-speed switch
- IBM MTM 8831-H40 – HDR IB high-speed switch
- IBM MTM 8831-S52 1G management switch
- IBM MTM 8831-00M – 100 GbE high-speed network switch
- Cisco Nexus 9000 200 GbE high-speed network switch. For more information, see Cisco Nexus 9300-GX Series Switches Data Sheet in Cisco documentation.

Planning for cables

Connections for IBM Storage Scale System 6000

Care must be taken to note the orientation of each IBM Storage Scale System 6000 so that the interconnect cables are properly connected.

The following figure shows the ports on IBM Storage Scale System 6000.

Figure 27. Orientation of ports on IBM Storage Scale System 6000

Table 20. Ports on IBM Storage Scale System 6000

Item number	Feature	Description
1	Power supply unit	Provides power supply to the system
2	Management network and SSR access	Provides two RJ45 10 GbE ports, one for management (top port) and one for SSR access (bottom port)
3	BMC or host	Provides one RJ45 1 GbE port for BMC connection
4	USB mini-C	Reserved for service
5	Mini HDMI	Provides video display connection
6	USB A	Provides keyboard connection

Supported host interface adapters, cables, and transceivers

Each IBM Storage Scale System 6000 server canister has eight Gen 5 x16 PCIe slots to support optional host interface adapters. The host interface adapters can be supported in any of the interface slots. The following tables provide an overview of the supported host interface adapters, also the cables and transceivers compatible with them.

Table 21. Summary of supported host interface adapters

Protocol	Feature code	Ports	FRU part number	Quantity supported per enclosure
CX7 NDR 200 Gb, dual VPI	AJQS	2 per adapter	01LL984	Maximum quantity of 8, between the two adapters. Order increments allowed for each adapter feature = 2 (2, 4, 6, or 8).
CX7 NDR 400 Gb, single port, in PCIe form factor	AJQQ	1 per adapter	01LL983	

Table 22. Summary of supported active optical cables for host interface adapters - InfiniBand protocol

Description	Length	Feature code	Used with switch/ adapter
NVIDIA HDR 200 Gb QSFP56 MMF. For more information, see NVIDIA documentation.	5 m	EBBB	CX-7 dual port VPI QSFP112
	30 m	EBBF	
NVIDIA splitter, twin-port, HDR 400 Gb/s to 2x 200 Gb/s, OSFP to 2x QSFP56 For more information, see NVIDIA documentation.	5 m	AJR8	NDR TOR to CX-7 dual port VPI
	30 m	AJR9	

Table 23. Summary of supported passive optical cables for host interface adapters - InfiniBand and Ethernet protocols

Description	Length	Feature code	Used with switch/ adapter
NVIDIA, MMF, MPO12 APC to MPO12 APC	5 m	AJQV	NDR200 (CX-7 only) Single: InfiniBand / Ethernet
	30 m	AJQY	
NVIDIA, MMF, MPO12 APC to 2xMPO12 APC	5 m	AJR2	NDR200 (CX-7 only) Splitter: InfiniBand / Ethernet
	30 m	AJR5	
For more information, see NVIDIA documentation.			

Table 24. Summary of supported transceivers for host interface adapters				
Description	**Length**	**Feature code**	**Protocol**	**Used with switch/ adapter**
Single port NDR OSPF transceiver, 400 Gbps, flat-top single MPO/ APC, 8 Watts. For more information, see NVIDIA documentation.	30 m	AJQU	• InfiniBand • Ethernet	CX-7, single port OSFP.
Single port NDR QSFP transceiver, 400 Gbps, flat-top MPO, 850 nm MMF, SR4. For more information, see NVIDIA documentation.	Up to 30 m	AJR1	• InfiniBand • Ethernet	CX-7 dual port VPI QSFP112.
OSFP: octal small form factor pluggable. QSFP: quad small form factor pluggable.				

Planning for adapters

Each IBM Storage Scale System 6000 server has two PCIe interface slots to support extra adapters.

The following two adapters are supported for use in the IBM Storage Scale System 6000:

- FC: AJQQ - CX7 NDR 400 GB, single port, in PCIe form factor
- FC: AJQS - CX7 NDR 200, dual VPI

Supported environment

Refer to the IBM support website for up-to-date information about the supported environment for your system.

Environmental topics can include updates about the following items:

- Host attachments
- Switches
- Firmware levels
- Other support hardware

Planning worksheets: Customer task

Customers are responsible for completing the system planning worksheets. The customer then provides the worksheets to the IBM SSR when they install and configure the system.

Installation worksheet

This worksheet is intended for customers as part of the TDA process to outline the items that are required to be implemented by the SSR during setup.

Important: Click here to download the blank worksheet.

Customer task

Customer fills out the desired IP addresses.

SSR task

The SSR uses this information to properly set the desired IP addresses on each canister and the management server (if applicable).

Note: The management IP addresses must be on the same subnet as the management server bridge interface. If you already have an existing management server, set IP addresses on the same subnet (all must be reachable).

Note: This document does not describe about the high-speed network. Customers must have that in place and ready to fill in (or update) the `/etc/hosts` file on the management server node during the software installation. For reference, see *IBM Storage Scale Install Complete Guide* and *IBM Storage Scale System Deployment Guide*. The high-speed interfaces are on a separate switch and must be defined on a different subnet.

- For instance, if your management interfaces are all 192.168.21.X/24, your high-speed hosts should be any other network. For instance, 172.16.0.0/24. You should use /24 (255.255.255.0), if possible.

Prerequisite (for SSR)

You must be connected to the SSR access port of the target canister (top or bottom).

Figure 28. SSR access port in the Utility Node

Figure 29. Management port in the Utility Node

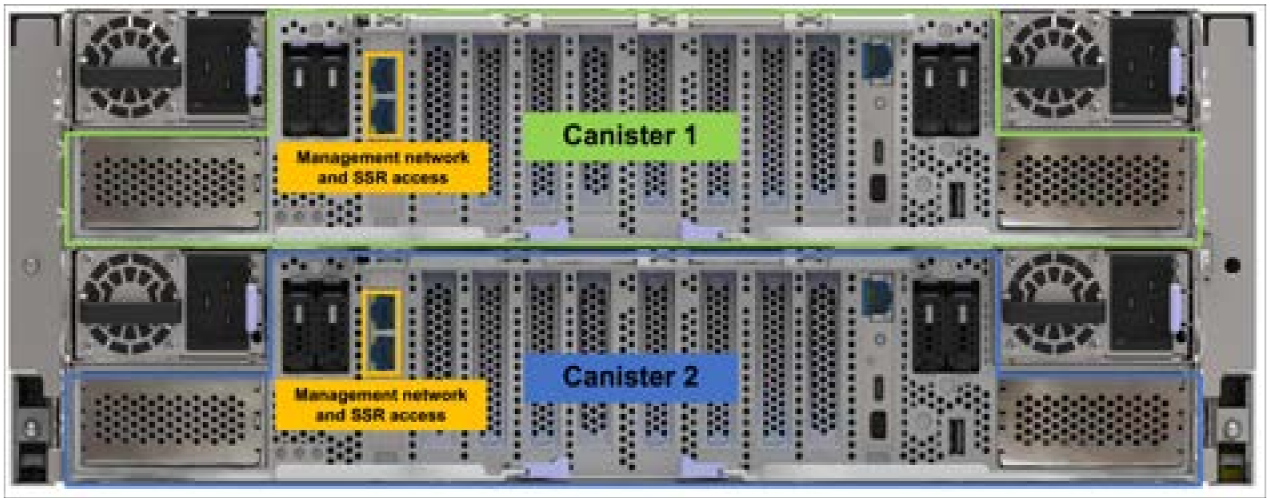


Figure 30. SSR and management access ports in the IBM Storage Scale System 6000

The following figure shows the cabling for IBM Storage Scale Systems.

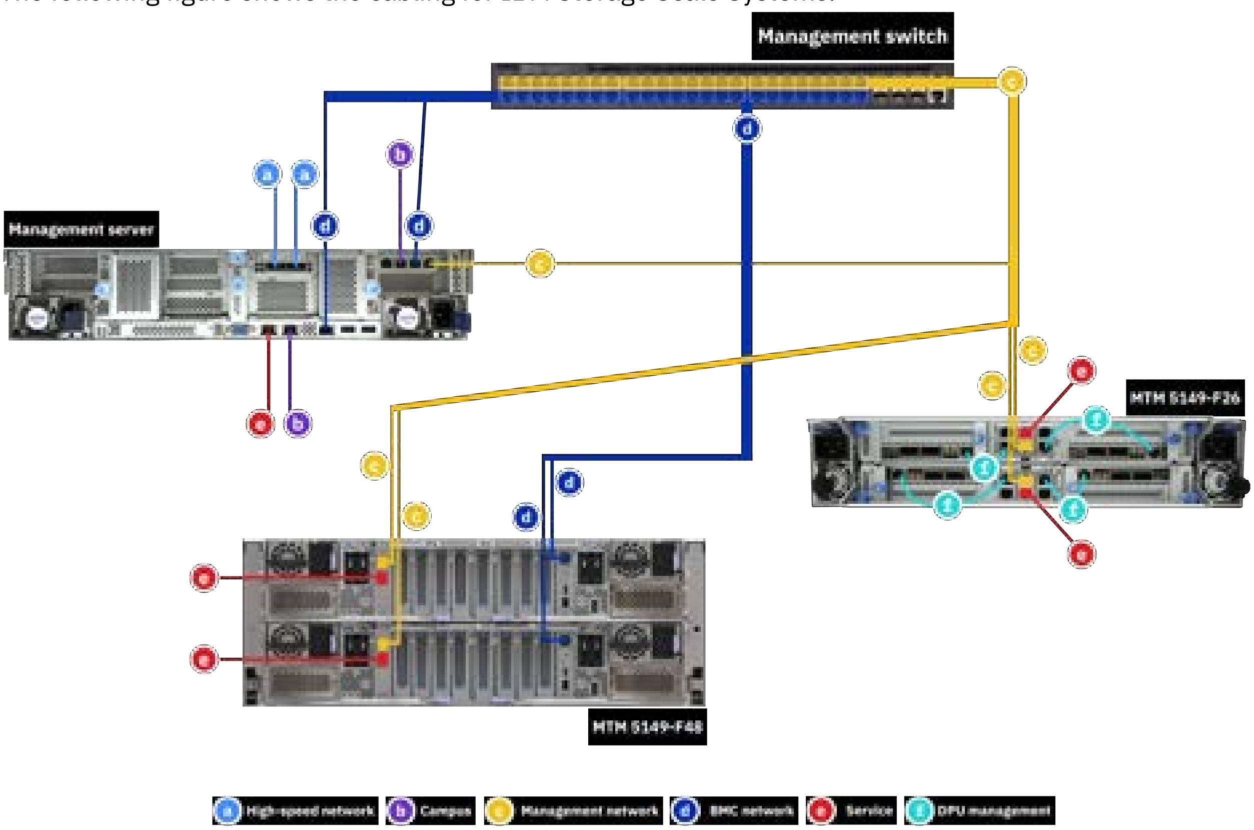


Figure 31. BMC and management network connections

SSR / Customer:

- Ensure that the management switch is in the rack and powered on.
- Ensure that there is one Ethernet cable from the management switch (if there are VLANs, connect from the management VLAN) to port (management port) on each canister as shown in the previous figures.
- If a management server node is received with IBM Storage Scale System 6000, connect two Ethernet cables from the management switch (if there are VLANs, connect from the management VLAN) to port C11-T1 (top port) and to port C11-T2 (second from top).

SSR: General guidance

- For each IBM Storage Scale System 6000 canister in the flow, connect your laptop point-to-point to the SSR port as shown in the previous *SSR access port* figure. You need to use this port to access each canister, perform hardware checks, and set the management IP address.

Reference:

- SSR user ID is **essserv1**
- SSR default password for Power nodes is the server serial number
- SSR default password is the IBM Storage Scale System 6000 serial number
  - Add ‘A’ for left or top canister
  - Add ‘B’ for right or bottom canister
- Other than the SSR access port, there are no default IP addresses set on any canister
- All management interfaces should be connected to the same VLAN switch
- Default physical Power9® Protocol or IBM Storage Scale System 5000 I/O node ports
  - C11-T1 = Management Interface (set by the SSR) - Logical port: `enP1p8s0f0`
- Default physical Power9 management server ports
  - C11-T1 = Management Interface (set by the SSR) - Logical port: `mgmt`
  - C11-T2 = FSP (BMC) interface (set by the SSR) - Logical port: `mgmt`
- Default physical IBM Storage Scale System 6000 ports (per canister)
  - Port 1 – Management port (set by the SSR) - Logical port: `mgmt`
  - The SSR connects a laptop to the SSR port when performing the IBM Storage Scale System 6000 tasks. You need to move the cable when you work on a different canister.

Note: The worksheet has default IP addresses filled out in case they were not provided by the customer during the TDA process. Before setting any default IP addresses have the customer confirm that they are not currently in use by the associated networks.

Customers must fill in the following values so that the SSR is able to perform the required networking tasks. If there are more than one building-block per system type being installed, you might need to add the corresponding rows.

Recommendations:

- Keep all management interfaces on 192.168.x.x/24 (netmask 255.255.255.0).
- Keep all BMC/FSP (HMC1) interfaces on 10.0.0.x/24 (netmask 255.255.255.0).

Note: The management server has an additional FSP connection at C11-T2, which is visible to the operating system.

Important: All IP addresses must on the same subnet. For example:

- All management interfaces on 192.168.x.x/24
- All FSP interfaces on 10.0.0.x/24

IBM Storage Scale System 3500 notes:

The IBM Storage Scale System 3500 has dedicated management (f1) and BMC (f0) interfaces. The management interfaces are connected to the yellow network, while the BMC interfaces are connected to the green trunk ports.

In IBM Storage Scale System 3500, proper switch configuration and VLAN tag should be set to route traffic from the BMC to the blue service/FSP network.

The SSR connects to each canister through a dongle to the bottom USB port. This allows point to point connectivity to the SSR’s laptop.

Utility Node / IBM Storage Scale System 6000 / IBM Storage Scale System 5000 notes:

The IBM Storage Scale System Utility Node has a dedicated SSR port (via Ethernet connection).

The SSR must set IP addresses for both the management interface and the BMC interface. For IBM Storage Scale System Utility Node protocol, it is management or BMC. For management server, it is campus (optional) or BMC.

Campus or remote connection notes:

A Power9 management server campus connection is highly recommended to be set prior to deployment (C11-T3). This allows remote access to the management server and ensures you will not lose a connection when starting the container. Optionally, space is also allocated to set a campus connection on the HMC2 port. This will allow remote access to the FSP which aids the recovery of the node (console/ power control) in case of an outage.

The SSR is provided the commands to set the campus interface (C11-T3) and HMC2 port connection to the customer campus connection. If C11-T3 (Power9 management server) is not in place or the IP address not provided to the SSR at time of Install Complete may still, consider the task complete. It will be up to the customer/LBS to set this connection before deployment can begin.

For the IBM Storage Scale System Utility Node, it is highly recommended for the SSR to set a campus connection on the host. This is in addition to the BMC connection.

IBM Storage Scale System 3500 building-block 1 (lower position in frame)

Note: IBM Storage Scale System 3500 only applies to IBM Storage Scale System 6.1.3.0 and later.

	IP address	**Netmask**
IBM Storage Scale System 3500 canister 1 management interface (left)	192.168.45.30	255.255.255.0
IBM Storage Scale System 3500 canister 1 BMC interface (left)	10.0.0.121	255.255.255.0
IBM Storage Scale System 3500 canister 2 management interface (right)	192.168.45.31	255.255.255.0
IBM Storage Scale System 3500 canister 2 BMC interface (right)	10.0.0.122	255.255.255.0

IBM Storage Scale System 3500 building-block 2 (higher position in frame)

Note: IBM Storage Scale System 3500 only applies to IBM Storage Scale System 6.1.3.0 and later.

	IP address	**Netmask**
IBM Storage Scale System 3500 canister 1 management interface (left)	192.168.45.32	255.255.255.0
IBM Storage Scale System 3500 canister 1 BMC interface (left)	10.0.0.123	255.255.255.0
IBM Storage Scale System 3500 canister 2 management interface (right)	192.168.45.33	255.255.255.0
IBM Storage Scale System 3500 canister 2 BMC interface (right)	10.0.0.124	255.255.255.0

IBM Storage Scale System 5000 building-block 1 (lower position in frame)

	IP address	Netmask
IO node 1 management interface (bottom node in building-block)	192.168.45.21	255.255.255.0
IO node 1 FSP (HMC1 port) interface (bottom node in building-block)	10.0.0.101	255.255.255.0
IO node 2 management interface (top node in building-block)	192.168.45.22	255.255.255.0
IO node 2 FSP (HMC1 port) interface (top node in building-block)	10.0.0.102	255.255.255.0

IBM Storage Scale System 5000 building-block 2 (higher position in frame)

	IP address	Netmask
IO node 1 management interface (bottom node in building-block)	192.168.45.23	255.255.255.0
IO node 1 FSP (HMC1 port) interface (bottom node in building-block)	10.0.0.103	255.255.255.0
IO node 2 management interface (top node in building-block)	192.168.45.24	255.255.255.0
IO node 2 FSP (HMC1 port) interface (top node in building-block)	10.0.0.104	255.255.255.0

IBM Storage Scale System 5000 Power9 protocol nodes

	IP address	Netmask
Protocol node 1 management interface (bottom-most)	192.168.45.50	255.255.255.0
Power9 protocol node 1 FSP (HMC1 port) interface (bottom-most)	10.0.0.110	255.255.255.0
Protocol node 2 management interface (top)	192.168.45.51	255.255.255.0
Power9 protocol node 2 FSP (HMC1 port) interface (top)	10.0.0.111	255.255.255.0

Note: If there are additional building-blocks or Power9 protocol nodes, please add more tables.

Power9 management server (should be only one)

	IP address	Netmask	Gateway
Management server management interface	192.168.45.20	255.255.255.0	N/A
Management server FSP (HMC1 port) interface	10.0.0.100	255.255.255.0	N/A

	IP address	Netmask	Gateway
Management server FSP (C11-T2) interface	10.0.0.1	255.255.255.0	N/A
Management server campus interface (C11-T3)	X.X.X.X	X.X.X.X	X.X.X.X
Management server campus interface (C11-T3) gateway	X.X.X.X	X.X.X.X	X.X.X.X

IBM Storage Scale System Utility Node management server (5149-23E)

	IP address	Netmask	Gateway
Management server (host) campus interface	X.X.X.X	255.255.255.0	X.X.X.X
Management server (host) BMC interface	10.0.0.130	255.255.255.0	N/A
Management server (host) campus interface gateway	X.X.X.X	X.X.X.X	X.X.X.X

IBM Storage Scale System Utility Node protocol (5149-23E)

Protocol node is sold in pairs, so two are listed. Additional nodes may be purchased individually.

Node 1 is lower in the frame, node 2 is higher.

	IP address	Netmask
Protocol (host) management interface	192.168.45.80	255.255.255.0
Protocol (host) BMC interface	10.0.0.140	255.255.255.0
Protocol (host) management interface	192.168.45.81	255.255.255.0
Protocol (host) BMC interface	10.0.0.141	255.255.255.0

IBM Storage Scale System 6000 building-block 1 (lower position in frame)

	IP address	Netmask
Canister 1/A management interface (top node in building block)	192.168.45.60	255.255.255.0
Canister 1/A interface (top node in building block)	10.0.0.60	255.255.255.0
Canister 2/B management interface (bottom node in building block)	192.168.45.61	255.255.255.0
Canister 2/B interface (bottom node in building block)	10.0.0.61	255.255.255.0

IBM Storage Scale System 6000 building-block 2 (higher position in frame)

	IP address	Netmask
Canister 1/A management interface (top node in building block)	192.168.45.62	255.255.255.0
Canister 1/A interface (top node in building block)	10.0.0.62	255.255.255.0
Canister 2/B management interface (bottom node in building block)	192.168.45.62	255.255.255.0
Canister 2/B interface (bottom node in building block)	10.0.0.62	255.255.255.0

Additional notes for customer / LBS:

The root password should be set for the **essserv1** SSR ID to execute commands. When prompted, set the password.

Account type	Account ID	Password
Linux OS	root	

Recommended HMC2 port campus connection (Power9 only). Consider cabling this port to a public network and setting a campus IP. This will allow remote recovery/debug of the management server in case of an outage. The gateway may be needed to for this IP address to work properly.

Management server HMC2 port IP	HMC2 port
Management server HMC2 port IP gateway	

The VLAN tag is required to route traffic on the BMC interface to the service network. This interface is connected to the **green** trunk ports on the switch. This is required only on IBM Storage Scale System 3200 and IBM Storage Scale System 3500. The default is 101.

IBM Storage Scale System 3500 VLAN tag	VLAN TAG	101 (default)

The IBM Storage Scale license that the customer has purchased. The options are Data Access Edition (DAE) or Data Management Edition (DME). When deciding which edition to download and copy to the /`serv` directory, you will need this information.

The IBM Storage Scale license field is only need for IBM Storage Scale System 6.1.3.0 or later.

IBM Storage Scale license	DAE or DME	

Note: Write down any vital information that should be shared with the customer / LBS that you encountered during execution of the Install Complete app.

Chapter 5. Installing

This chapter covers rack and rails installation specifications and process. Also, find the references about configuration, IP addresses, and VLAN tag.

The Service Support Representatives (SSR) can refer to the *IBM Storage Scale Install Complete Guide* to perform a hardware checkout and to set the management interface IP, BMC interface IP, and VLAN tag.

Note: An IBM intranet connection is required to access this content.

Note: Much of the information in this section is intended only for IBM authorized service providers. Customers need to consult the terms of their warranty to determine the extent to which they must attempt any IBM Storage Scale System 6000 maintenance.

Tasks to be done by the Customers or an IBM SSR are marked in this chapter.

Detailed installation steps (SSR task)

This section is intended for IBM authorized service personnel only.

IBM Storage Scale System 6000 system view

Components required to set up EMS

For information on the components required to set up EMS, see Before Installation (Sections 1 through 8) on the *IBM Worldwide Customized Installation Instructions (WCII)* website.

IBM Storage Scale System 6000 system network connection diagram

For information on the IBM Storage Scale System 6000 network connection diagram, see Section 2.3.1 - ESS network connection diagrams on the *IBM Worldwide Customized Installation Instructions (WCII)* website.

Technical and Delivery Assessments (TDAs)

For information on the Technical and Delivery Assessments (TDAs), see Section 3.1 - ESS - Technical and Delivery Assessments (TDAs) on the *IBM Worldwide Customized Installation Instructions (WCII)* website.

Rack details

Prepare the rack for installation work

For information on preparing the rack for installation work, see Section 10.2 - Prepare the rack for installation work on the *IBM Worldwide Customized Installation Instructions (WCII)* website.

Mark EIA locations for rail mounting

For information on verifying the position and installation of new system racks, see Section 10.3 - Mark EIA locations for rail mounting on the *IBM Worldwide Customized Installation Instructions (WCII)* website.

Installing the management server in a rack

For the instructions about how to install an management server on a rack, see Installing the EMS server in a rack and Planning for racks.

Unpacking IBM Storage Scale System 6000 (IBM SSR task)

Before you unpack IBM Storage Scale System 6000, ensure that you have reviewed and followed all the related instructions.

The IBM Storage Scale System 6000 system and related parts are shipped preinstalled in the enclosures. The following list details the components that come preinstalled in the system:

- Two 2U server canisters with memory feature codes
- Four PSUs (80 Plus Titanium)
- Six 4U fan modules
- 24 or 48 drives
- Network adapters (depends on the order)
- Interface cables for network adapters (if ordered)
- Rail kit
- Left and right bezel ear caps
- Two cable management assemblies (CMAs)

Note: You might need a box knife to unpack the IBM Storage Scale System 6000 system.

CAUTION: To lift an assembled enclosure, it is necessary to use a mechanical lift or involve four people. If the enclosure is unpacked and dismantled, it is necessary to involve two people to lift the enclosure.

Perform the following steps to unpack the IBM Storage Scale System 6000 system:

1. Cut the box tape and open the lid of the shipping carton.
2. Remove the rail kit box and set them aside in a safe location.
3. Remove the cable management assembly kit box and set them aside in a safe location.
4. Lift the front and rear foam packing pieces from the carton.
5. Remove the four corner reinforcement pieces from the carton.
6. Using the box knife, carefully cut the four corners of the carton from top to bottom.
7. Fold the sides and back of the carton down to uncover the rear of the IBM Storage Scale System 6000 system.

   If necessary, carefully cut along the lower fold line of the sides and remove them.
8. Carefully cut the raised section of the foam packing away from the rear of the enclosure.
9. Carefully cut open the bag covering the rear of the enclosure.
10. To take the system out of the shipping carton, align the lift plate of the mechanical lift with the rear of the enclosure and slide the enclosure onto the lift.

Installing support rails for the IBM Storage Scale System 6000 (IBM SSR task)

Before you install the IBM Storage Scale System 6000 system, you must first install the support rails for it.

To install the support rails for the IBM Storage Scale System 6000 system, you must first prepare the rails for installation by completing steps 1 and 2; then, complete steps 3-5 to install the outer rail assemblies into the rack.

1. Prepare both the left and right rails for installation by following the next substeps:

   a) Pull out the inner rail until it is fully extended. See **a** in the following figure.

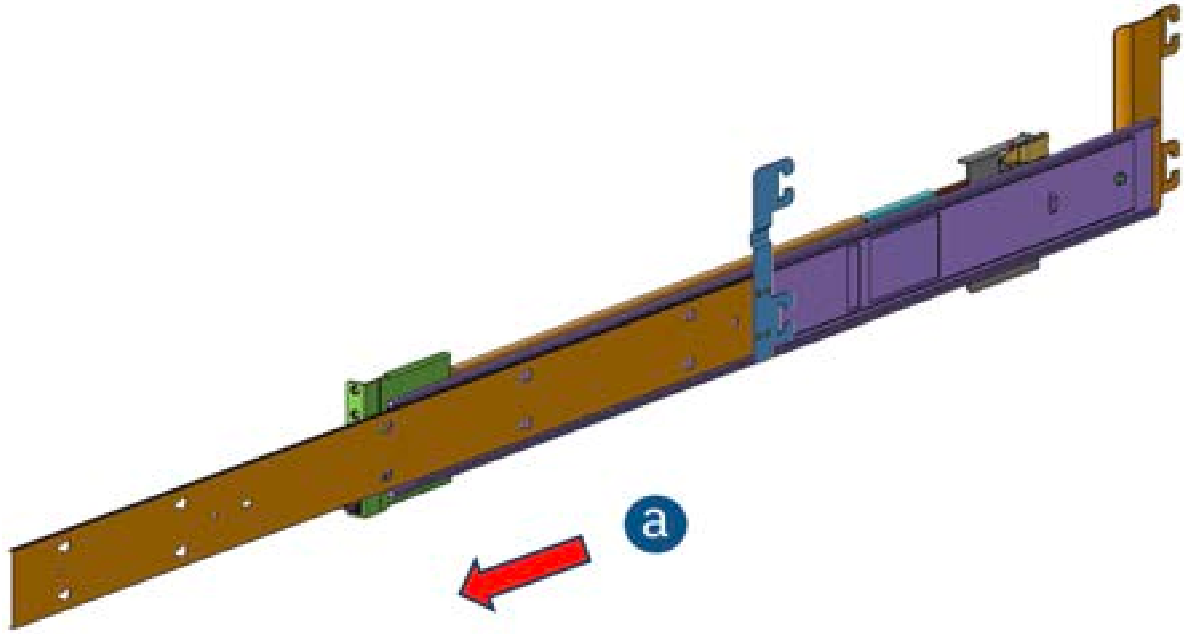


Figure 32. Pulling the inner rail

b) Slide and hold the release tab in the direction shown in **a** in the next figure.

c) Remove the inner sliding rail from the outer mounting rail. See **b** in the following figure.

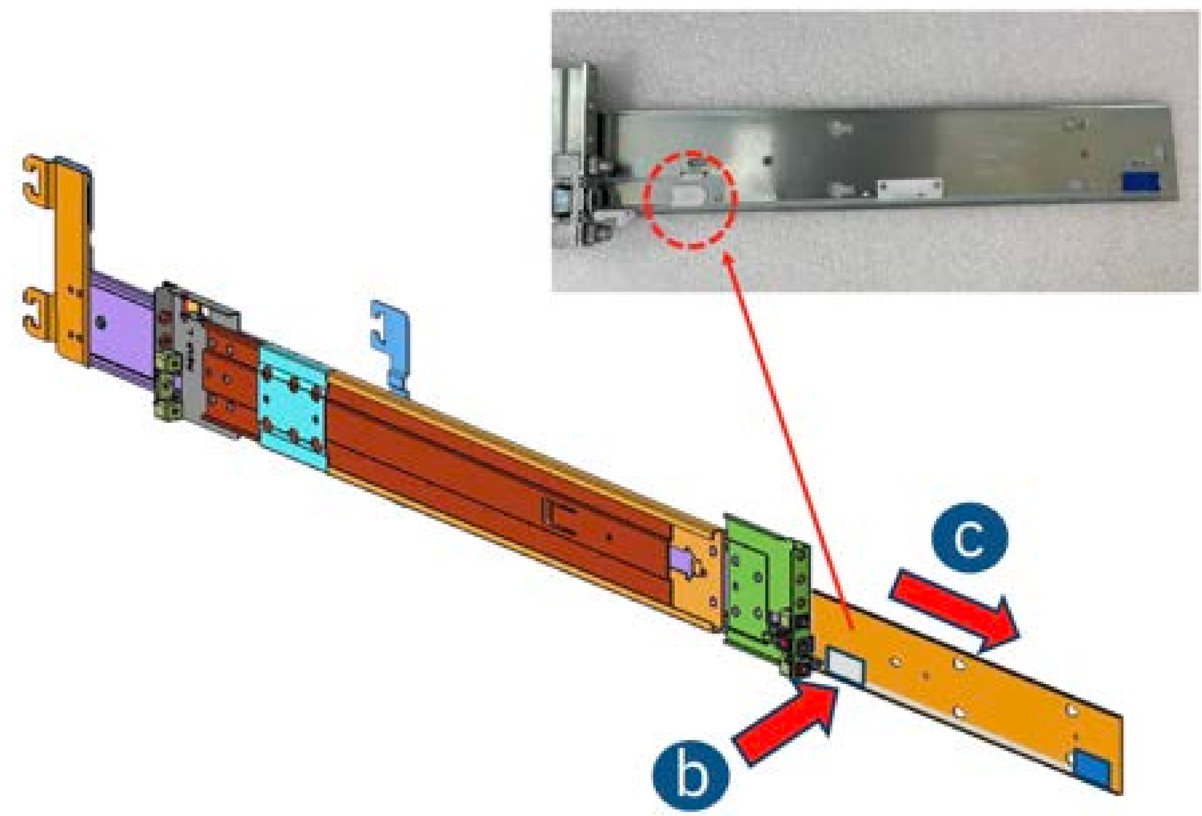


Figure 33. Holding the release tab

2. Install the sliding rails.

   a) Align the notches on the inner rail with the pins on the side of the enclosure. See **a** in the following figure.

   b) Slide the rail towards the rear of the system to lock it in place. See **b** in the following figure.

c) Fix the rail with 1x M4 screw. One screw is required per inner bracket. See **c** in the following figure.

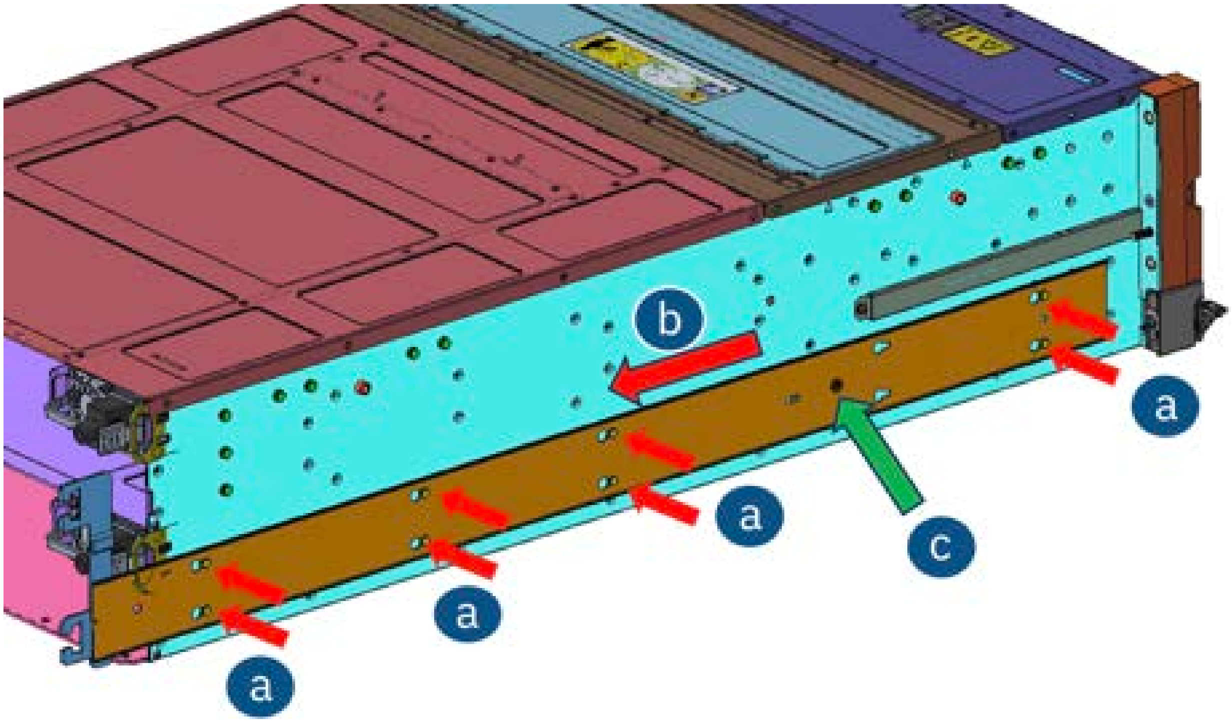


Figure 34. Installing the sliding rails

3. Mount the rails to the rack posts. For reference about the pins location in the front and the rear rack posts, see the following figure.

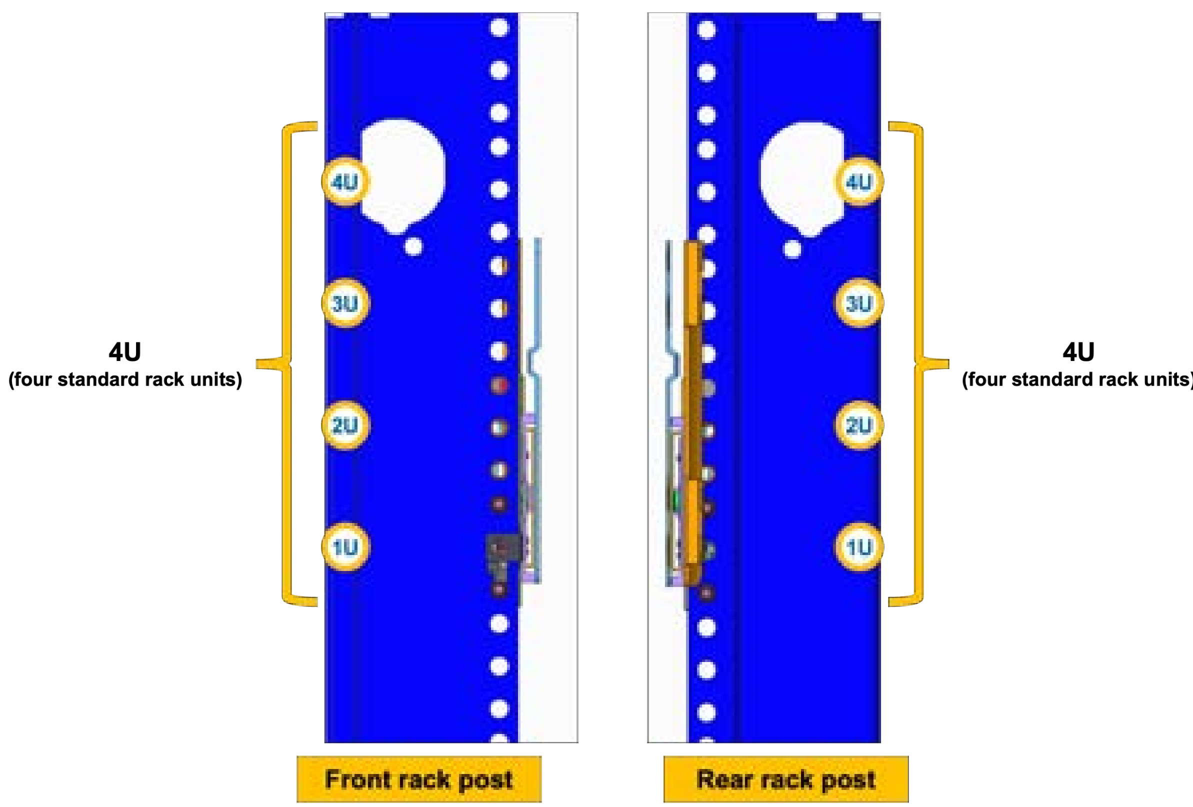


Figure 35. Rear and front rack posts

a) Retract and hold the front retention bracket. See **a** in the following figure.

b) Insert the pegs on the mounting flange into the rack holes. See **b** in the following figure.

c) Release the front retention bracket. See c in the following figure.

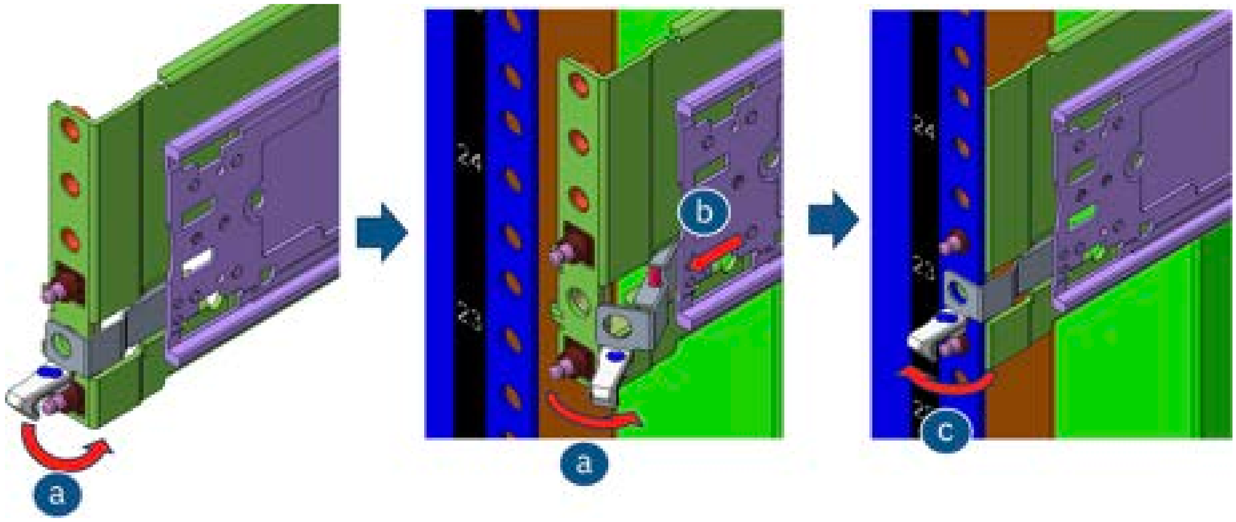


Figure 36. Mounting the front left rail to the rack posts

4. Release the front retention bracket.
   a) Retract and hold the rear retention bracket. See a in the following figure.
   b) Insert the pegs on the mounting flange into the rack holes. See b in the following figure.

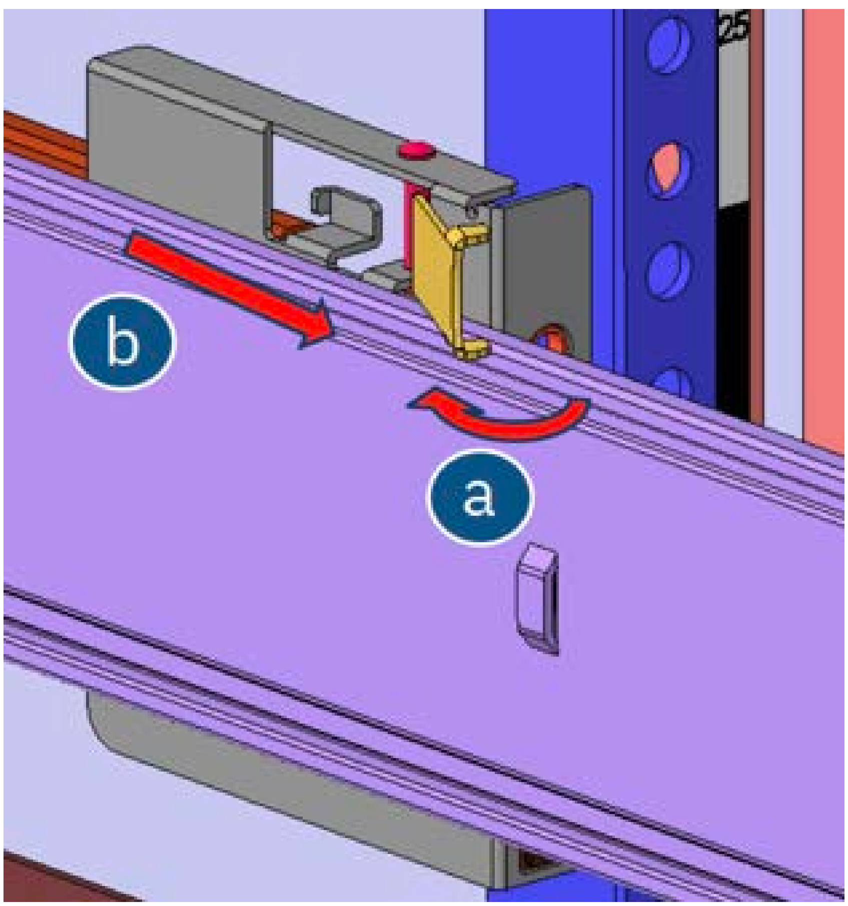


Figure 37. Releasing the retention bracket of the rear right rail

5. Fix the system in the rack.
   a) Release the rear retention bracket. See a in the following figure.
   b) Fix the rails on the rack post by using 2x M5 screws. See b in the following figure.

Note: Make sure that both rails are mounted at the same vertical position on both sides of the rack.

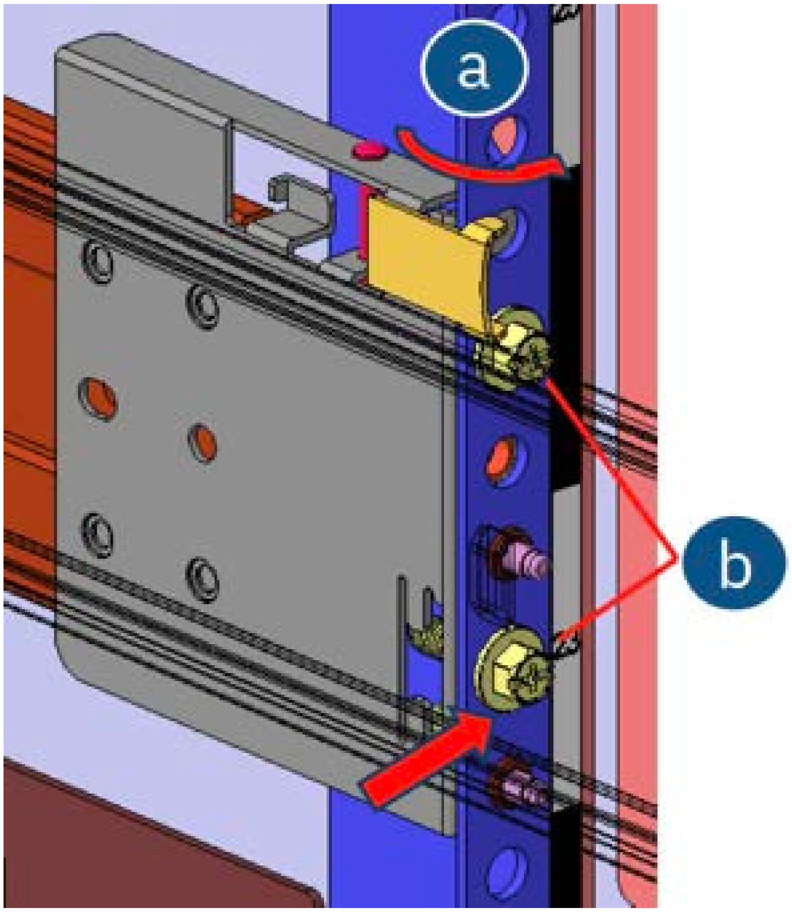


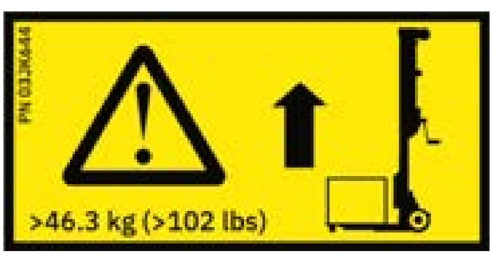
Figure 38. Fixing the system in the rack

For enclosure installation, see "Installing enclosures (IBM SSR task)" on page 60.

Installing enclosures (IBM SSR task)

Following your enclosure location plan, install the IBM Storage Scale System 6000 system.

CAUTION:

- 

  To lift an assembled IBM Storage Scale System 6000 system, you must use a mechanical lift or involve four people. Do NOT attempt to reduce the enclosure weight by temporarily removing server components from the enclosure.
- Install an IBM Storage Scale System 6000 system only onto the IBM Storage Scale System 6000 system rails supplied with the enclosure.
- Load the rack from the bottom up to ensure rack stability. Empty the rack from the top down.

To install an enclosure, complete the following steps:

1. Lift the system by using a mechanical lift or involving four people.

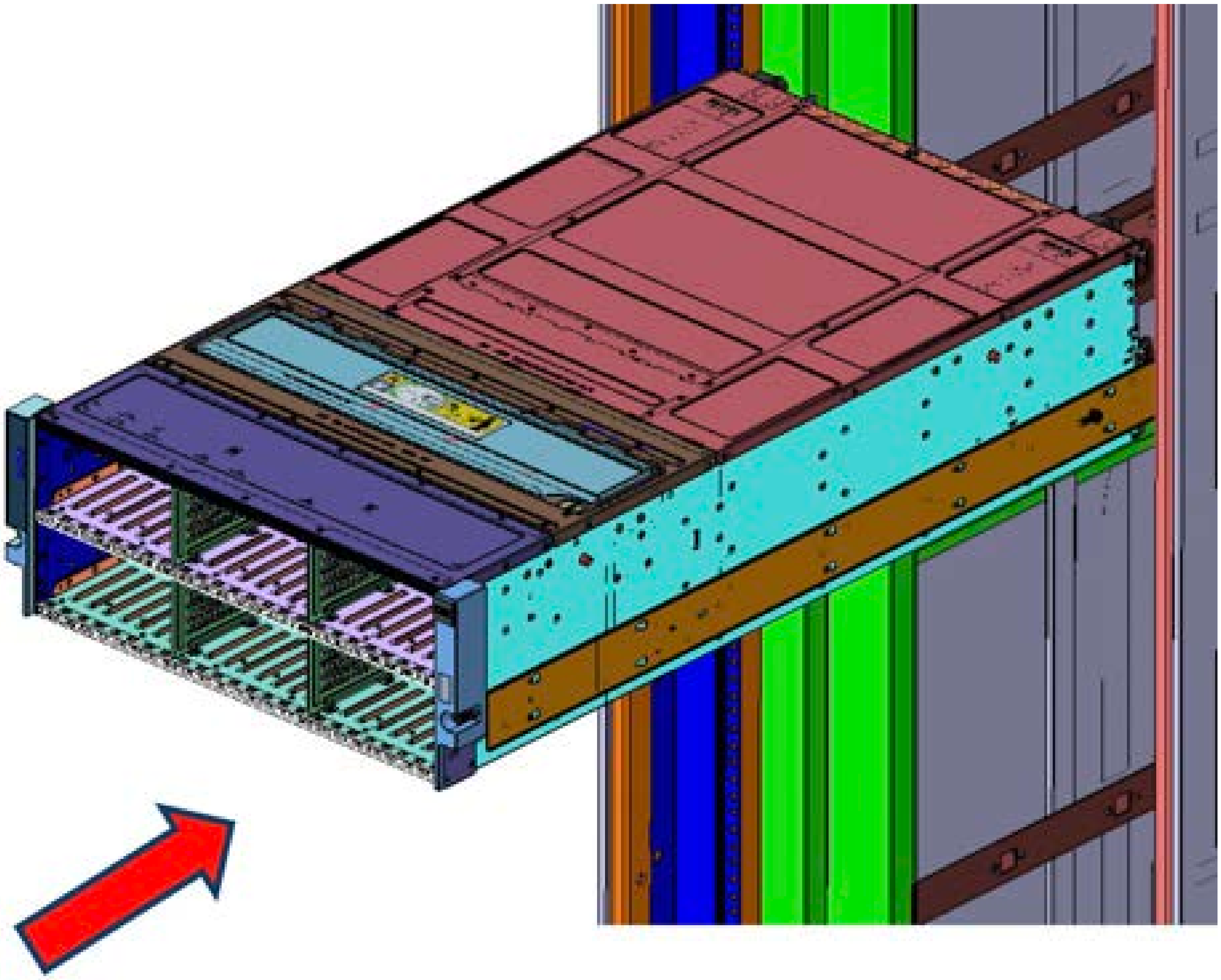


Figure 39. Lifting the system by using a mechanical lift

2. On both sides, engage the inner rail with the outer rail.
3. Slide the system until it stops.
4. Pull the release tab on the inner rail to release the lock. Then, push the system further into the rack.

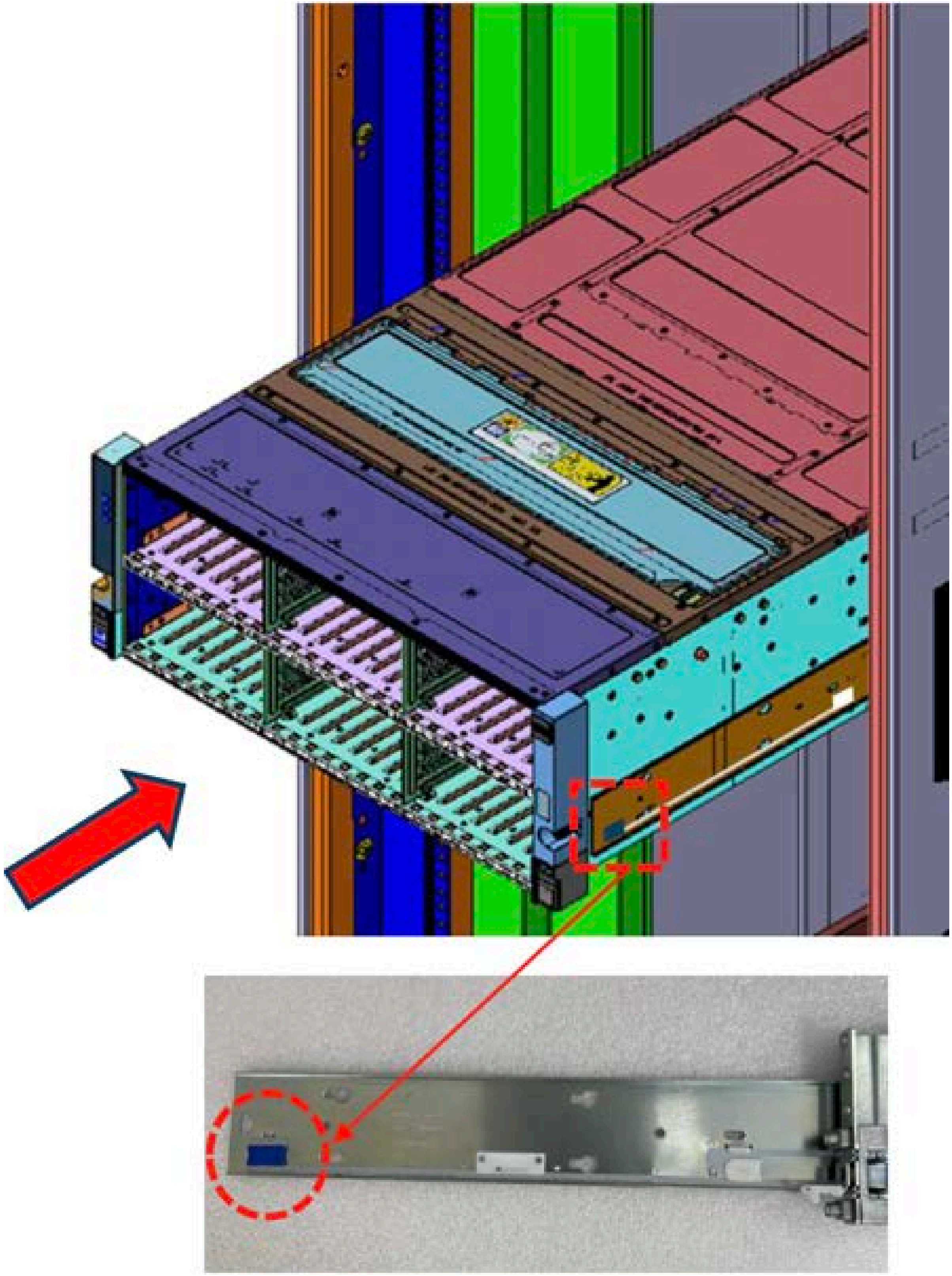


Figure 40. Release tab of the inner rail

5. On each ear, use the slam latch to lock the system assembly on the rack by installing one M5 screw. See the following figure for reference.

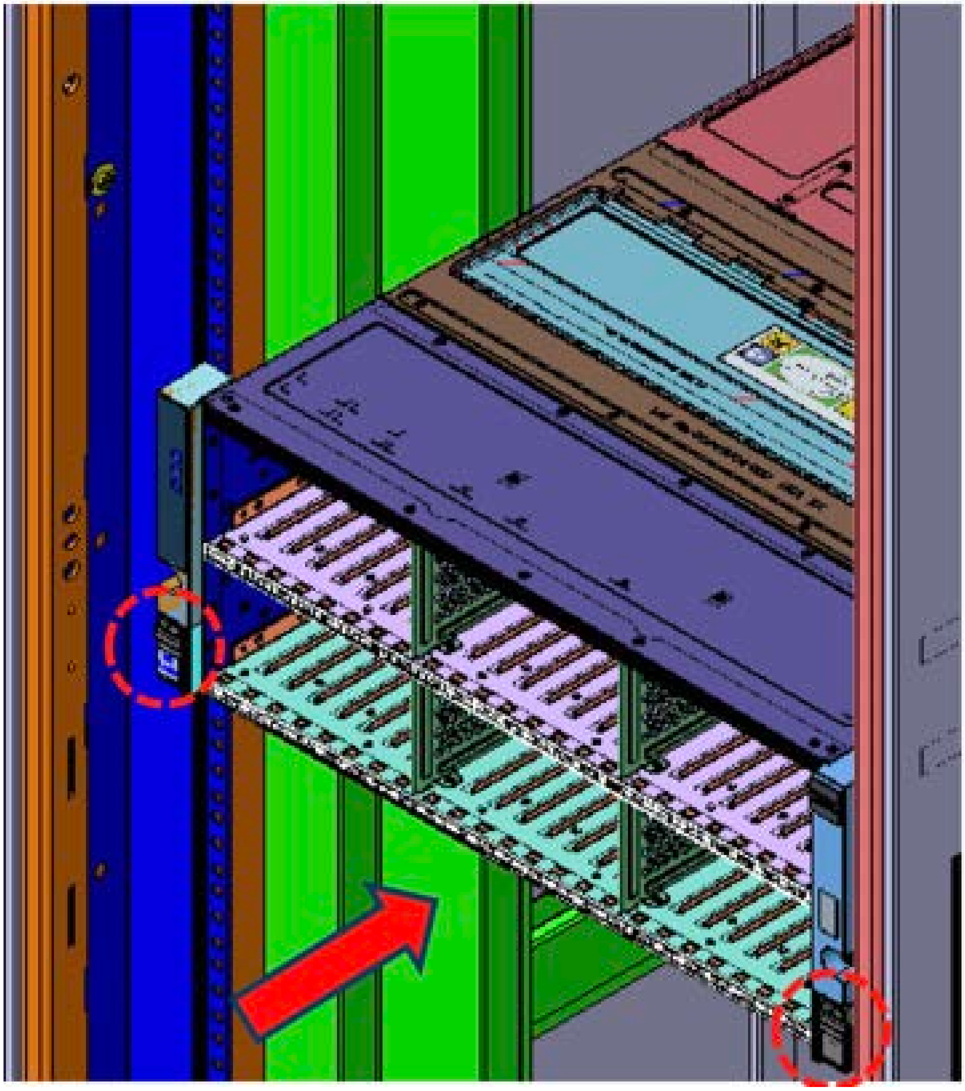


Figure 41. Slam latches on the ears of the system

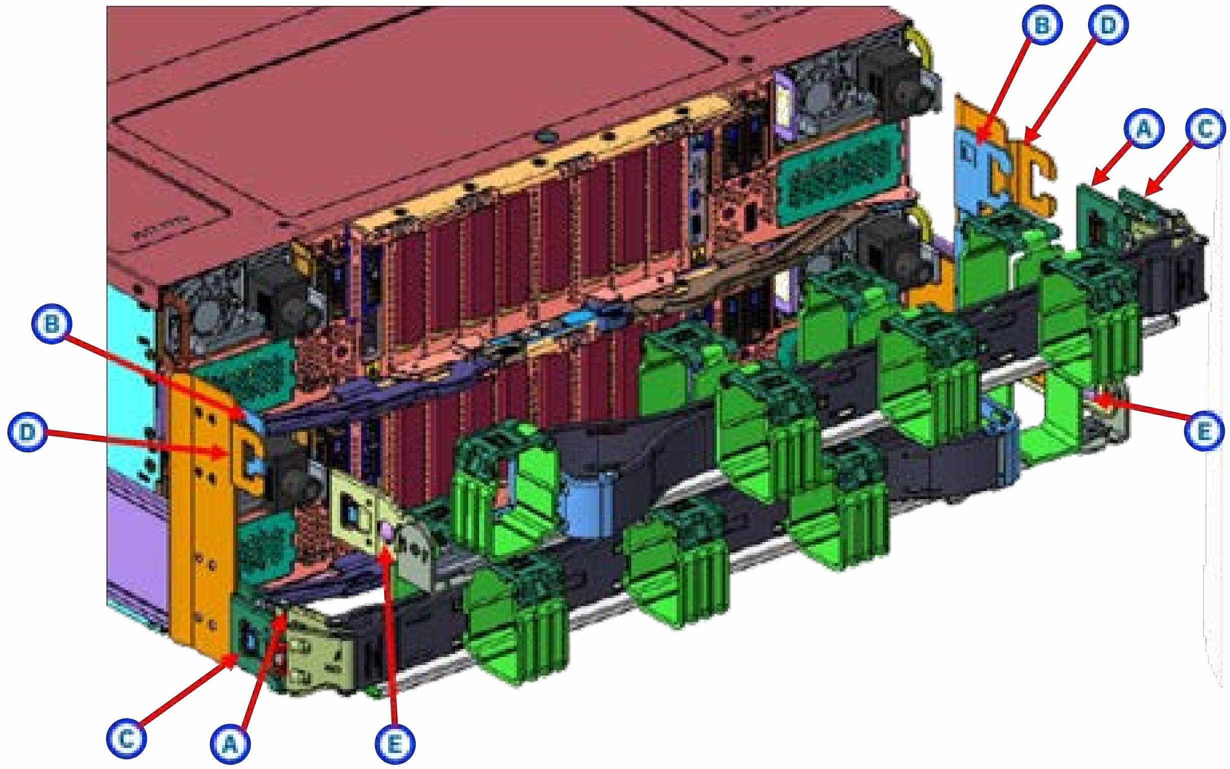
Installing the cable management arm (IBM SSR task)

Each canister of the IBM Storage Scale System 6000 requires a cable management arm (CMA); a total of two CMAs is required per enclosure. When working on replacing hot-swappable components, use the CMAs to smoothly pull and push the enclosure.

The CMA aids in better routing and securing of the system’s cabling. The CMA enables the storage enclosure easily slid in and out of the rack for drive installation or replacement without disconnecting the cables from the PSU or server canisters. A properly installed CMA prevents cable tangling and interference with other components in the rack, allowing for smooth operation of the rails.

Note: If the CMAs are preinstalled, the CMAs would be secured by a strap to the cross bar during transportation to avoid damage. This strap must be removed before installation.

1. Verify the direction of the CMA as shown in the following figure.



A - Inner CMA connector

B - Inner rail connector

C - Outer CMA connector

D - Outer rail connector

E - CMA body connector

Figure 42. CMA parts and orientation

2. Install the CMA in the bottom canister:

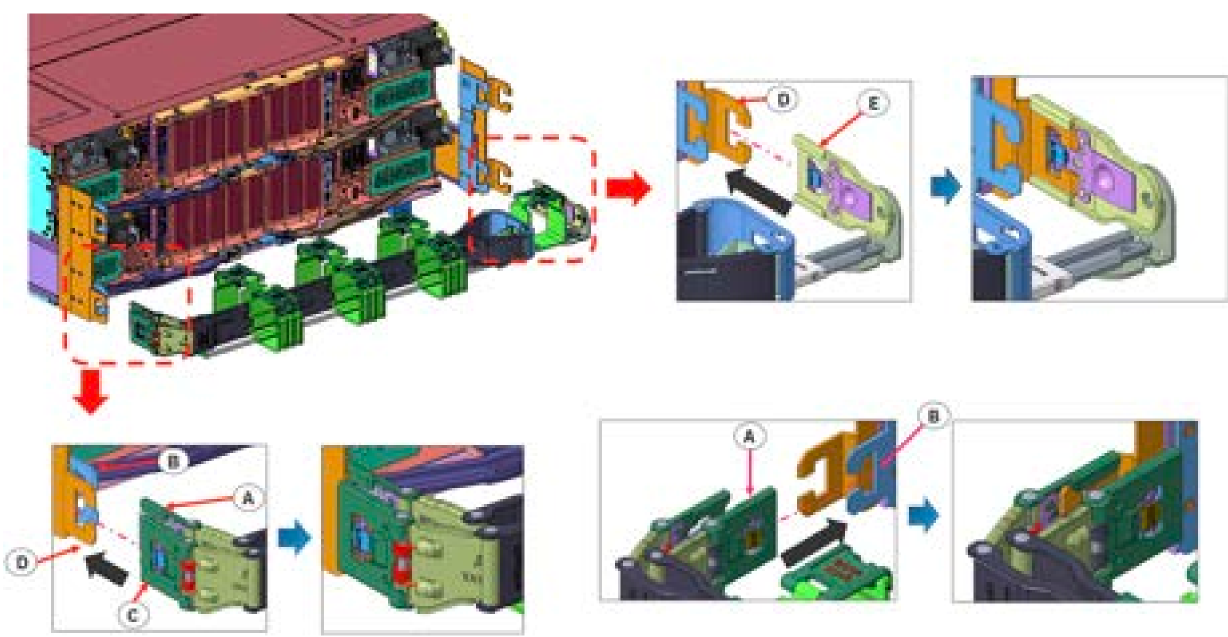


Figure 43. Installing the CMA in the bottom canister

a) Insert the CMA body connector (E) into the outer rail connector (D).

b) Insert the inner CMA connector (A) into the inner rail connector (B).

c) Insert the outer CMA connector (C) into the outer rail connector (D).

3. Install the CMA in the top canister:

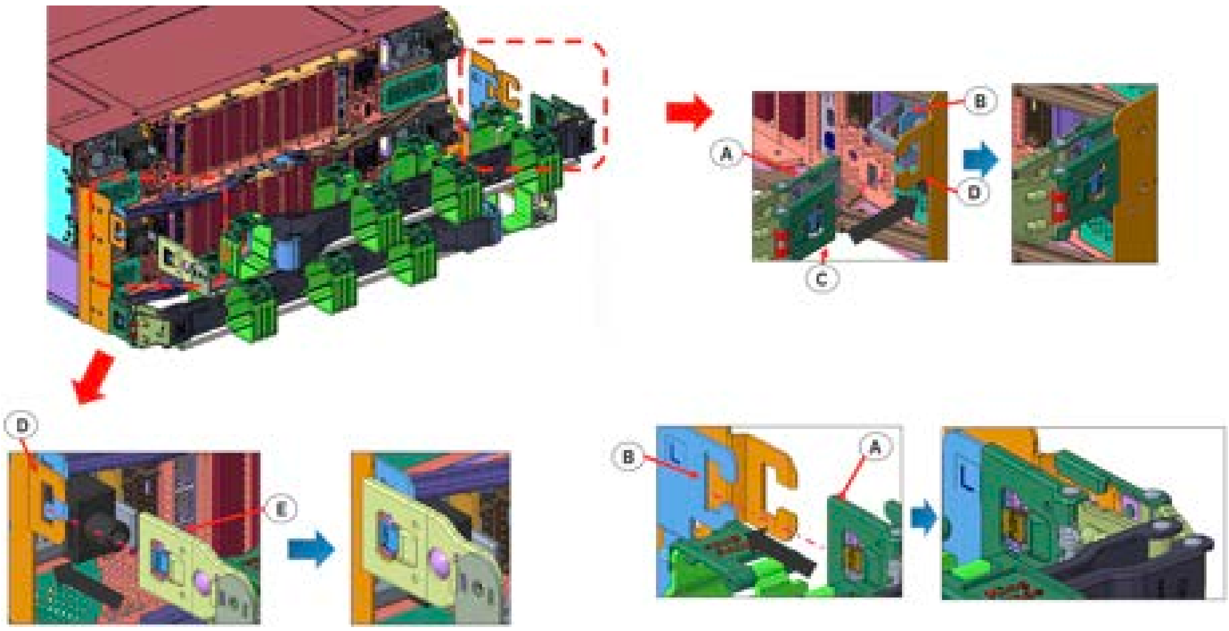


Figure 44. Installing the CMA in the top canister

a) Insert the inner CMA connector (A) into the inner rail connector (B).

b) Insert the outer CMA connector (C) into the outer rail connector (D).

c) Insert the CMA body connector (E) into the outer rail connector (D).

4. Connect the cables to the appropriate ports on the server canister operator panels and the jacks for the power modules.

CAUTION: Do not connect the power cords to a power module yet.

5. Starting with the thickest gauge cable first, run each cable through the loops on the CMA.
6. Test the CMA to ensure smooth operation. Ensure that the cables do not have pinching or bindings.

Note: When you pull an enclosure from the rack, the rails lock in the service position. To release the rails from the service position, press the release tab on each inner member rail and simultaneously push the enclosure into the rack.

Configuring remote code load

Remote code load (RCL) is a service that enables IBM service personnel to securely connect and complete code updates on IBM Storage Scale System from a remote location. The RCL service is the preferred method for delivering code updates, which does not require an onsite visit from IBM service personnel. It provides a secure and efficient alternative to on-premises microcode upgrades.

RCL for IBM Storage Scale System 6000 (MTM 5149-F48) requires IBM Storage Scale System RCL service container and API server container to run on the IBM Storage Scale System Utility Node. You need to configure these containers on the IBM Storage Scale System Utility Node before updating the IBM Storage Scale System 6000 nodes. For more information, see *Configuring Remote Code Load* in the IBM Storage Scale System : Software deployment guide.

Configuring data acceleration tier

Starting with IBM Storage Scale System 7.0.0.0, data acceleration tier (DAT) are supported on a fully populated IBM Storage Scale System 6000 with performance configuration (48 TLC NVMe drives).

DAT for IBM Storage Scale System 6000 addresses the IOPS performance issues that constrain real-time AI inferencing.

In an IBM Storage Scale System 6000, DAT uses the IBM Storage Scale System technology that is called asymmetric data replication, which provides a *performance* data replica for fast reads and a *GNR* data replica for data resiliency.

The following DAT deployment option is supported.

Decentralized DAT deployment
: Deploys the performance pool on NVMe storage that is physically present on the client machines. This deployment option maintains the GNR pool on the IBM Storage Scale System 6000, thus providing data resiliency by using the erasure encoding feature of IBM Storage Scale RAID.

Restrictions for data acceleration tier

Before you enable a DAT file system on an IBM Storage Scale System 6000, consider these restrictions for decentralized DAT deployments.

The following list comprises the restrictions that apply to a DAT file system on an IBM Storage Scale System 6000 with IBM Storage Scale System 7.0.0.0.

- Red Hat Enterprise Linux (RHEL) is the only supported operating system for DAT deployment.
- The open-source version of OpenFabrics Enterprise Distribution (OFED) is not supported for the deployment of DAT in an IBM Storage Scale System 6000. The only supported frameworks for DAT deployments are NVIDIA DOCA and NVIDIA MOFED.
- IBM Storage Scale native REST API cannot deploy DAT.
- The mapping between a DAT file system and an IBM Storage Scale System 6000 is 1 to 1, within a single IBM Storage Scale System building block. Meaning, one IBM Storage Scale System 6000 per DAT file system, and one DAT file system per IBM Storage Scale System 6000.
  - An IBM Storage Scale System 6000 does not support more than one DAT file system.
  - For a decentralized DAT deployment, disks on the clients that comprise the performance pool disks are the only additional storage that can be configured for a DAT file system.
  - With IBM Storage Scale System 7.0.0.0, a file with a performance replica in a DAT file system can have only one GNR data replica. To corroborate that you comply with this restriction, use `mmlsfs` *<file_system>* and make sure that `-r 1` is the default number of data replicas.
- DAT file systems require a fully populated IBM Storage Scale System 6000 with performance configuration (48 TLC NVMe drives). The hybrid configuration is not supported, which means that the IBM Storage Scale System 6000 *must not have* any external storage enclosures.
- DAT file systems do not support a remote storage cluster.

Decentralized DAT deployment

In a decentralized DAT deployment, the performance pool is created on NVMe storage that is physically present on the client machines.

Before you start a decentralized DAT deployment, read "Best practices for a decentralized DAT deployment" on page 67.

Optional: Enable protection information for a decentralized DAT deployment

Enable the protection information feature on the NVMe drives of the performance pool on the decentralized-DAT clients to support data checks on DAT file system reads.

Note:

- The protection information feature must be enabled on the NVMe drives of the performance pool before you create the decentralized-DAT file system.
- A minor performance impact might occur when you enable protection information.
- Data checks are enabled by default in the GNR pool of a decentralized-DAT file system. No additional steps are required.

Example: Enabling protection information

Note: The NVMe device name and LBA index that are used in this example can vary by manufacturer and model.

Run the following three commands for all the NVMe drives that are part of the performance pool.

1. Check the available LBA formats.

   ```
   nvme id-ns /dev/nvme0n1
   ```

   Choose an LBA with `Metadata Size: 8 bytes` and your preferred data size.

   Note: This chosen LBA is used for the `-l` argument in the next step.

   Example output for nvme `id-ns`

   ```
   LBA Format 1 : Metadata Size: 8   bytes - Data Size: 4096 bytes - Relative
   Performance: 0x2 Good
   LBA Format 2 : Metadata Size: 0   bytes - Data Size: 512 bytes - Relative
   Performance: 0x1 Better
   ```

2. Format the NVMe drives for protection information.

   ```
   nvme format /dev/nvme0n1 -i 2 -m 0 -l 1
   ```

3. Verify that protection information is enabled.

   In the LBA section of the output, look for `in use`.

   ```
   nvme id-ns /dev/nvme0n1
   ```

The expected output is similar to the following sample.

```
LBA Format 1 : Metadata Size: 8   bytes - Data Size: 4096 bytes - Relative Performance: 0x2
Good
```

Best practices for a decentralized DAT deployment

For optimal results of a DAT file system on IBM Storage Scale System 6000, follow these proven or standard configurations.

For small I/O workloads, direct I/O contributes to get the best possible performance

Particularly for small I/O workloads, applications that use direct I/O mode contribute to achieve the best possible performance.

For information about requirements and restrictions that apply to direct I/O with IBM Storage Scale, see Using Direct I/O on a file in a GPFS file system and Considerations for the use of direct I/O (O_DIRECT) in IBM Storage Scale in IBM Storage Scale documentation.

Configure NSD checksum

In IBM Storage Scale, enable its NSD checksum by setting this configuration option as follows: `nsdCksumTraditional=yes`. When you enable NSD checksum, you enable additional data checks (network transfers) on DAT file system reads.

Note: A minor performance impact might occur when you enable the `nsdCksumTraditional` configuration option.

Creating a decentralized-DAT file system

Create the GNR pool and the performance pool, specify the required replicas, and mount the DAT file system.

Remember: Only a fully populated IBM Storage Scale System 6000 with performance configuration (48 TLC NVMe drives) can support DAT.

The following procedure is an example of how to create a decentralized-DAT file system that consists of a GNR pool and a performance pool.

1. Enable the cluster to use remote direct memory access (RDMA) for IBM Storage Scale.
2. Create a GNR file system that uses the 48 drives of the IBM Storage Scale System 6000.

   To complete this step, follow the standard deployment procedure for an IBM Storage Scale System 6000.
3. Create the pool stanza file and the NSD stanza file for the DAT performance pool.

   Note: In the following example of a stanza file, the write affinity and failure group settings force writes to the performance pool to occur on the disks of the client-local performance pool (instead of striping the performance replica over performance pool disks on all clients). Therefore, readings from the performance pool are local on each client.

   Remember: The next snippet is an example. Customize the fields of your stanza files according to your own configuration.

   Example stanza file

   ```
   %pool:
   pool=perf
   allowWriteAffinity=yes
   writeAffinityDepth=1
   performancePool=yes
   layoutMap=cluster
   blockSize=8M

   %nsd:
   device=/dev/nvme0n1
   server=fscc-sr655v3-17
   nsd=node17_NSD1
   usage=dataOnly
   pool=perf
   failureGroup=1,0,1

   %nsd:
   device=/dev/nvme1n1
   server=fscc-sr655v3-17
   nsd=node17_NSD2
   usage=dataOnly
   pool=perf
   failureGroup=1,0,1
   ```

   For information about performance replica placement and migration, see mmchpolicy command and mmchattr command in IBM Storage Scale documentation.
4. Create the stanza files for pool NSDs for the DAT performance pool.

   Remember: Make sure that the NVMe drives that are used for NSDs are formatted with protection information.
5. Create the performance pool NSDs and add them to the file system.

   ```
   mmcrnsd -F /<stanza_file> -v no
   mmvdisk filesystem add --fs <file_system> --nsd <comma-separated_list_of_NSDs>
   --pools
   perf,performancePool=yes,allowWriteAffinity=yes,writeAffinityDepth=1,layoutMap=cluster,blockS
   ize=8M
   ```
6. Specify one performance replica for files in this DAT file system.

```
mmchfs <file_system> --perf-replicas 1
```

7. Mount the DAT file system on all nodes.

```
mmmount <file_system> -N All
```

To verify the configuration, follow these steps on the client machine where you mounted the decentralized-DAT file system.

1. Use the **mmlsfs** command to verify that the performance replica, performance pool, and GNR pool were created.

   Example output for mmlsfs

   ```
   # mmlsfs <file_system>
   flag                value                              description
   ------------------- ----------------------------------
   ----------------------------------
   …
   --perf-replicas     1                                   Additional data replicas for
   performance
   …
   -P                  system;perf                         Disk storage pools in file system
   …
   -d                  RG001LG001VS001;RG001LG002VS001;    Disks in file system
                       RG001LG003VS001;RG001LG004VS001;
                       node1_perf_NSD1;node1_perf_NSD2;
                       node23_perf_NSD1;node23_perf_NSD2
   ```

   In the output, look for these values:

   - 1 performance replica (1 in the previous example output)
   - 1 GNR pool (`system` in the previous example output)
   - 1 performance pool (`perf` in the previous example output)
   - Performance pool NSDs (`node1_perf_NSD1`, `node1_perf_NSD2`, `node23_perf_NSD1`, `node23_perf_NSD2` in the previous example output)

   Remember: The names of the GNR pool NSDs (`RG...` in the previous example output) are set by the **mmvdisk** or the **essrun** command. But the names of the performance pool NSDs (like `node1_perf_NSD1` in the previous example output) are defined by the user in the NSD stanzas.

2. Use the **mmlspool** command to verify the configuration.

   Example output for mmlspool

   ```
   # mmlspool  <file_system>  perf  -L
   Pool:
     name                    = perf
     poolID                  = 65537
     blockSize               = 4 MB
     usage                   = dataOnly
     maxDiskSize             = 23.50 TB
     layoutMap               = cluster
     allowWriteAffinity      = yes
     writeAffinityDepth      = 1
     blockGroupFactor        = 1
     performancePool         = yes
   ```

   In the output, look for these values:

   - `usage = dataOnly`
   - `performancePool = yes`

   Remember: Some of the other fields are defined by the user in the NSD and pool stanzas.

Administering DAT file systems

After the deployment of a DAT file system, other procedures are necessary to administer it.

Deleting a decentralized-DAT file system

Follow this procedure to delete a decentralized-DAT file system.

1. Delete the decentralized-DAT file system.

```
mmvdisk filesystem delete --fs <file_system> --confirm
```

2. Delete the virtual disks set.

```
mmvdisk vdiskset delete --vs <vdisk_set>
```

3. Revert the definition of the virtual disks set.

```
mmvdisk vs undefine --vs <vdisk_set> --confirm
```

4. Delete the recovery group.

```
mmvdisk recoverygroup delete --rg <recovery_group> --confirm
```

5. Delete the performance pool NSDs.

```
mmdelnsd "<semicolon-separated_list_of_NSDs>"
```

Chapter 6. Monitoring the system using LEDs

The section provides information about LED-based monitoring of IBM Storage Scale System 6000. The chapter guides customers to use LEDs for monitoring and maintenance of the system.

The following components of IBM Storage Scale System 6000 support LED-based monitoring:

- System status
- Server canisters
- Power supply units (PSUs)
- Drive
- PCIe slots

System status LEDs

System status LEDs provide information about the status of the IBM Storage Scale System 6000.

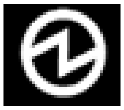
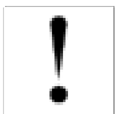
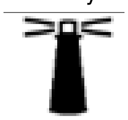
In the front of the system, three status LEDs are available per enclosure in the left ear, as shown in the following figure.

Figure 45. System status LEDs on an IBM Storage Scale System 6000 enclosure

From the top of the enclosure's left ear, the following LEDs are supported:

1. The first LED provides the status of the PSUs.
2. The second LED alerts about faulty field-replaceable units (FRUs).
3. The third LED corresponds to the `Identify` status.

The following table describes the system status LEDs in the IBM Storage Scale System 6000.

Table 25. Description of the system status LEDs			
LED type	**Color**	**States**	**Description**
Power 	True green	Off	No external power to the enclosure.
		Slow blink	The enclosure is in a stand-by state.
		Solid	AC power to the enclosure and at least one canister is not in stand-by.
Fault 	Amber	Off	No isolated FRU failures on the enclosure.
		Solid	The enclosure has one or more isolated FRU failures and they require service or replacement.
Identify 	Blue	Off	The enclosure is not being identified.
		Solid	The enclosure has been set to `Identify`.

Server canister status LEDs

These LEDs are used to monitor the status of server canisters and Ethernet ports of the IBM Storage Scale System 6000.

The IBM Storage Scale System 6000 server has three status LEDs on the operator panel of each server canister. In addition, two Ethernet port LEDs per Ethernet port (four LEDs) and two BMC port LEDs are part of each server canister's status LEDs.

The following images show the server canister LEDs included in the IBM Storage Scale System 6000.

Figure 46. IBM Storage Scale System 6000 server canister LEDs

The following figure shows a close-up view of server canister LEDs.

Figure 47. Close view of server canister LEDs

Ethernet and BMC ports status LEDs

These LEDs are used to monitor Ethernet and BMC ports of the IBM Storage Scale System 6000.

The Ethernet ports and the BMC ports on the rear of server canister are for standard RJ45-style connectors. A close view of these port LEDs is shown in the following figure.

Figure 48. Close view of IBM Storage Scale System 6000 Ethernet and BMC port LEDs

These LEDs indicate speed and link activity as described in the following table.

Table 26. IBM Storage Scale System 6000 Ethernet and BMC port status LEDs

LED	Color	States	Description
State	Green	Off	The RJ45 link is not connected.
		Flashing	Network activity on the port.
		Solid	No activity on the port. However, a link is established.

Table 26. IBM Storage Scale System 6000 Ethernet and BMC port status LEDs (continued)			
LED	**Color**	**States**	**Description**
Status	Green	Off	No link is established.
		Slow blink	Indicates that the port speed is at a lower data rate than the maximum for the port. Note that blink rates may indicate specific speeds that the port supports, other then the maximum.
		Solid	Indicates that the port speed is at the maximum data rate for the port.

Server canister status LEDs

These LEDs are used to monitor server canisters of the IBM Storage Scale System 6000.

The table describe the server canister LEDs included in the IBM Storage Scale System 6000.

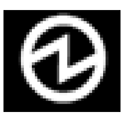
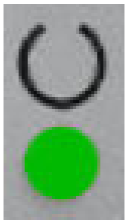
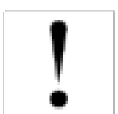
Table 27. Description of the canister status LEDs			
LED type	**Color**	**States**	**Description**
Power 	True green	Off	No external power to the canister.
		Slow blink	The canister is in a stand-by state.
		Fast blink	During POST, indicates that the BIOS is in POST.
		Solid	The canister is in a power on state.

Table 27. Description of the canister status LEDs (continued)			
LED type	**Color**	**States**	**Description**
Status 	True green	Off	The controller RAID stack is not operational.
		Blink	The node is in a candidate or service state. So, the node is not able to perform I/O in a cluster. It is safe to remove the node.
		Fast blink	The canister is performing destage activity.
		Solid	The controller is active and joined a cluster. The canister has state data that must be preserved on shutdown. Normally, this means that it is active in the cluster, handling I/O; but a transient starting state is also being covered. It is possible to get held in starting state if a quorum cannot be established; then, the alert lights.
Fault 	Amber	Off	No isolated FRU failures on the canister.
		Blink	The canister is being identified by using an identify command.
		Solid	The canister has one or more isolated FRU failures and they require service or replacement.

PSU status LEDs

These LEDs are used to monitor power supply units (PSUs) of the IBM Storage Scale System 6000.

The PSU status LEDs are visible from the rear of the enclosure. The following figure provides a close view of the power module status LED.

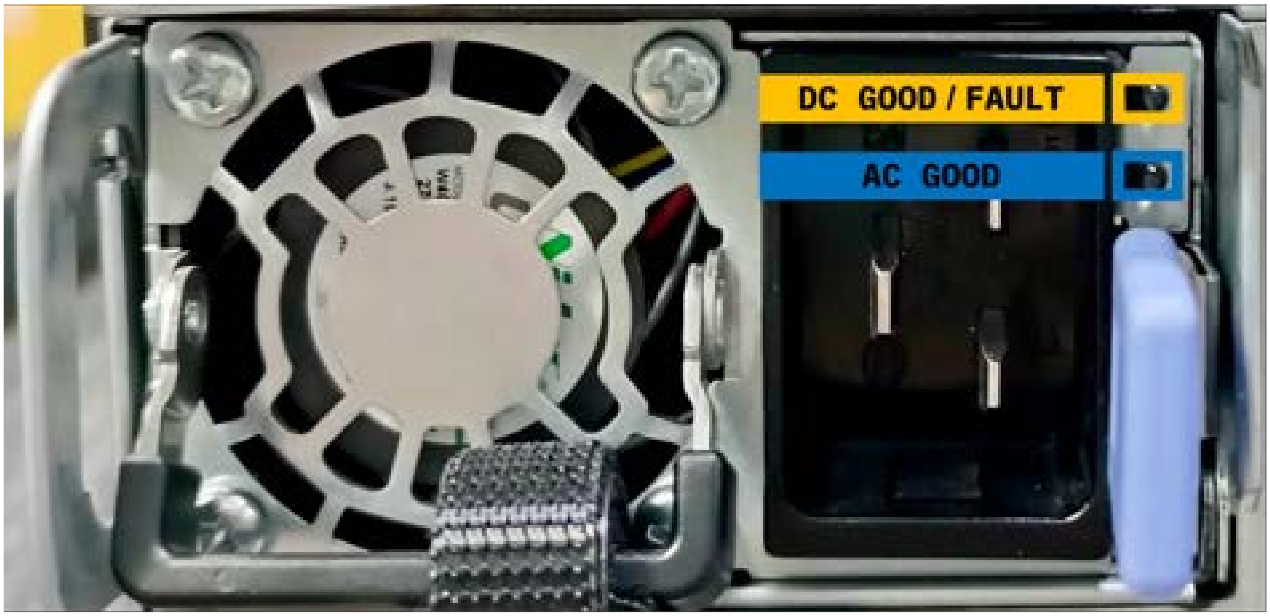


Figure 49. Close view of theIBM Storage Scale System 6000 PSU status LEDs

The following table describes bi-color PSU status LEDs.

Table 28. IBM Storage Scale System 6000 PSU status LEDs

LED	Green color	Amber color	Description
DC GOOD and FAULT indicators[1]	Solid	Off	Represents the DC GOOD signal. The PSU is operating normally.
	Off or solid	Blinking	The PSU is being identified.
	Blinking	Off	The PSU is at standby state. The blinking rate is 1 Hz.
	Solid	Solid	Represents the FAULT signal. Declares a fault in the PSU; except for a 12 V overcurrent fault.
AC GOOD indicator	Solid	Off	Represents the AC GOOD signal. The emergency power-off warning (EPOW) signal is inactive. Indicates that the PSU is receiving normal input power.
	Off	Off	The PSU is not receiving proper input power.
	Blinking	Off	Indicates input OC.

[1] The amber LED can be set to be on, off, or to blink by using an I2C command. In a redundant configuration, the host system can control the amber fault LED when AC is removed from the PSU.

Drive status LEDs

These LEDs are used to monitor drives of the IBM Storage Scale System 6000.

Each drive carrier assembly of IBM Storage Scale System 6000 includes a set of LEDs that are visible from the top of the drive carrier. On system startup, the green LED illuminates automatically.

In the following figure of the drive status LEDs on a drive carrier, the green LED is on.

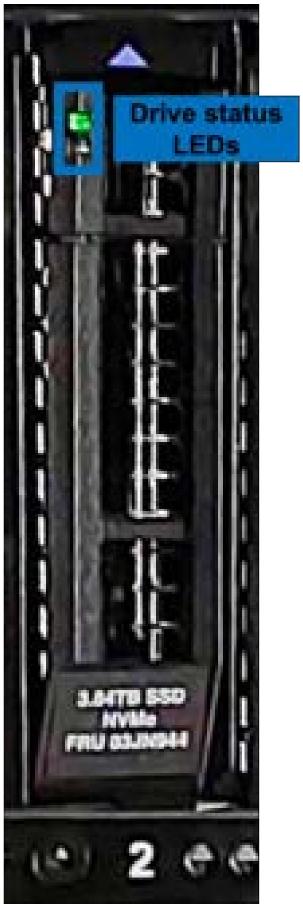


Figure 50. Drive status LEDs on an IBM Storage Scale System 6000 drive carrier

The following table describes the drive carrier assembly LEDs supported in the IBM Storage Scale System 6000.

Table 29. IBM Storage Scale System 6000 drive status LEDs

LED	Color	States	Description
Activity	True green	Off	The drive is powered off or is not online.
		Blink	The drive is being powered on or spun up.
		Flashing	Activity on the drive.
		Solid	The drive is powered on and ready, but with no activity.

Table 29. IBM Storage Scale System 6000 drive status LEDs (continued)

LED	Color	States	Description
Fault	Amber	Off	No isolated FRU failures on the drive.
		Blink	The drive is being identified by using an identify command.
		Solid	The associated drive has been isolated and is requiring service or replacement.

Appendix A. Slot placement summary

In an IBM Storage Scale System 6000, each slot number has a predefined usage and needs a specific setup for the VM to use it.

The following table and illustration represent each slot and its usage:

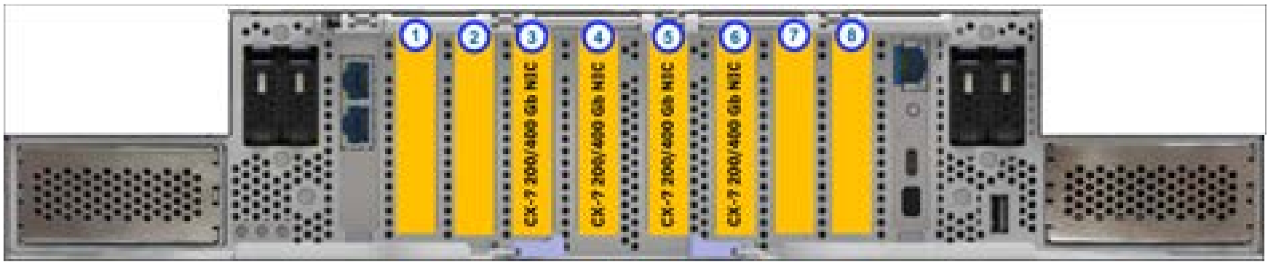


Figure 51. Slot placement summary

Table 30. Slot configuration

Slot configuration	**1** **x8**	**2** **x8** **x16 if 1NP**	**3** **x16**	**4** **x16**	**5** **x16**	**6** **x8** **x16 if 7 NP**	**7** **x8**	**8** **x8**	**Remarks**
1 network (N)				N					
2 network (N)				N	N				
3 network (N)			N	N	N				
4 network (N)			N	N	N	N			
1 - 4 network / 2 SAS	S		N	N	N	N		S	

Appendix B. PCIe 5 x16 dual-port NDR 200 Gb/s InfiniBand and Ethernet ConnectX-7 VPI adapter (FC: AJQS)

Learn about the specifications and operating system requirements for feature code (FC) AJQS adapter.

Overview

FC AJQS is a low-profile adapter. PCIe Gen 5 x16 dual-port high data rate NDR 200 GB InfiniBand (IB) ConnectX-7 adapter is a PCI Express (PCIe) generation 5 (Gen 5) x16 adapter. This adapter can be used in a Gen 5 x16 PCIe slot in the IBM Storage Scale System 6000 system. The adapter enables higher HPC performance with new Message Passing Interface (MPI) offloads, such as MPI Tag Matching and MPI AlltoAll operations, advanced dynamic routing, and new capabilities to perform various data algorithms. The NDR 200 Gb adapter supports connecting to a NDR 200 Gb switch by using NDR 200 Gb cables such as Active Optical Cables (AOC). The NDR 200 Gb adapter also supports enhanced data rate (EDR) optical cables when connecting to EDR 100 Gb or HDR 200 Gb InfiniBand switches. For more information, see "Supported host interface adapters, cables, and transceivers" on page 45.

Note: The Virtual Protocol Interconnect (VPI) feature is supported on this adapter. The adapter may be used as InfiniBand or Ethernet adapter.

Figure 52. 2-port HDR 200 Gb InfiniBand ConnectX-7 adapter (FC AJQS)

Specifications

Adapter FRU number
: 01LL984

I/O bus architecture
: The adapter is a Gen 5 cable and runs on Gen 5 PCIe x16 IBM Storage Scale System 6000.

Slot requirement
: For more information about slot priorities, maximums, and placement rules, see Appendix A, "Slot placement summary," on page 79.

Thermal requirement
If you have a IBM Storage Scale System Utility Node, you might be required to set the thermal mode of the system to a setting other than the default setting, depending on your system, adapter, and cable type.

Cables
For this host bus adapter (HBA), IBM offers Active Optical Cables (AOC) cables up to 30m. QSFP112-based transceivers are included on each end of these cables.

Cable matrix
The NDR cables that are listed in the below table support connecting an NDR two port 200 Gb HBA.

Table 31. ConnectX-7 cable matrix

Feature code	Description
AJR8	NVIDIA AOC splitter, InfiniBand dual-port HDR, 400 Gb/s to 2x 200 Gb/s, OSFP to 2x QSFP56, 5m
AJR9	NVIDIA AOC splitter, InfiniBand dual-port HDR, 400 Gb/s to 2x 200 Gb/s, OSFP to 2x QSFP56, 30m

Voltage
3.3 V, 12 V

Form factor
Short, low-profile.

Attributes and features of the adapter

- PCIe Gen 5.0 x16 host interface
- Optimized to provide accelerated networking for cloud, artificial intelligence, and conventional enterprise workloads.
- Delivers various software-defined, hardware-accelerated capabilities for networking, storage, and security.
- Two QSFP112-based transceivers for 200 Gb/s connectivity
- Total bandwidth of 400 GB/s
- InfiniBand NDR200 or HDR (speeds up to 200 Gb/s)
- Enhanced InfiniBand and Ethernet networking
- Ethernet connectivity that supports: speeds up to 200 GbE; NRZ (10 G, 25 G) and PAM4 (50 G, 100 G); RDMA over Converged Ethernet (RoCE).
- Storage accelerations like NVMe over Fabrics (PVMe-oF) and NVMe over TCP (NVMe/TCP).
- Support for various management and control protocols, such as NC-SI, MCTP over SMBus, MCTP over PCIe, Serial Peripheral Interface (SPI) to flash.
- Remote boot over InfiniBand or over iSCSI.

PCIe single port

For information about the single port CX7 NDR 400 Gb VPI adapter, see the Summary of supported host interface adapters table in "Connections for IBM Storage Scale System 6000" on page 44.

Appendix C. Best practices for space management in IBM FlashCore Modules

Learn best practices to manage your IBM Storage Scale System 6000 (MTM 5149-F48) with IBM FlashCore Modules (FCMs) and how to provision the correct amount of capacity for your system.

The IBM Storage Scale System 6000 is compatible with IBM FlashCore Modules, which have inline, always-on compression. Because the FCMs compress data, they can have an effective capacity that is significantly larger than their actual physical capacity. The amount of compression varies depending on the type of files that are stored in the MTM 5149-F48.

An MTM 5149-F48 with FCMs can compress data and maintain high performance because of the unique ability of FCMs to compress data without using host resources. To make the most out of this benefit, you must appropriately provision and monitor the MTM 5149-F48; and know whether the data to be stored is encrypted or compressed by a host application. Before you provision your system, use the ESS Comprestimator tool to estimate the data compressibility factor.

Select the appropriate set size for a virtual disk based on the estimated compression ratio for your data. Monitor the percentage of physical space that declustered arrays use, to prevent the following issues:

- With IBM Storage Scale System 6.2.3.2 through 7.0.0.x, monitor to avoid the throttling safeguards that are built into the IBM Storage Scale System 6000. If the safeguards are reached, act immediately to prevent further throttling or the stoppage of I/O write operations.
- Starting with IBM Storage Scale System 7.0.1.x, monitor to avoid the out-of-space (OOS) safeguard that is built into the IBM Storage Scale System 6000. If the system is getting close to the OOS safeguard, act immediately to prevent the stoppage of I/O write operations.

Based on the results of the ESS Comprestimator tool about the estimated compression ratio for your data, select the appropriate set size for a virtual disk.

Make sure that TRIM is properly enabled to optimize space recovery on the FCMs. If an FCM fails, quickly replace the drive.

If you have questions about these *Best practices for space management in IBM FlashCore Modules*, contact IBM Support.

Key best practices and considerations

The following list summarizes the best practices and considerations that you must know during the configuration of your IBM Storage Scale System 6000 with IBM FlashCore Modules.

- If a host application encrypts the data that is sent to the MTM 5149-F48 or if the IBM Storage Scale System software encryption is enabled, the FCMs are unable to compress the data and do not provide more effective capacity than physical capacity.
- If data can be compressed, FCMs have a capacity advantage over the industry-standard NVMe drives. Many users get a good compression ratio if their data is not already encrypted by a host application and is not already in a compressed file format (like `.ZIP` or `.TAR`).
- IBM developed the ESS Comprestimator tool to help users to estimate the compression that they can expect from the FCMs. Based on this information, system administrators can properly provision the storage on an MTM 5149-F48.
- The amount of data compression, which is known as the *compression ratio*, must be closely monitored when FCMs are being used because data profiles can change over time and alter the compression ratio.
- Set up TRIM on the IBM Storage Scale System 6000 that uses FCMs. For more information, see Managing TRIM support for storage space reclamation in the *IBM Storage Scale RAID: Administration Guide*.

- If an FCM fails, replace the faulty drive. And change the settings to make sure that the system alerts a replacement is needed when a single FCM fails. Failing to do so might result in issues that are reported as `out of space` errors.
- An MTM 5149-F48 with IBM FlashCore Module 4 (FCM 4) needs a minimum firmware level of IBM Storage Scale System 6.2.3.2. Keep updating the minimum firmware level as this guidance changes over time.

Data storage in IBM FlashCore Modules

FCMs use hardware to compress the data, which is much more efficient than data compression through software.

IBM designed the FCMs to be unique SSDs with three desirable properties:

- Compression (see "The compression property of IBM FlashCore Modules" on page 84)
- Encryption (see "Encryption for IBM FlashCore Modules" on page 89)
- Endurance

The compression property of IBM FlashCore Modules

Data compression is valuable because it makes it possible to store more data in the same physical space size, when compared with noncompressed data. FCMs help you to optimize capacity, while also minimizing the negative impact on performance; whereas software compression can significantly impact performance.

An MTM 5149-F48 with FCMs can compress data so that the logical capacity is greater than the physical capacity. IBM Storage Scale Systems are provisioned based on the logical capacity, not the physical capacity of the drives. When administrators are configuring the system capacity, they must know about the compression ratio, the logical to physical storage ratio. Different types of data compress at different ratios or may not compress at all. To estimate the compression ratio of data, use the ESS Comprestimator tool, provided by IBM at https://github.com/IBM/ess_comprestimator/releases.

Before you decide to use FCMs, consider the following compression limitations.

- If a host application encrypts data that is sent to the MTM 5149-F48 or if the IBM Storage Scale System software encryption is enabled, FCMs cannot compress the data and do not provide more logical capacity than physical capacity; meaning, the compression ratio is 1x.
- If all the data that is sent to the MTM 5149-F48 is composed of files that are already in a compression format (for example, `.ZIP` or `.TAR`), the compression ratio that FCMs provide is significantly reduced.

In these two cases, FCMs offer little to no compression benefit. Therefore, in these cases, FCMs are likely not the right choice for your solution.

Remember: If the IBM FlashCore Modules can compress the data, FCMs can offer a capacity benefit over standard NVMe drives.

Monitor the compression ratio over time when you use FCMs, because data types can change and affect the amount of compression. In extreme cases, the compression ratio changes so much that the logical to physical space ratio becomes significantly different than the compression ratio that was originally determined. When this happens, the system is at risk of using all the physical space of the disks. In IBM Storage Scale Systems that use all the disks space, data loss is a possibility. But the MTM 5149-F48 has safeguards to prevent the use of too much physical space; if the FCMs are using too much physical space, the system throttles the I/O and eventually stops writing data to the disks. For more information, see "Space warnings and safeguards" on page 90 to consult recommended best practices to prevent data throttling and disruption of write operations.

Starting with IBM Storage Scale System 7.0.1.x: The I/O operations are not throttled if the FCMs are using too much physical space and are approaching the safeguard where the system eventually stops writing data to the disks.

TRIM for space management in IBM FlashCore Modules

Enable TRIM and automatic space reclaim so that FCMs can free the allocated blocks that are not storing any user data.

FCMs are constantly managing data that is stored in them, to optimize endurance and free blocks that are allocated but are not storing any user data. This process is commonly known as *garbage collection* in the storage industry. Even though data can be deleted at the file system level of the IBM Storage Scale System 6000, the FCMs do not free the space that is used by that data if the system settings are not set correctly.

The TRIM feature contributes to performance enhancement by enabling storage controllers to reclaim free space. IBM Storage Scale supports a TRIM operation to enhance the performance of declustered arrays (DAs) that are based on flash. TRIM also allows the storage controller to reclaim physical blocks that are no longer being used (space that is freed by file system) and can be erased for storage of new data.

Enable TRIM

Use the following commands to enable TRIM at the physical disk level, within a DA.

- During the creation of a recovery group, run the `mmvdisk recoverygroup create` command with the `- -trim-da nvme` option for a DA.
- During the creation of a file system, run the `mmvdisk filesystem create` command with the `--trim auto` option to enable TRIM for the file system that you are creating.

You can find more information on this process at Managing TRIM support for storage space reclamation in the *IBM Storage Scale RAID: Administration Guide*.

The auto reclaim process sets how often TRIM runs. Running TRIM more frequently optimizes free space, but it also can affect system performance. By default, auto reclaim is enabled on FCMs machines that have TRIM enabled, and manages TRIM operations that are based on the system use of physical space. To configure auto reclaim, use the **`backgroundSpaceReclaim`** parameter.

Getting notified when an IBM FlashCore Module fails

Replace a faulty FCM drive quickly after it fails, because the physical space of the DA gets reduced when an FCM failure occurs. The system default is to notify that a replacement is needed when two FCMs fail. To help you get a faulty FCM replaced quickly, change that default configuration to make sure that you are notified after one FCM fails.

Follow the next steps to change the default notifications.

1. Identify the DAs that are a member of the recovery group.

```
# mmvdisk rg list --rg <recovery_group_name> --da

declustered   needs          vdisks       pdisks          capacity
   array     service  type    BER     trim  user log  total spare rt  total raw free raw
background task
-----------  -------  ----  -------  ----  ---- ---  ----- ----- --  --------- --------
---------------
DA1          no       NVMe  enable   no       4   9     12     1  2     75 TiB   48 GiB
scrub 14d (47%)
DA2          no       NVMe  enable   no       4   0     13     2  2     38 TiB   14 GiB
scrub 14d (15%)
```

2. For each of the listed DAs (in the previous example, DA1 and DA2), use the next command to set the replacement threshold to 1.

```
mmvdisk recoverygroup change --recovery-group <recovery_group_name> --da
<declustered_array_name> --replace-threshold 1
```

3. To verify that the replacement threshold field (labeled `rt` in the following example) is updated to 1, run `mmvdisk rg list --rg <recovery_group_name> --da` again.

```
# mmvdisk rg list --rg <recovery_group_name> --da

declustered   needs                              vdisks       pdisks           capacity
   array     service  type    BER     trim  user log  total spare rt  total raw free raw
background task
-----------  -------  ----  -------   ----  ---- ---  ----- ----- --  --------- --------
---------------
DA1           no      NVMe  enable    no       4   9     12     1  1     75 TiB   48 GiB
scrub 14d (47%)
DA2           no      NVMe  enable    no       4   0     13     2  1     38 TiB   14 GiB
scrub 14d (15%)
```

Space provisioning for IBM Storage Scale System 6000

Based on how important the compression ratio is for your storage, select the appropriate set size for a virtual disk set.

Starting with IBM Storage Scale System 7.0.1.x: The distinction between a standard capacity configuration and a maximum capacity configuration is not used.

An IBM Storage Scale System 6000 can be configured in one of two ways, namely: standard capacity configuration or maximum capacity configuration.

Standard capacity configuration

A standard capacity configuration is for someone who wants to easily manage the MTM 5149-F48 and is less concerned about whether the maximum compression ratio is achieved by the FCMs. This user monitors the compression ratio less frequently and does not proactively push the system toward the edge of its capacity limits.

For a standard capacity, configure the set size of the vdisk set according to the values in the *Standard capacity configuration* column of the next table. Assuming that the compression ratio is stable, you can prevent I/O from being throttled by using one of the set sizes that are listed in the next table.

Maximum capacity configuration

A maximum capacity configuration is for someone that closely tracks the compression ratio that is being achieved by the FCMs in the system. To get the maximum effective capacity from the FCMs, they monitor the compression ratio daily; and even more frequently if data is being migrated onto the MTM 5149-F48.

For a maximum capacity, configure the set size of the vdisk set according to the values in the *Maximum capacity configuration* column of the next table. Assuming that the compression ratio is stable, you can prevent I/O from being throttled by using one of the set size values that are listed in the table. After 15% to 20% of the data is loaded on the DA, verify the actual compression ratio by using the method for compression ratio monitoring that is described in "Space usage monitoring for IBM Storage Scale System 6000" on page 88.

When a new MTM 5149-F48 is set up, during the definition of the vdisk set, configure its `set-size`. Before the set up, run the ESS Comprestimator tool to estimate the compressibility of the data. To provision the system, use Table 32 on page 87 to configure the set size of the vdisk set, based on the compression ratio estimate from the ESS Comprestimator tool.

Note: If IBM Storage Scale data encryption is to be used, the data compression rate is ~1x, regardless of the compressibility of the original data.

"Physical space usage in IBM FlashCore Modules, declustered arrays, and set sizes" on page 90 explains more about this recommendation, how DAs work, and reasons for the selected `set sizes`.

Table 32. IBM Storage Scale 6.2.3.2 through 7.0.0.x: Provisioning an IBM Storage Scale System 6000 based on estimated compression ratios		
Compression ratio estimate from ESS Comprestimator	Standard capacity configuration Max set-size % when defining a `vdiskset`	Maximum capacity configuration Max set-size % when defining a `vdiskset`
1.0	21%	22%
1.25	26%	27%
1.5	31%	33%
1.75	36%	38%
2.0	42%	44%
2.25	47%	49%
2.5	53%	55%
2.75	57%	60%
3.0	63%	66%
3.25	68%	71%
3.5	72%	77%
3.75	78%	80%
4.0 or greater	80%	80%

Table 33. Starting with IBM Storage Scale 7.0.1.x: Provisioning an IBM Storage Scale System 6000 based on estimated compression ratios	
Compression ratio estimate from ESS Comprestimator	Maximum capacity configuration Max set-size % when defining a `vdiskset`
1.0	24%
1.25	30%
1.5	36%
1.75	42%
2.0	48%
2.25	54%
2.5	60%
2.75	66%
3.0	72%
3.25	78%
3.5	80%
3.75	80%
4.0 or greater	80%

Space usage monitoring for IBM Storage Scale System 6000

It is important to monitor the compression ratio of data over time because data types can change and affect the amount of compression.

Starting with IBM Storage Scale System 7.0.1.x: The distinction between a standard capacity configuration and a maximum capacity configuration is not used.

During normal operation, a system that is using the standard capacity configuration should be monitored for the compression ratio and the percentage of the physical space that is used in the DA every few days. With the maximum capacity configuration, the user should monitor for the compression ratio at least daily. For both configuration types, the user must closely monitor the compression ratio as data is initially loading on the system, to verify that the ESS Comprestimator tool made a valid estimate.

IBM Storage Scale System 6.2.3.2 through 7.0.0.x: Use the `mmvdisk` command to monitor space

When the MTM 5149-F48 uses IBM Storage Scale System 6.2.3.2 or later, the **`mmvdisk`** command with **`--da`** shows the DA percentage of total, free physical space and the compression ratio, as highlighted in the next figure.

To compute the current DA percentage of physical space used, use the formula: `DA % physical space used = (total physical capacity - free physical capacity) / total physical capacity`. To avoid the throttling safeguard, the DA percentage of physical space that is used must be less than 60%.

CAUTION: If the DA percentage of physical space that is used is approaching 60%, act immediately to avoid the system from throttling I/O. For actions to take in this situation, see "Space warnings and safeguards" on page 90.

Compare the compression ratio reported in the output of `mmvdisk rg list -rg` *`<recovery_group_name>`* `--da` to the value that you used based on the Table 32 on page 87. And, from the next list, follow the action that applies to the compression ratio.

- If the compression ratio is +/- 10% of the value that you used based on the table, no action is needed.
- If the compression ratio is much higher than the value that you used based on the table, you might need to reconfigure the DA set size to enable more logical capacity.
- If the compression ratio is much lower than the value that you used based on the table, an action is needed to correct the DA set size.

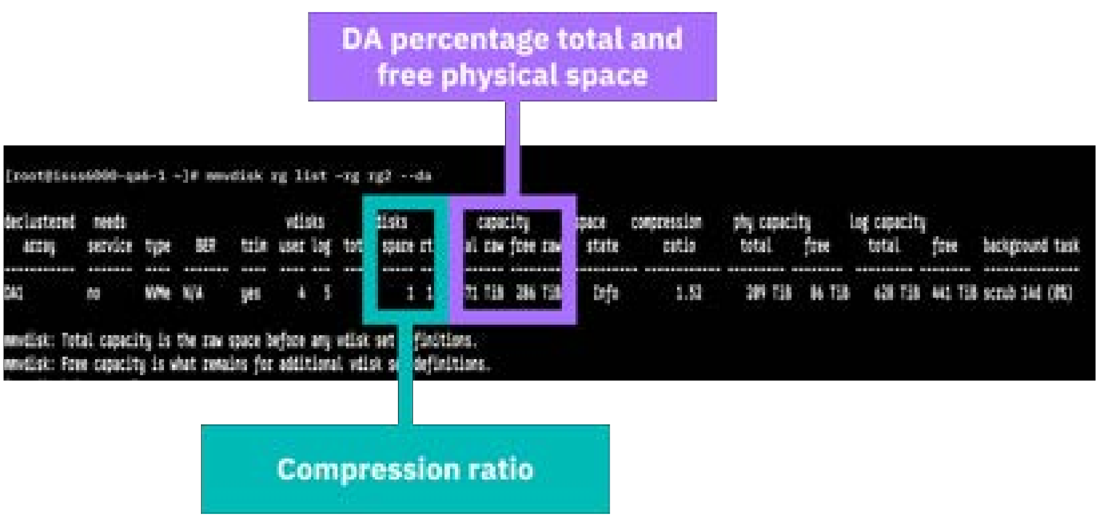


Figure 53. Sample output for `mmvdisk rg list -rg <recovery_group_name> --da` in an MTM 5149-F48 with releases 6.2.3.2 through 7.0.0.x

For more information, see mmvdisk command in the *IBM Storage Scale RAID: Administration Guide*.

Starting with IBM Storage Scale System 7.0.1.x: Use the GUI to monitor space

IBM Storage Scale System 7.0.1.x onward, the GUI shows the DA percentage physical space that is used and the free physical space.

CAUTION: To avoid the out-of-space (OOS) threshold, the DA percentage of physical space that is used must be less than 75%. If the used physical space is approaching 75%, act immediately to avoid the system from preventing I/O writes. For actions to take in this situation, see “Space warnings and safeguards” on page 90.

Compare the compression ratio reported in the GUI to the value that you used based on the table Table 33 on page 87. And, from the next list, follow the action that applies to the compression ratio.

- If the compression ratio is +/- 10% of the value that you used based on the table, no action is needed.
- If the compression ratio is much higher than the value that you used based on the table, you might need to reconfigure the DA set size to enable more logical capacity.
- If the compression ratio is much lower than the value that you used based on the table, an action is needed to correct the DA set size.

In the following figure, the DA percentage of used physical space and the compression ratio are displayed in the highlighted row within the GUI screen.

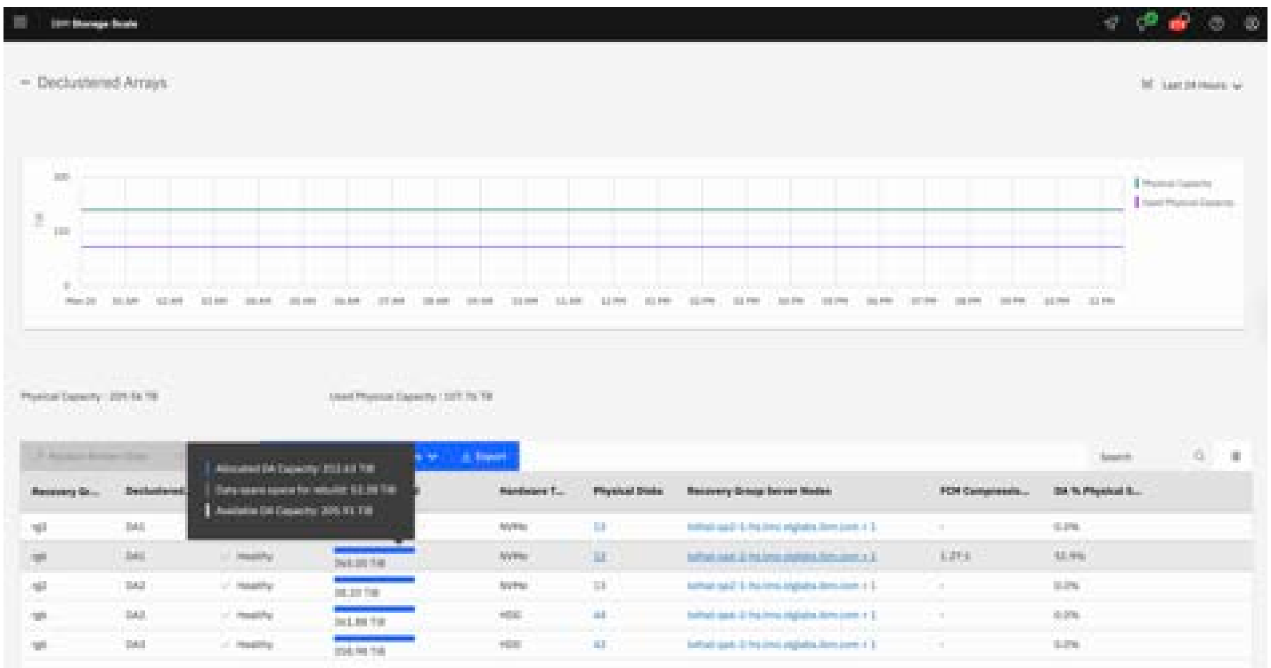


Figure 54. Space usage report through the GUI of an MTM 5149-F48 with release 7.0.1.x or later

Encryption, physical space usage, and space warnings

Use these best practices for physical space usage in FCMs and DAs. Understand the considerations for the correct set sizes, and learn how to troubleshoot the different FCM space warnings that MTM 5149-F48 reports.

Encryption for IBM FlashCore Modules

FCM encryption is enabled by using the self-encrypting drive (SED) function. Support for SED FCMs in IBM Storage Scale System 6000 requires FCMs firmware version 4.4.4.122 (or higher) and IBM Storage Scale System 7.0.0.0 (or higher).

Physical space usage in IBM FlashCore Modules, declustered arrays, and set sizes

MTM 5149-F48 supports FCMs of 38.4 TB usable (TBu) capacity. This drive size reflects the total physical capacity of the drive in TB. The physical capacity TB (decimal) can also be shown as TiB (binary). The total physical and logical capacity of a 38.4 TB FCM is shown in the next table.

Table 34. FCM capacity

FCM type	Total physical capacity in TB	Total physical capacity in TiB	Total logical capacity in TB	Total logical capacity in TiB
38.4 TB	38.4	34.9	115.2	104.7

In the MTM 5149-F48, the total logical capacity that is presented is greater than the total physical capacity because it is assumed that the FCMs compress data.

IBM Storage Scale System 6.2.3.2 through 7.0.0.x: The total physical capacity of the FCM is reduced because 30% of the space is set aside for emergency space, spare space, GNR operations, and performance or TRIM (see Figure 56 on page 92). In comparison, industry standard drives also set about the same amount of space aside. Also, as a safeguard against running out of space, the MTM 5149-F48 throttles FCMs when they reach 60% of their capacity. When an FCM reaches 70% use of its total physical capacity, this condition is known as an out of space (OOS).

Starting with IBM Storage Scale System 7.0.1.x: The total physical capacity of the FCM is reduced because 25% of the space is set aside for emergency space, spare space, GNR operations, and performance or TRIM (see Figure 56 on page 92). When an FCM reaches 75% use of its total physical capacity, this condition is known as an out of space (OOS).

FCM drives are used to make one or more DAs. Through a DA, space is allocated on an MTM 5149-F48; but the FCM monitors and manages the physical space. The virtual disks (vdisks) in the FCM DAs use the underlying physical capacity of the FCMs. The capacity must be configured so that the underlying physical space that is required by the vdisks for data can be provided as the vdisks get written to.

Table 32 on page 87 in "Space provisioning for IBM Storage Scale System 6000" on page 86 considers the compression ratio of the data and the usable space in the FCM to recommend a set size for the vdisk set that is defined in the DA. The maximum set size percentage when you are defining a vdisk set is the percentage of the total logical capacity that is shown in Table 34 on page 90 in "Physical space usage in IBM FlashCore Modules, declustered arrays, and set sizes" on page 90.

Remember: In an IBM Storage Scale System 6000, the total logical capacity is assumed to be greater than the total physical capacity because of compression.

Example

- If a DA is made up of 48 x 38.4 TB (34.9 TiB) in FCMs, the total physical capacity is 48 x 34.9 TiB = 1674 TiB.
- The total logical capacity of the maximum DA is 48 x 104.7 TiB = 5026 TiB.
- The total logical capacity is reduced by the amount of spare and GNR reserved space. With two spares and GNR the reduction is: 5026 TiB - 225 TiB = 4801 TiB.
- If the compression ratio is 2.0x, a user should configure the maximum set size to 48%. So, the total logical capacity should be set to 4 801 TiB x 48% = 2308 TiB of logical capacity. If 8+2P RAID is used, this reduces the logical capacity by another 20%; meaning: 1843 TiB would be the final logical capacity with such RAID configuration.
- In this example, the DA provides 1843 TiB of logical capacity, although the total physical capacity is 1674 TiB.

Space warnings and safeguards

When an FCM experiences an OOS condition, further write operations to the file system cannot be performed anymore. As a result, the file system on these vdisks might fail and it might unmount, thus creating a loss of access condition. MTM 5149-F48 has a system safeguard to make sure if the DA is full, data is not lost.

With IBM Storage Scale System 6.2.3.2 through 7.0.0.x

When an OOS condition occurs, the MTM 5149-F48 notifies the user that the used physical space is becoming a concern. The system generates event notifications and triggers certain actions when the space usage reaches predefined thresholds. The space usage information is summarized into five space usage states (see Figure 55 on page 91).

- **Good.** This state means that the system is healthy, and that no action is required.
- **Informational.** Triggered at 55% of physical space used, means that the system is healthy, and that you must start planning actions.
- **Warning.** Triggered at 60% of physical space used, means that a low space condition is happening. An action must be taken to avoid the possibility of an OOS. The write throughput is limited to 10 GB/s per MTM 5149-F48 I/O canister (approximately 40% of the maximum write throughput). This provides more time for the administrator to take a preventive action.
- **Critical.** Triggered at 65% of physical space used, means that a critical space condition is happening. An action must be taken to avoid the imminent possibility of an OOS. The write throughput is limited to 1 GB/s per MTM 5149-F48 I/O canister (approximately 5% of the maximum write throughput). Decreasing write I/O to a low level gives more time for the administrator to take immediate action.
- **OOS.** Triggered at 70% of physical space used, means that an out of space condition is happening. The file system panics and might be unmounted. The file system unmount might be avoided if the metadata is replicated (and write operation of the replica is successful) or if the FCM DA contains only data vdisks.

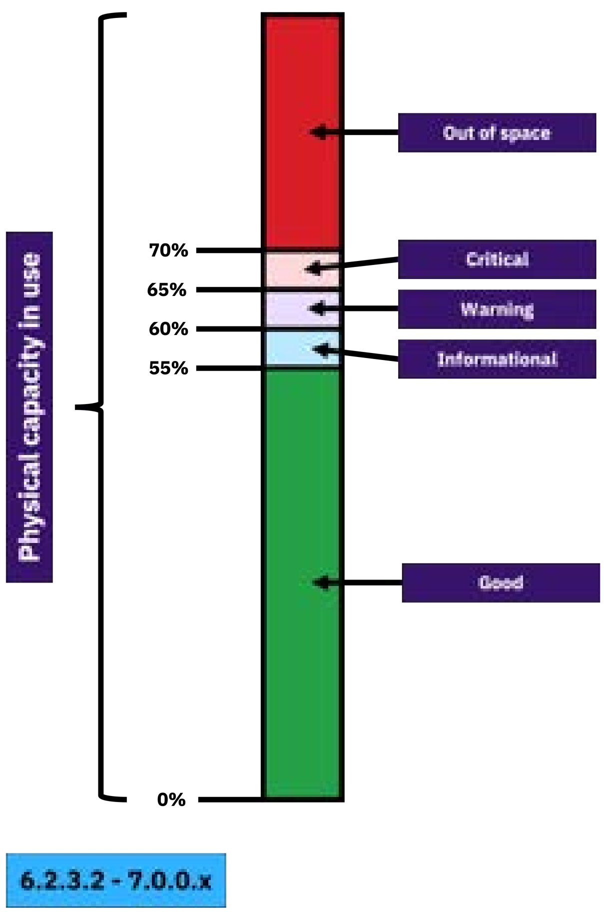


Figure 55. IBM Storage Scale System 6.2.3.2 through 7.0.0.x: Physical space usage in an FCM or a DA, and safeguard thresholds

Starting with IBM Storage Scale System 7.0.1.x

When an OOS condition occurs, the MTM 5149-F48 notifies the user that the used physical space is becoming a concern. The system generates event notifications and stops write I/O actions when 75% of the raw physical space is in use. The space usage information is summarized into five space usage states (see Figure 56 on page 92).

- **Good.** This state means that the system is healthy, and that no action is required.
- **Informational.** Triggered at 60% of physical space used, means that the system is healthy, and that you must start planning actions.
- **Warning.** Triggered at 65% of physical space used, means that a low space condition is happening. An action must be taken to avoid the possibility of an OOS.
- **Critical.** Triggered at 70% of physical space used, means that a critical space condition is happening. An action must be taken to avoid the imminent possibility of an OOS.
- **OOS.** Triggered at 75% of physical space used, means that an out of space condition is happening. The file system panics and might be unmounted. The file system unmount might be avoided if the metadata is replicated (and write operation of the replica is successful) or if the FCM DA contains only data vdisks.

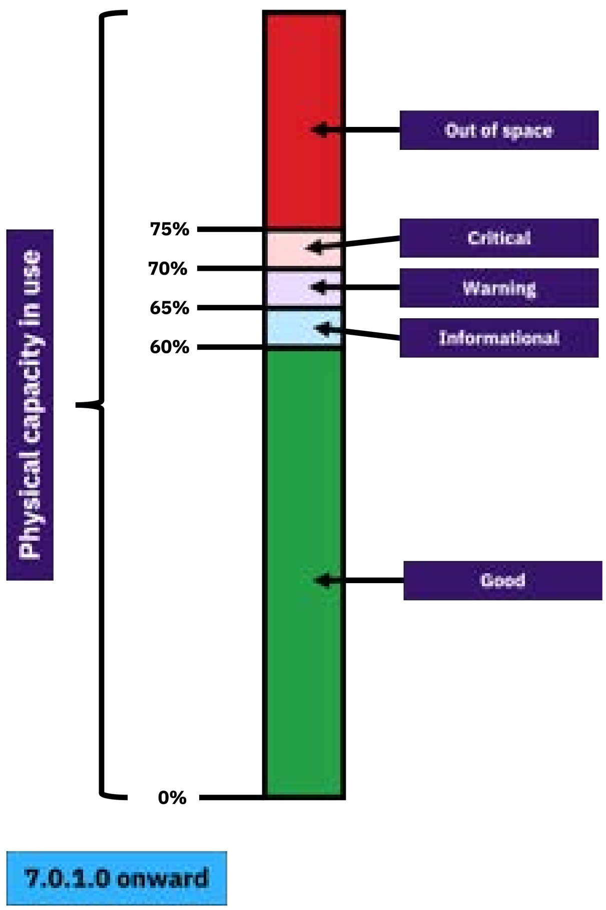


Figure 56. Starting with IBM Storage Scale System 7.0.1.x: Physical space usage in an FCM or a DA, and safeguard thresholds

Troubleshooting an OOS state

When your system is approaching the OOS state of capacity use, free capacity by following these three actions:

1. Delete unneeded files and objects from the vdisk.
2. Start migrating files off the vdisk to a different location.

3. Run a manual space reclaim (TRIM).

Warning: If these actions do not help to free space, contact IBM Support immediately for further assistance.

Appendix D. Best practices for space management of industry-standard NVMe drives

Learn best practices to manage your IBM Storage Scale System 6000 (MTM 5149-F48) with industry-standard NVMe drives and how to provision the correct amount of capacity for your workload.

The IBM Storage Scale System 6000 is compatible with industry-standard NVMe drives. These drives do not compress data. Before you provision your system, use the IBM Storage Modeller tool to determine how to configure the drives.

Example of how to use the IBM Storage Modeller tool

1. In a web browser, go to the tool site and sign in.

   https://www.ibm.com/tools/storage-modeller/
2. Create a new project by clicking **New Project +**.
3. Enter the information that the form requests: a name and a description for your project. Click **Submit**.
4. When you land on the **Products** page, click **Add Product +**.
5. Go to the **IBM Storage Scale** section and click **Scale System 6000**.
6. In the **Scale System 6000** pane that appears at the page bottom, click **Add Product +**.
7. In the **New Product** dialog, go to **Solution Group name** and click the **+** button.
8. In the **Add Solution group** dialog, enter a descriptive name for your solution group. Click **Submit**.
9. When you land back in the **Scale System 6000** pane, click **Submit**.
10. When you land on the **Scale System 6000** page, click the **Capacity** tab.
11. In the **NVMe Enclosures** section, use the drop down menus to select the applicable information for the following fields.

    - **ISS NVMe drives**
    - **Quantity**
    - **RAID type**

    To save the information, click **Configure**.

When you land back on the **Scale System 6000** page, you see a table similar to the one presented in the next example, where you can see the following information.

- The drives configuration that you previously entered. See **1** in the next figure.
- A note that states that, starting with IBM Storage Scale System 7.0.1.0, if you use the default **`essrun`** configuration, the capacity matches the effective capacity that is shown in the IBM Storage Modeller tool. See **2** in the next figure.
- The effective capacity, based on the information that you entered in the tool. See **3** in the next figure.
- A checkbox that allows you to over-provision the system, thus leaving spare disk space. See **4** in the next figure.

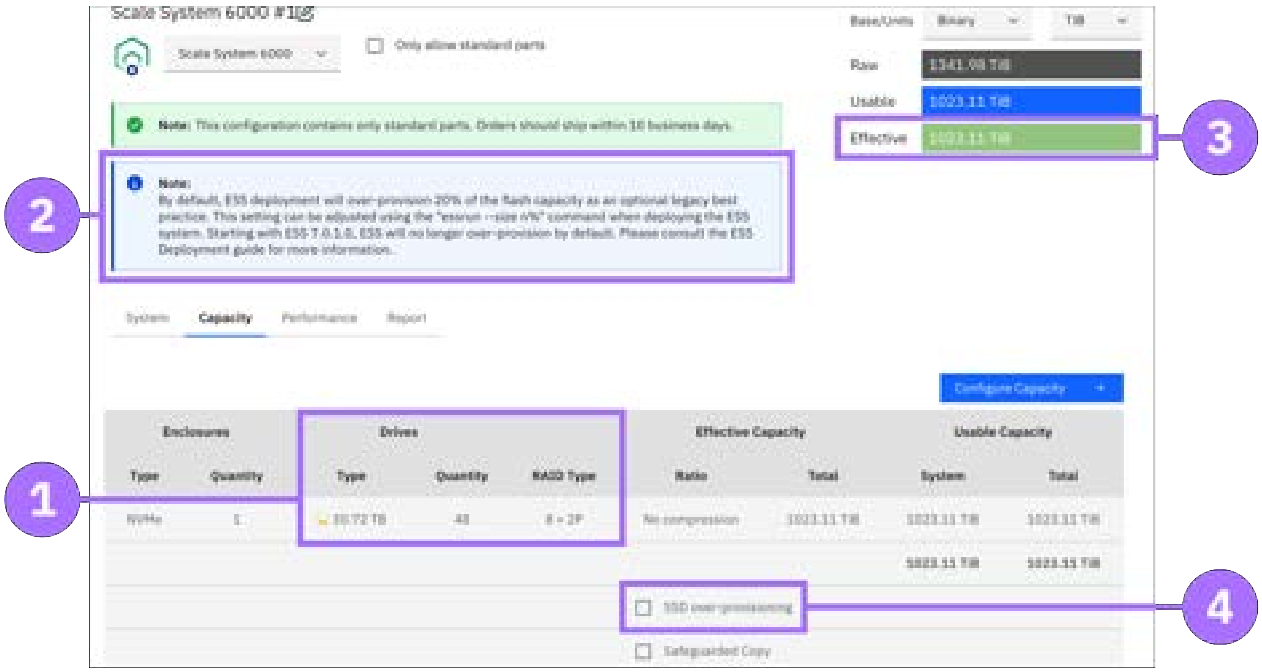


Figure 57. IBM Storage Modeller tool

SSD over-provisioning checkbox

When you select the **SSD over-provisioning** checkbox, the effective capacity of the system is reduced to help ensure that the used disk space does not negatively impact workload performance. With IBM Storage Scale System 7.0.1.0 onward, you must use the `essrun -- size n%` command during IBM Storage Scale System 6000 configuration if you want the configured capacity to match the value that is shown in the IBM Storage Modeller tool when the **SSD over-provisioning** box is checked.

Appendix E. Switches for IBM Storage Scale System 6000

These switches can be installed in the same rack as the IBM Storage Scale System solution, in a similar way as other supported high-speed data switches.

MTM 5149-N64 switch

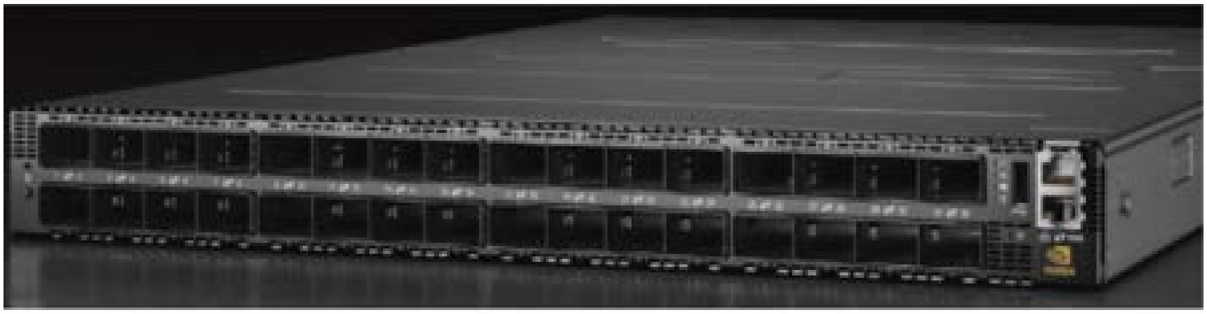


Figure 58. Ports view of the MTM 5149-N64 switch

Figure 59. PSUs view of the MTM 5149-N64 switch

MTM 5149-S05 switch

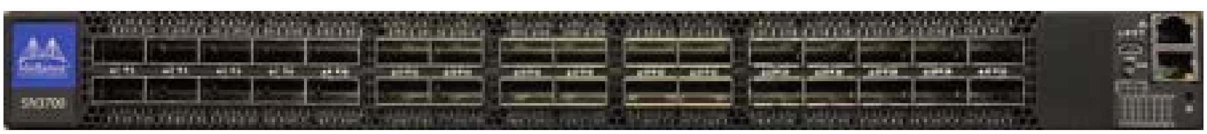


Figure 60. Ports view of the MTM 5149-S05 switch

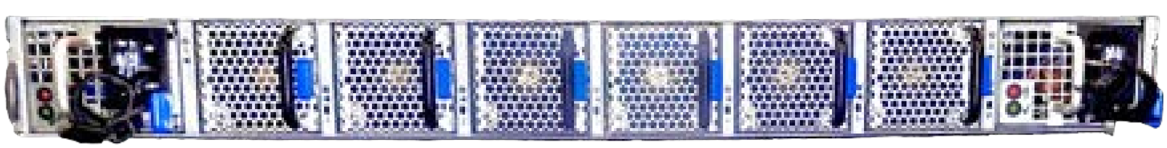


Figure 61. PSUs view of the MTM 5149-S05 switch

MTM 5149-S54 switch

Figure 62. Ports views of the MTM 5149-S54 switch

Figure 63. PSUs view of the MTM 5149-S54 switch

The MTM 5149-N64 switch

This topic describes the key features and specifications of the NVIDIA/Mellanox NDR switch, IBM MTM 5149-N64.

The 5149-N64 switch introduction

The NVIDIA/Mellanox NDR switch, IBM MTM 5149-N64 switch, is a managed 64-port switch that provides 400 Gb/s InfiniBand bandwidth per port in a 1U standard chassis design. The MTM 5149-N64 switch is the next generation after MTM 8831-H40, and can be installed in the same rack as the ESS solution, similar way as other supported high-speed data switches.

The following figures show the ports side and the power side of a 5149-N64 switch.

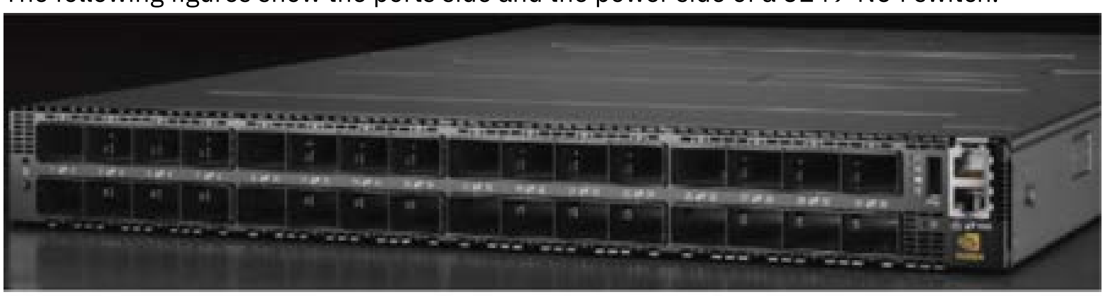


Figure 64. 5149-N64 switch ports

Figure 65. 5149-N64 switch power

For more information about the 5149-N64 switch, see NVIDIA documentation. The following table lists the IBM MTM and the NVIDIA system model:

Table 35. IBM MTM and NVIDIA system model	
IBM MTM	**NVIDIA system mode**
5149-N64	QM9700

Key hardware and software features

Key hardware features

Feature	Quantity
Form factor	U1
NDR 400 Gb/s InfiniBand ports	64
Octal small form factor pluggable (OSFP) connectors	32
USB management ports	1
RJ45 management ports	1
RJ45 management ports (UART)	1
Power supply units	2 (1+1 redundant)
Fan modules	7 (6+1 redundant)

Key software features

The software features of the 5149-N64 switch are as follows:

- On-board subnet manager
- NVIDIA MLNX-OS software package
- Management access via:
  - Command line interface (CLI)
  - Web based graphical user interface (GUI)
  - Simple Network Management Protocol (SNMP)
  - JavaScript Object Notation (JSON)

Note: Installing and upgrading switch software is a customer responsibility. Updates must be installed only upon the direction of IBM Support.

The IBM MTM 5149-N64 switch views

The following image labels the 32 OSFP ports and highlights features on the port side of 5149-N64 switch. Port numbers are printed on the switch between the top and bottom ports.

Figure 66. Port view of the 5149-N64 switch

The following image highlights features on the power side of the 5149-N64 switch:

Figure 67. Power side view

Labels of the 5149-N64 switch

The details of the 5149-N64 switch labels are provided in this section.

Chassis label

The chassis label is affixed to a pull out tab on the left of the system on the power side. The chassis label contains the part number, serial number, and GUID of the 5149-N64 switch.

Figure 68. Chassis labels

Switch labels

On the bottom of the switch you can find the regulatory label, and the MTM label, as shown in the following figure:

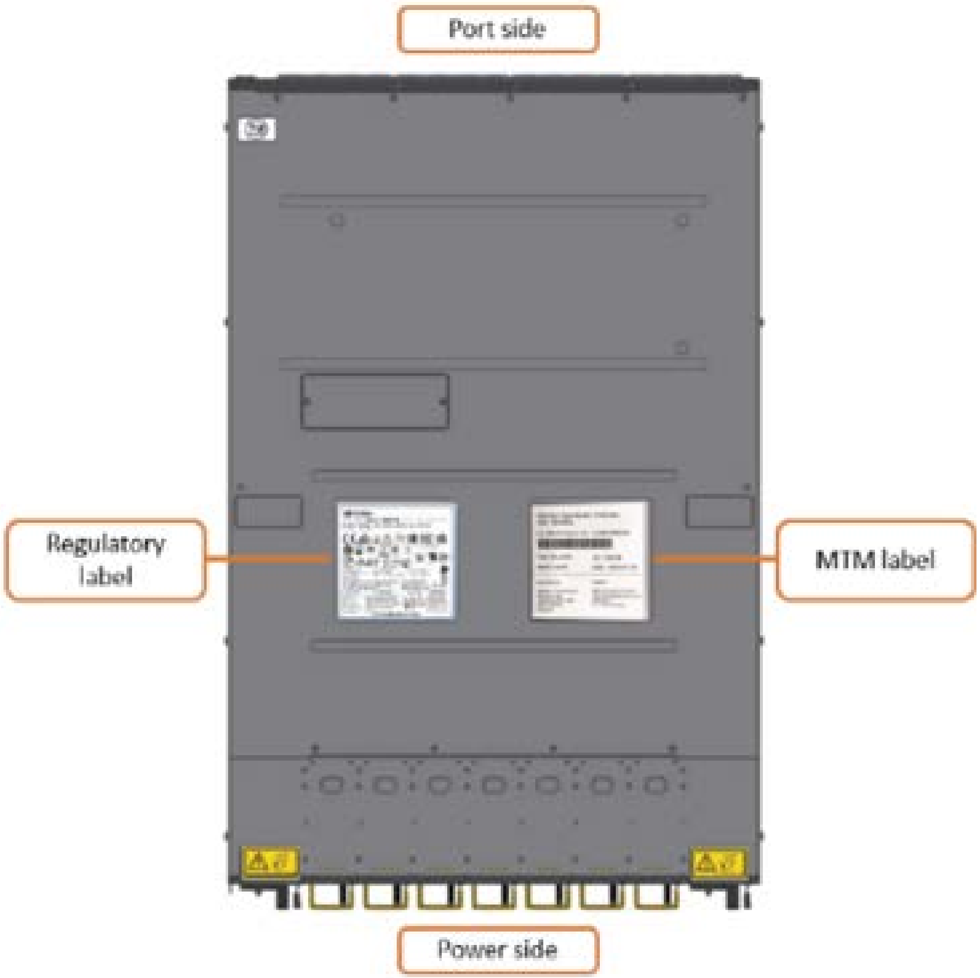


Figure 69. Switch labels

Specifications of the 5149-N64 switch

The following specifications provide brief information about the 5149-N64 switch.

System dimensions and weight

System weight and dimensions are listed in the following table:

Height **(1U rack standard)**	**Width** **(19" rack standard)**	**Depth**	**Weight**
4.36 cm (1.7 in)	43.8 cm (17 in)	66.04 cm (26 in)	14.5 kg (31.97 lb)

Note: Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be installed. Installation locations at 32U and higher require two people to lift the switch. For installation locations less than 32U, one person can safely lift the switch.

Air temperature

Air flows from the power side of the switch to the port side (power-to-connector (P2C) air flow configuration). Temperature requirements for the switch are listed in the following table:

Air flow direction	Operating	Storage
Forward, power-to-connector (P2C)	0° C to 35°C (32°F to 95°F) 10%-85% non-condensing humidity	-40° to 70°C (-40°F to 158°F) 10%-90% non-condensing humidity

Important: Regardless of the system specifications about absolute maximum temperature, the nonoperating environmental conditions of the system solid-state devices must adhere to the *JEDEC JESD218 Enterprise-class SSD* requirements for data retention endurance. These requirements must be met particularly for the system storage, transit, or shipping maximum temperatures. To best maintain the data retention of the system solid-state devices, keep an *optimal* condition for nonoperating temperatures by making sure that the system does not remain for longer than 3 months at temperatures that exceed 40°C.

Power

There are two hot swappable PSUs in the switch. Power requirements for each PSU are listed in the following table:

Input	Frequency
200-240 VAC	50/60 Hz

Service indicators

Service indicators for the switch are implemented by using LEDs to provide status information.

LEDs provide a visual method to identify, clarify, and resolve system issues. The system-level LEDs indicate the overall switch health and status. LEDs on individual modules, such as the power supplies, indicate the health and status of the specific module.

LEDs of the 5149-N64 switch

System LEDs

The following figure shows system LEDs on the port side of the 5149-N64 switch:

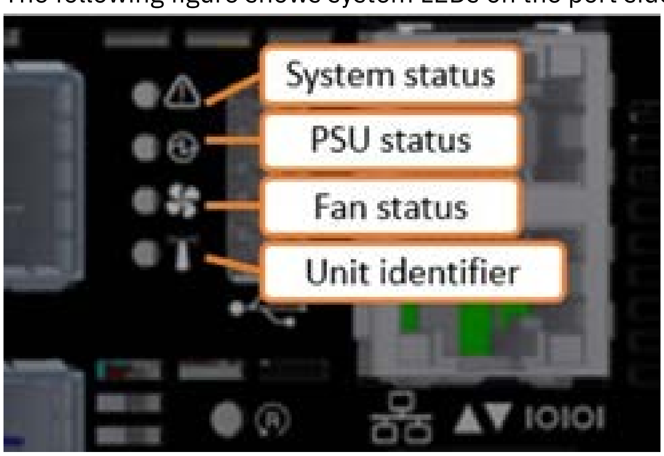


Figure 70. LEDs on the port

The following table lists the port side LED states:

LED	State	Description	Action required
System status LED	Solid green	The system is up and running normally.	-
	Flashing green	The system is booting up.	Wait up to five minutes for the end of the booting process.
	Solid amber	A major error has occurred. For example, corrupted firmware or an overheated system.	If the system status LED shows amber five minutes after starting the system, unplug the system and call your IBM support for assistance.
Power supply units status LED	Solid green	All plugged (one or two) power supplies are running normally.	-
	Solid amber	One or both of the power supplies are not operational or not powered up or the power cord is disconnected.	Make sure the power cord is plugged in and active. If the problem resumes, the FRUs might be faulty, and should then be replaced.
Fan status LED	Solid green	All fans are up and running.	-
	Solid amber	Error, one or more fans are not operating properly.	The faulty FRUs should be replaced.
Unit identification LED	Blue	On	-

Fan module LEDs

Each fan module has its own status LED. LED indicators for fans 2 and 3 are highlighted in the following figure:

Figure 71. Fan LEDs

State	Description	Action required
Solid green	This fan unit is operating as expected.	-
Solid amber	This fan unit is missing or not operating properly.	The fan unit should be replaced.

Power supply unit LEDs

Each PSU has its own status LED, which reflect the system PSU state. The LED indicator for PSU 1 is highlighted in the following figure:

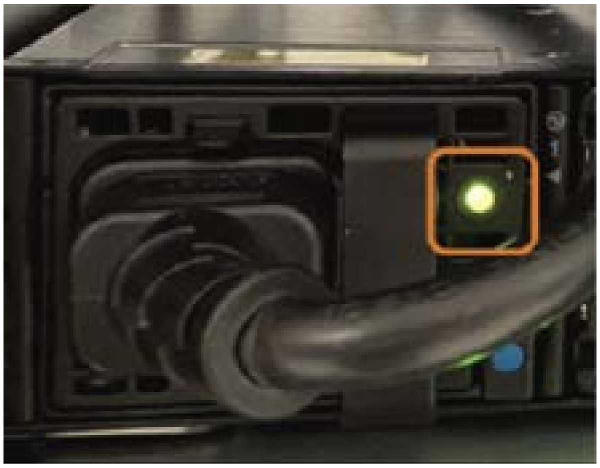


Figure 72. Power supply unit LEDs

State	Description	Action required
Solid green	All PSUs are connected and running normally.	-
Flashing green	Only 12VSB on (PSU off), PSU is in Smart-on state.	Contact IBM support.
Solid amber	AC cord unplugged or AC power lost on one PSU while the other power supply still has AC input power.	• Make sure that both PSU cords are connected and fully seated. • Determine why power was lost in the electrical outlet that the PSU is plugged into.
	PSU failure (including voltage out of range and power cord disconnected).	Check voltage. If OK, contact IBM support.
Flashing amber	Power supply warning events where the power supply continues to operate; high temp, high power, high current, or slow fan.	Contact IBM support.
Off	No AC power to all power supplies. Power might not be applied to the electrical outlets, or one or both of the PSUs might be faulty.	• Make sure that both PSU cords are connected and fully seated. • Determine why power was lost in the electrical outlet that the PSU is plugged into.

Port LEDs

Each OSFP port can be divided into 2 dual-lane ports, for a total of 4 ports.

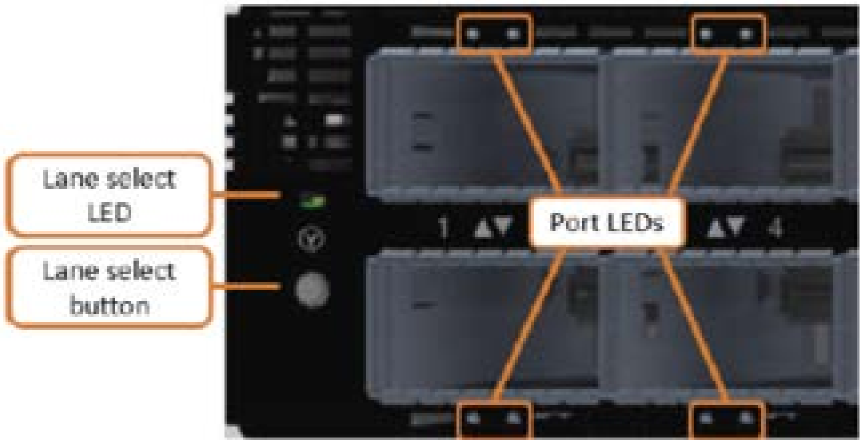


Figure 73. Port LEDs

The following table lists port LED states:

State	Description	Action required
Off	Link is down.	Check the cable.
Solid green	Link is up with no traffic.	-
Flashing green	Link is up with traffic.	-
Solid amber	Physical connection established, logical connection is pending.	Wait for the logical link to raise. Check that the subnet manager (SM) is up.
Flashing amber	There is a problem with the link.	Check that the SM is up.

The 5149-N64 switch installation

This switch is installed by the customer or by IBM, depending on the deployment.

The 5149-N64 switch is typically installed in an ESS system. ESS systems can be purchased pre-racked or unracked. SSRs are responsible for installing the5149-N64 switch in an unracked ESS system.

For installing a system requires planning to facilitate a smooth installation, initialization, and configuration.

Planning for the 5149-N64 switch

For IBM ESS deployments, IBM is responsible for installing the switch, whether done by manufacturing for the racked solutions or by the SSR onsite for the unracked solutions.

Customers must complete a Technical and Delivery Assessment (TDA) that the assigned SSR then reviews with them. In conjunction with the TDA, planning information is available in IBM Docs. The SSR and the customer must also review and complete all of the planning information in IBM Docs. Before scheduling the trip onsite to perform the installation, the SSR should attend meetings with the customer to perform planning tasks.

Pre-install TDA checklist

TDA checklists ensure that the customer prepares the site and gathers information that is needed to deploy a system. The IBM sales team or business partner helps the customer complete the TDA checklists, with an assigned IBM quality practitioner or technical account manager (TAM) managing the communications.

Note: TDA completion is mandatory for these systems and must be completed before arriving onsite to install the system.

It is the responsibility of the customer, the sales team or business partner, and the IBM quality practitioners or TAM to ensure that the TDA is completed so that the SSR can use information in the TDA checklist to install and initially configure the solution.

Two types of TDA checklists are as follows:

Pre-sale TDA checklist
: Provides a roadmap of questions associated with a configuration and proposal for IBM products and solutions. The pre-sale TDA is not relevant to SSRs or to system installations.

Pre-install TDA checklist
: Provides a roadmap of questions that assess site readiness and prepares the environment for the installation. Information in the pre-install TDA checklist is used the installation and initial configuration.

TDAs are online forms, but can be exported in an .xlsx or .csv format, saved, and printed.

TDA checklists are in the IMPACT website (this site requires an IBM ID). The IBM practitioner creates the pre-install checklist. Customers, their business partner, IBM quality practitioner or TAM, and the assigned SSR can access this site to look at and to work on the TDA.

Installation overview

Switch package contents

The switch package contains the following contents.

Important: If parts are missing or damaged, do not proceed with the installation. Contact your next level of support.

- One switch
- One rail kit

  Includes system rails that attach to the switch, and rack rails that attach to the rack.
- Four power cables

  Type C14 to C15
- Two PSU cable retainers
- 32 dust caps
- One environmental notice user guide
- One repair identification tag

Installing the 5149-N64 switch

Complete this procedure to install the 5149-N64 switch.

Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be installed. Installation locations at 32U and higher require two people to lift the switch. For installation locations less than 32U, one person can safely lift the switch.

1. Attach the left and right system rails to the switch.
   a) Align the switch and the system rails so that the system chassis pins align with the button holes on the rails and press the rails to the switch.
   b) Slide the rails toward the power side of the switch.

Figure 74. The 5149-N64 switch and rails alignment

The following figure shows a closeup of a rail secured to the switch:

Figure 75. A rail secured to the 5149-N64 switch

2. Attach the rack rails to the rack.
   a) Insert the rack rails into the back of the rack at an angle, then rotate the rails to align with the front rack post.

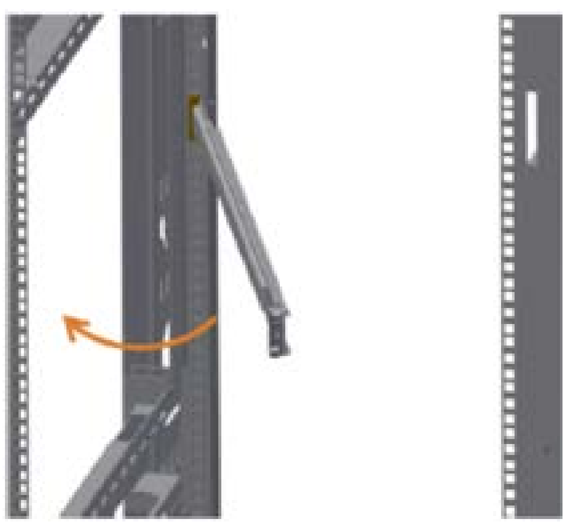


Figure 76. Inserting rack rails into the back of the rack

A rack rail that is inserted into the back of the rack is shown in the following figure:

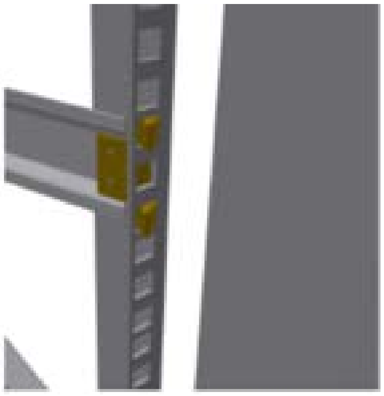


Figure 77. A rack rail that is inserted into the back of the rack

b) Extend the telescopic rail to the front of the rack.

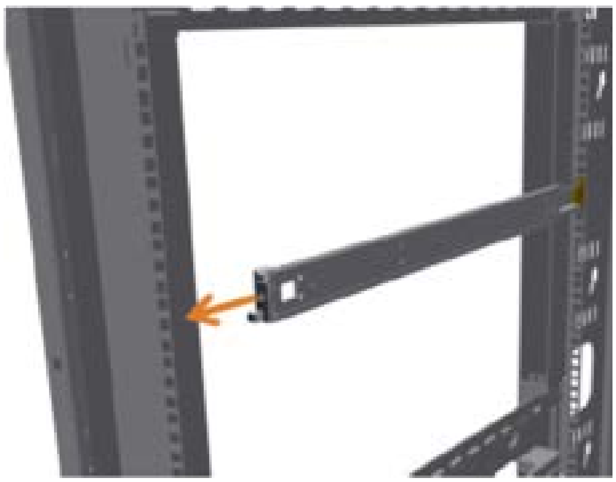


Figure 78. Extending the telescopic rail

c) Insert the rail latches into the rack slots. The following figure shows that the latch is fully inserted. A correctly secured rack rail is shown in the following figure:

Figure 79. Fully inserted latch into the rack slots

3. Mount the 5149-N64 switch.

   a) Slide the switch into the channel on the rack rails. Push the switch until you hear the locking mechanism click on both sides.

Figure 80. Sliding the switch into the channel

b) Tighten the two captive screws at the front of the mounted switch.

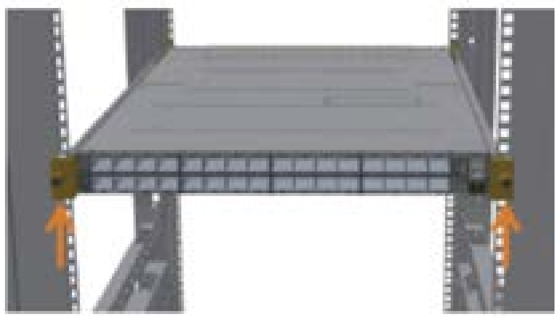


Figure 81. Tightening the two captive screws

Note: Unused OSFP cages must be closed with dust caps that are supplied with the system.

MTM 5149-N64 switch service

Service procedures and a list of replaceable parts are provided for the MTM 5149-N64 switch service.

Key replaceable parts for MTM 5149-N64 switch

The following table lists key replaceable parts for the switch. Because part numbers can change over time, in a repair situation, verify that the correct CRU or FRU is obtained.

Part	FRU part number	CRU or FRU	Concurrent replace
NVIDIAQuantum-2 based NDR InfiniBand switch, 64 NDR ports, 32 OSFP port	01LL978	FRU	No
Air duct and rack mount rail kit	01LL977	CRU Tier 1	No
NVIDIA800 Gb NDR twin port transceiver	01LL927	CRU Tier 1	No
PSU (P2C airflow)	01LL976	CRU Tier 1	Yes
Fan unit (P2C airflow)	01LL975	CRU Tier 1	Yes

Replaceable parts

Each replaceable part is assigned one of three designations. Most of the parts for this switch are classified as Tier 1 CRU. None of the parts for this switch are Tier 2 CRU.

For more information, see the following table.

Replacement designation	Description
CRU -Tier 1	Replacing a Tier 1 CRU is a customer responsibility, but dependent on if they purchased a service upgrade: • Under the standard warranty – If an IBM representative installs a Tier 1 CRU at the customer's request, the customer is charged for the installation. – If an IBM representative installs a Tier 1 CRU at IBM Support's request, the customer is not charged for the installation. • Under an IBM Storage Expert Care service upgrade: The customer can request that IBM install the CRU at no additional cost.
CRU -Tier 2	Replacing a Tier 2 CRU can be completed by the customer or can be installed at the customer's request by an IBM representative. If a Tier 2 CRU is replaced by an IBM representative: • If the product is under warranty: There is no charge for the part or for the installation if the product. • If the product is no longer under warranty: The customer is charged for the part and the installation.
FRU	Replacing a FRU must be performed by an IBM service personnel or an IBM authorized warranty service provider.

Replacing a customer replaceable unit

Use this procedure for servicing a customer replaceable unit (CRU).

Before you start a replacement, the customer must complete the following tasks:

- If the replacement procedure might put data at risk, the customer must back up the system or logical partition (including operating systems, licensed programs, and data).
- Review the replacement procedure for the part.
- Obtain the required tools for the replacement procedure.
- Inspect the replacement part. If any of the parts are incorrect, missing, or visibly damaged, they should contact the provider of the part or IBM Support.

The replacement best practices are as follows:

- For all replacements performed on an operational system, remove only one module from the enclosure at a time, unless directed otherwise by a replacement procedure.
- To reduce the amount of time required for the replacement, remove the failed module only when the replacement module is available, unpackaged, and ready for installation.
- Follow all recommended procedures for handling devices that are sensitive to electrostatic discharge (ESD).

1. Identify the system that contains the part to replace.
2. Attach the ESD wrist strap. The ESD wrist strap must connect to an unpainted metal surface until the service procedure is completed.
3. Locate the CRU part to be serviced.
4. Remove and replace the CRU part.

5. Verify that the installed part is working as expected.

Replacing an MTM 5149-N64 switch

Complete steps in this procedure to replace a switch.

1. Locate the faulty switch.

   The service ticket should include location information. Confirm that the switch location by looking for fault LEDs on the switch.
2. Back up the switch configuration, if possible. (Customer task)
3. Back up the switch firmware, if possible. (Customer task)
4. Label all data cables that connect to the switch.
5. Label all power cords that connect to the switch.
6. Disconnect the two power cords from their PDU connections.
7. Disconnect all data cables from the switch.

Important:

- Before removing the switch from the rack, ensure that the switch is powered down, with all network and power cables disconnected.
- Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be replaced. Replacement locations at 32U and higher require two people to lift the switch. For replacement locations lower than 32U, one person can safely lift the switch.

1. Unmount the system.
   a) Loosen the captive screws at the front of the system.

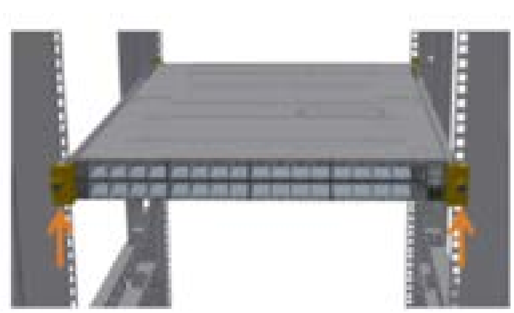
   

   Figure 82. Loosening the captive screws

   b) Pull out the system until the rails stop.

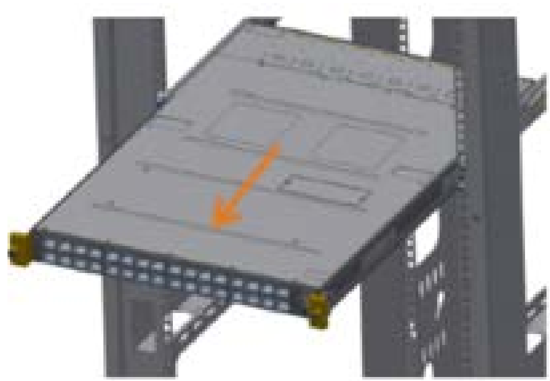
   

   Figure 83. Pulling out the system

   c) Press the spring latches on both sides of the system. Continue pulling out the system until the system rails are clear from the rack rails.

Figure 84. Pressing the spring latches

2. (Optional) Remove the rails from the system.
   a) Release the metal latches and slide the rails to move the system pins out of the smaller oval hole. The rail can then be removed from the system. The following figure shows the system pin and rail holes aligned so that the rail can be removed.

Figure 85. Releasing the metal latches

   b) Press the lock button, and pull the rail outside of the rack assembly.

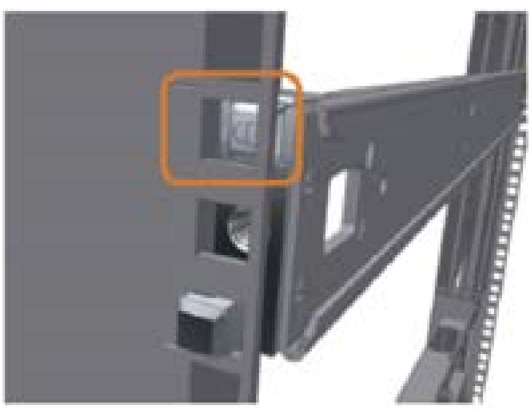


Figure 86. Pressing the lock button

3. Install the switch. For more information about installation, see "Installation overview" on page 107.

Replacing the power supply unit of an MTM 5149-N64 switch

The 5149-N64 switch has two replaceable power supply units (PSUs) in a redundant configuration. Only one healthy PSU is required for the switch to function.

Before removing a PSU, ensure that the other PSU is connected and shows a solid green LED.

1. Remove a power cord from a PSU.

2. Push the release latch while pulling the handle toward you. As the power supply unit unseats, the PSU status LED turns off.

Figure 87. Releasing latch while pulling the handle

3. Install the PSU.

CAUTION: Do not insert a power supply that has power cables connected. Disconnect power cords from outlets before starting.

a) Insert the PSU by sliding it into the opening, until a slight resistance is felt. The latch snaps into place, confirming the installation is correct.

b) Insert the power cord into the PSU connector.

c) Insert the other end of the power cord into an outlet of the correct voltage.

d) Wait for the PSU LED to illuminate solid green. If the LED does not turn solid green, restart the procedure, and ensure that the PSU is correctly seated.

Replacing a fan module of an MTM 5149-N64 switch

The 5149-N64 switch can work with one failed fan module. Operating with more than one failed fan module is not supported.

1. Remove a fan module.

a) Extract the fan module by pulling the left side of the handle. As the fan module unseats, the fan module status LED turns off.

Figure 88. Extracting a fan module

b) Remove the fan module from the switch chassis.

2. Insert a fan module.
   a) Insert the fan module by sliding it into the opening until slight resistance is felt. Continue pressing the fan module until it seats completely.
   b) Wait for the fan module LED to illuminate solid green. If the fan module LED does not turn solid green, re-start the procedure and ensure the fan module is correctly seated.

The MTM 5149-S05 switch

This NVIDIA switch, IBM MTM 5149-S05 switch, is a 1U, Ethernet, 200 GbE (50G PAM4 per lane), 32-ports switch that offers a switching capacity of 6.4 Tb/s, with port speeds spanning from 10 Gb/s to 200 Gb/s per port. The MTM 5149-S05 switch can be installed in the same rack as the IBM Storage Scale System solution, in a similar way as other supported high-speed data switches.

Note: If Licensee purchases and incorporates IBM MTM 5149-S05 switch ("Switch") as part of your IBM Storage Scale System, the Switch may not comply with all of the requirements included in IBM's Security and Privacy by Design practices: https://www.ibm.com/support/pages/ibm-security-and-privacy-design.

Overview of the MTM 5149-S05 switch

Use the MTM 5149-S05 switch as a management switch in data center networking, enterprise access networking, or campus access networking.

This open network switch includes the following components:

- 32 200-GbE ports
- 1 100/1000Mb management port
- 1 RJ45 serial port
- 1 micro-USB port
- 2 hot-swappable PSUs (1+1 redundant)
- 6 hot-swappable fans (N+1 redundant)
- Open Network Install Environment (ONIE) software that supports the installation of an open source or commercial compatible network operating system (NOS).

The following figures show the PSUs side and the ports side of an MTM 5149-S05 switch.

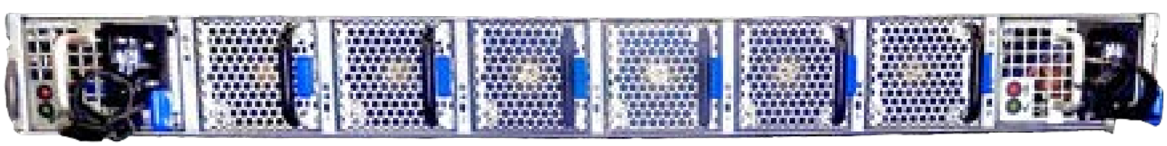


Figure 89. PSUs view of the MTM 5149-S05 switch

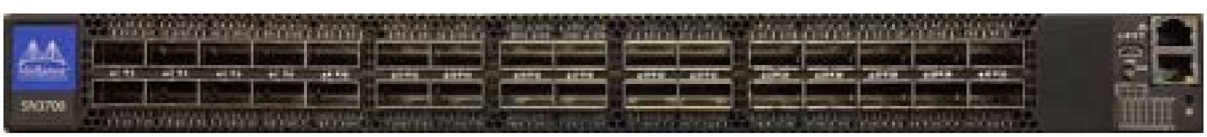


Figure 90. Ports view of the MTM 5149-S05 switch

For more information about the MTM 5149-S05 switch, see NVIDIA Spectrum SN3700 in NVIDIA documentation.

The following table lists the IBM MTM and the NVIDIA system model.

Table 36. IBM IBM and NVIDIA system model

IBM MTM	NVIDIA system model
5149-S05	NVIDIA Spectrum SN3700

Key features and components of the MTM 5149-S05 switch

The key benefits and components of this entry-level open networking switch include features and components like 108 Gbps of switching capacity 61.6 Gbps in forwarding rate, 32 switch ports, and 3 management ports.

In the next table, consult the performance and capacity characteristics of an MTM 5149-S05 switch.

Feature	Performance and capacity
Switching capacity	6.4 Tb/s
Wire speed switching	8.33Bpps (Billion packets per second)
Lanes per port x maximum speed per lane	4x 50G PAM4
Latency	425 ns
CPU	Quad-core x86
System memory	8 GB
SSD memory	32 GB
Packet buffer	42 MB

Hardware components:

- 32 200-GbE ports
- 1 100/1000Mb management port
- 1 RJ-45 serial port
- 1 micro-USB port
- 2 hot-swappable PSUs (1+1 redundant)
- 6 hot-swappable fans (N+1 redundant)

Software features:

- ONIE software that supports the installation of an open source or commercial compatible NOS.
- Support to run third-party containerized applications on the switch system.

Views of the MTM 5149-S05 switch

The switch ports are accessible from one of the sides of the enclosure, the hot-swappable fans and the hot-swappable PSUs are placed on the contrary side of the enclosure.

Ports view (front of the enclosure)

The following figure shows the 32 200-GbE ports and the 3 management ports of an MTM 5149-S05 switch.

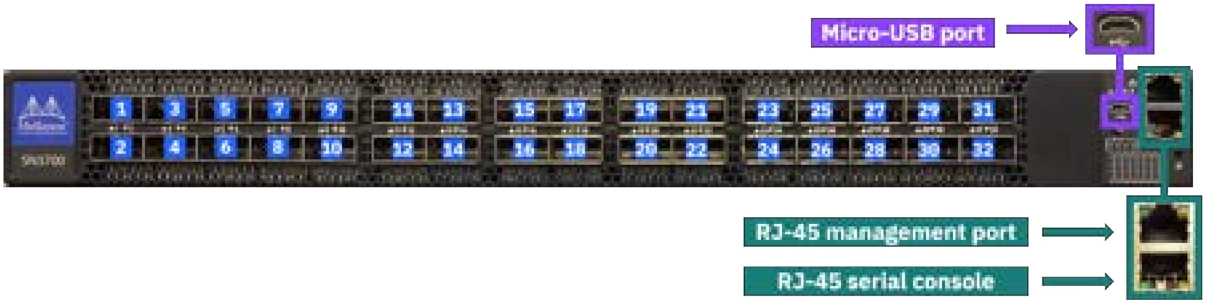


Figure 91. Ports of an MTM 5149-S05 switch

Port numbers are printed in pairs and placed between the top and bottom rows of switches, as highlighted in the following figure.

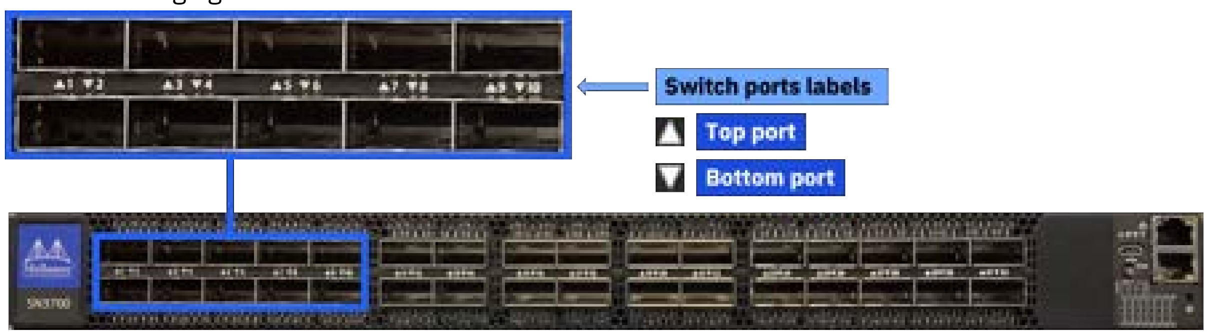


Figure 92. Switch port numbers of an MTM 5149-S05 switch

Fixed fans and PSUs view (rear of the enclosure)

The following figure highlights the placement of the six hot-swappable fans and the two hot-swappable PSUs of an MTM 5149-S05 switch.

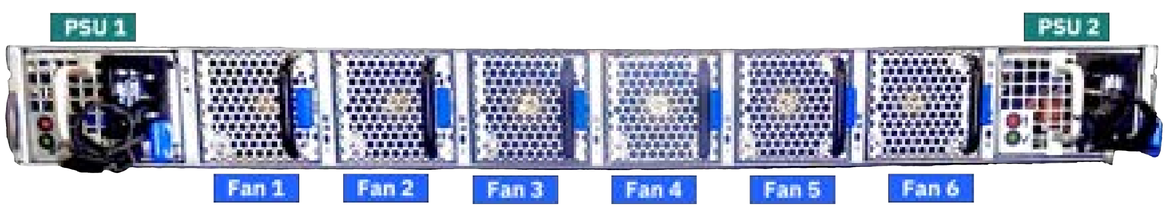


Figure 93. Hot-swappable fans and PSUs of an MTM 5149-S05 switch

Labels of the MTM 5149-S05 switch

Different details are provided in the labels of the MTM 5149-S05 switch and its PSUs.

PSUs labels

Each PSU has its labels at its top side.

Figure 94. PSUs label

Switch labels

On the bottom of the switch, you can find the regulatory label, and the MTM label, as shown in the following figure. These labels contain the part number, serial number, and GUID of the MTM 5149-S05 switch.

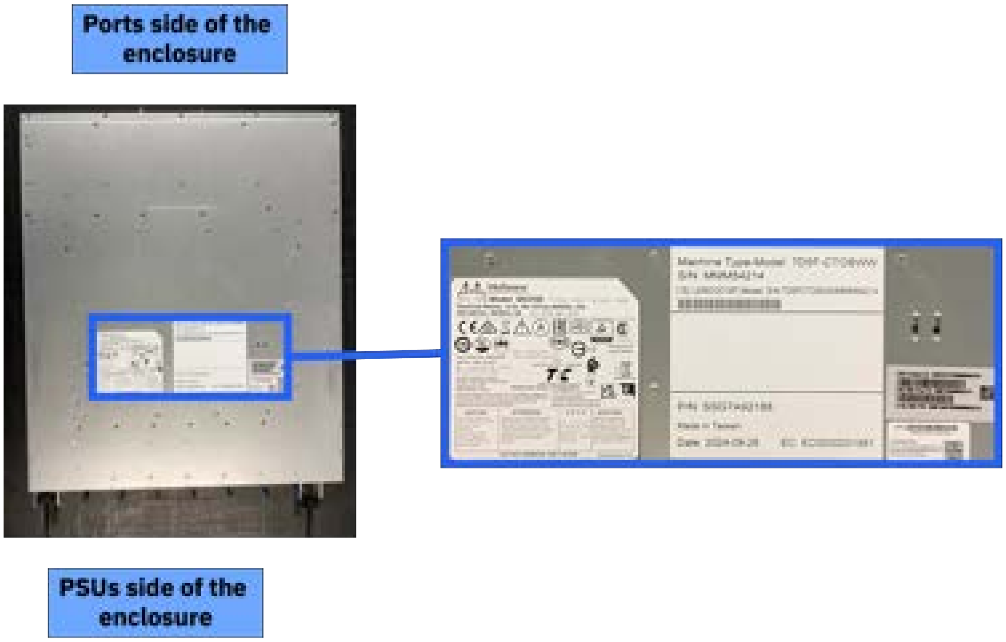


Figure 95. Labels on the bottom of the switch enclosure

Specifications of the MTM 5149-S05 switch

The following specifications provide brief information about the MTM 5149-S05 switch.

System dimensions and weight

System weight and dimensions are listed in the following table.

Height **(1U rack standard)**	**Width** **(19" rack standard)**	**Depth**	**Weight**
44 mm (1.72 in)	428 mm (16.84 in)	559 mm (22 in)	14 kg (30.8 lb)

Note: Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be installed. Installation locations at 32U and higher require two people to lift the switch. For installation locations less than 32U, one person can safely lift the switch.

Air temperature

Air flows from the PSUs side of the switch to the ports side (power-to-port air flow configuration). Temperature requirements for the switch are listed in the following table.

Air flow direction	**Operating**	**Storage**	**Relative humidity**
Power-to-connector (forward)	0°C to 40°C (32°F to 104°F)	-40°C to 70°C (-40°F to 158°F)	5% to 85%

Important: Regardless of the system specifications about absolute maximum temperature, the nonoperating environmental conditions of the system solid-state devices must adhere to the *JEDEC JESD218 Enterprise-class SSD* requirements for data retention endurance. These requirements must be met particularly for the system storage, transit, or shipping maximum temperatures. To best maintain the data retention of the system solid-state devices, keep an *optimal* condition for nonoperating temperatures by making sure that the system does not remain for longer than 3 months at temperatures that exceed 40°C.

Power

The switch has two redundant, load-sharing, hot-swappable 1100W AC PSUs that comply with 80 PLUS Gold.

Power requirements for each PSU are listed in the following table.

Input	AC input current	Frequency
100 VAC to 264 VAC	2.9 A to 4.5 A	50 Hz to 60 Hz

Service indicators: LEDs of the MTM 5149-S05 switch

Service indicators for the switch are implemented by using LEDs to provide status information.

LEDs provide a visual method to identify, clarify, and resolve system issues. The system-level LEDs indicate the overall switch health and status. LEDs on individual modules, such as the PSUs, indicate the health and status of the specific module.

System LEDs (switch ports side of the enclosure)

The following figures and tables describe the LEDs on the ports side of an MTM 5149-S05 switch.

Figure 96. LED indicators for system, fans, PSUs, and identifier of an MTM 5149-S05 switch

Table 37. Descriptions for the LED indicators for system, fans, PSUs, and identifier of an MTM 5149-S05 switch

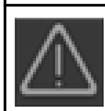
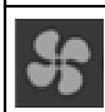
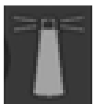
State LED	Color	Description	Action required
 System status LED	Solid green	The system is up and running normally.	
	Flashing green	The system is initializing.	Wait up to five minutes for the end of the process.
	Solid amber	An error occurred. For example, corrupted firmware or an overheated system.	If the system status LED shows amber five minutes after the system is started, unplug the system and call your IBM Support representative for assistance.
 Fan status LED	Solid green	All fans are up and running.	The faulty FRUs should be replaced.
	Solid amber	Error, one or more fans are not operating properly.	

Table 37. Descriptions for the LED indicators for system, fans, PSUs, and identifier of an MTM 5149-S05 switch (continued)			
State LED	**Color**	**Description**	**Action required**
 PSU LEDs	Solid green	All plugged power supplies are running normally.	
	Solid amber	One or both of the power supplies are not operational or not powered; or the power cord is disconnected.	Make sure that the power cord is plugged in and active. If the problem resumes, the FRUs might be faulty; and faulty FRUs must be replaced.
 Unit identifier LED	Solid blue	Turns on when identification is requested vis CLI.	
	Off	No identification in progress.	

Switch ports LEDs

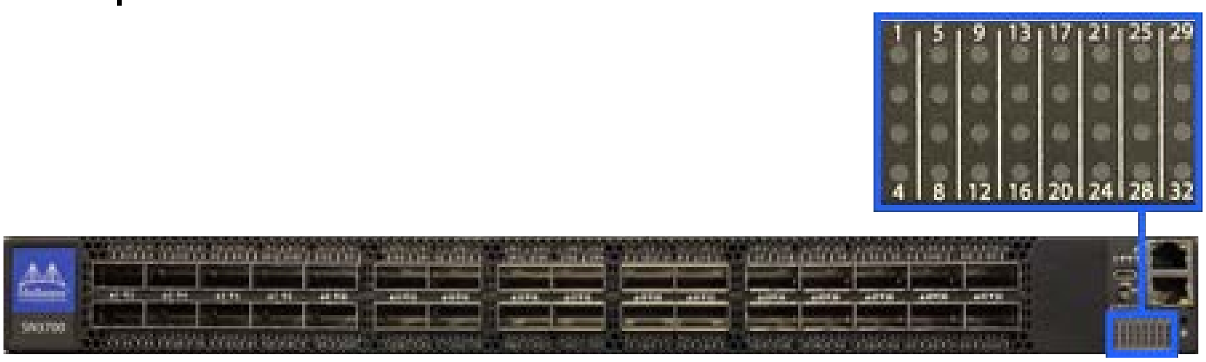


Figure 97. Switch port LEDs of an MTM 5149-S05 switch

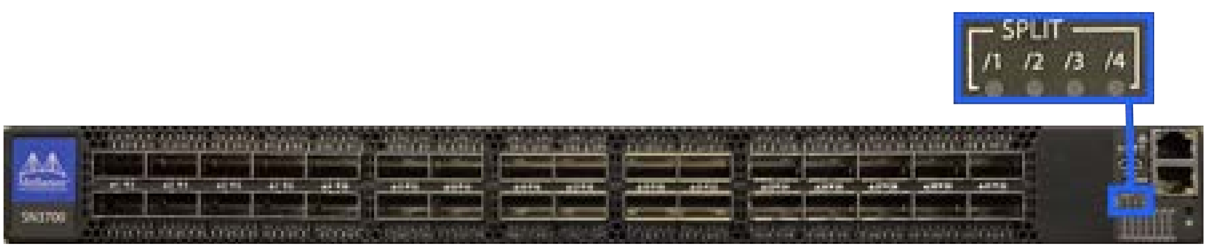


Figure 98. LED splitting options of an MTM 5149-S05 switch

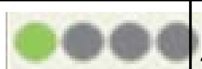
Table 38. Descriptions for the LED splitting options of an MTM 5149-S05 switch				
State	**State indication LEDs** **[/1 /2 /3 /4]**	**Module LED indication**	**LED color**	**State description**
0		Indication for all link types	Off	No 4X/2X/1X link was established on this QSFP module
			Solid green	At list one link was established: 4X/ 2XA/2XB/1XA /1XB/1 XC/1XD
			Flashing green	Traffic is running in linked ports
			Flashing amber	N/A
1		LED indication for 4X/2XA/1XA link types	Off	The link is down
			Solid green	The link is up with no traffic
			Flashing green	The link is up with traffic
			Flashing amber[1]	A problem with the link
2		LED indication for 4X/2XB/1XB link types	Off	The link is down
			Solid green	The link is up with no traffic
			Flashing green	The link is up with traffic
			Flashing amber[1]	A problem with the link
3		LED indication for 4X/1XC link types	Off	The link is down
			Solid green	The link is up with no traffic
			Flashing green	The link is up with traffic
			Flashing amber[1]	A problem with the link
4		LED indication for 4X/1XD link types	Off	The link is down
			Solid green	The link is up with no traffic
			Flashing green	The link is up with traffic
			Flashing amber[1]	A problem with the link

Table 38. Descriptions for the LED splitting options of an MTM 5149-S05 switch (continued)				
State	**State indication LEDs** **[/1 /2 /3 /4]**	**Module LED indication**	**LED color**	**State description**
[1] A flashing amber indicates that an action is required.				

Fan modules LEDs (PSUs side of the enclosure)

Each fan module has its own status LED.

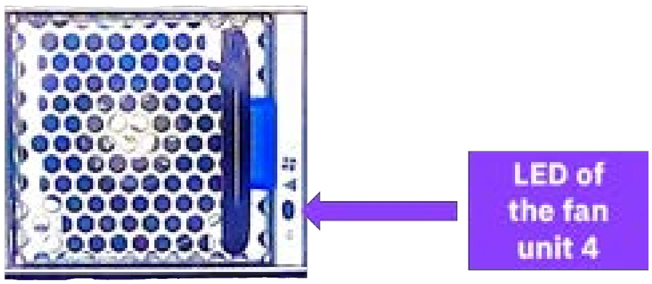


Figure 99. Example of a fan LED (fan unit 4)

Table 39. Descriptions for the fan LEDs of an MTM 5149-S05 switch		
State	**Description**	**Action required**
Solid green	This fan unit is operating as expected.	N/A
Solid amber	This fan unit is missing or not operating properly.	The fan unit should be replaced.

PSU modules LEDs

Each PSU has its own status LED, which reflects the system PSU state. The LED indicators are highlighted in the following figure.

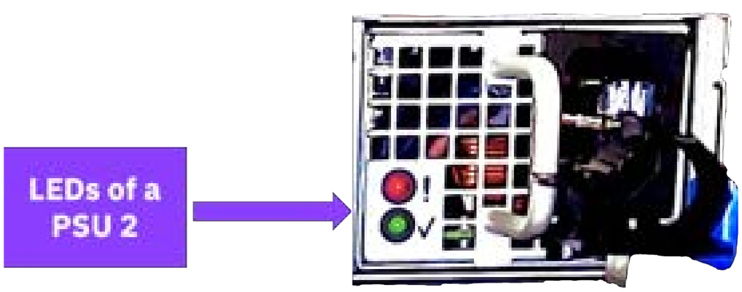


Figure 100. LED indicator for the PSUs of an MTM 5149-S05 switch

Table 40. Descriptions for the PSU LEDs of an MTM 5149-S05 switch		
State	**Description**	**Action required**
Solid green	All PSUs are connected and running normally.	N/A
Flashing green	AC present / 5VSB on (the PSU is off)	Contact IBM support.

Table 40. Descriptions for the PSU LEDs of an MTM 5149-S05 switch (continued)

State	Description	Action required
Solid red or amber	AC cord is unplugged or AC power is lost on one PSU while the other PSU still has AC input power.	• Make sure that both PSU cords are connected and fully seated. • Determine why power was lost in the electrical outlet that the PSU is plugged into.
	PSU failure (voltage, current, temperature, or fan-related issue).	Check the voltage. If OK, contact IBM support.
Flashing red or amber	PSU warning events where the PSU continues to operate.	Contact IBM support.
Off	No AC power to all power supplies. Power might not be applied to the electrical outlets, or one or both of the PSUs might be faulty.	• Make sure that both PSU cords are connected and fully seated. • Determine why power was lost in the electrical outlet that the PSU is plugged into.

MTM 5149-S05 switch installation

This switch is installed by the customer or by IBM, depending on the deployment. Installing a system requires planning to facilitate a smooth installation, initialization, and configuration.

The MTM 5149-S05 switch is typically installed in an IBM Storage Scale System. IBM Storage Scale Systems can be purchased preracked or unracked. SSRs are responsible for installing the MTM 5149-S05 switch in an unracked IBM Storage Scale System.

Planning and preinstallation TDA checklist for an MTM 5149-S05 switch

For IBM Storage Scale System deployments, IBM is responsible for installing the switch, whether done by manufacturing for the racked solutions or by the SSR onsite for the unracked solutions.

Customers must complete a technical and delivery assessment (TDA) that the assigned SSR then reviews with them. In conjunction with the TDA, planning information is available in IBM Docs. The SSR and the customer must also review and complete all of the planning information in IBM Documentation. Before scheduling the trip onsite to perform the installation, the SSR should attend meetings with the customer to perform planning tasks.

TDA checklists ensure that the customer prepares the site and gathers information that is needed to deploy a system. The IBM sales team or business partner helps the customer complete the TDA checklists, with an assigned IBM quality practitioner or technical account manager (TAM) managing the communications.

Note: TDA completion is mandatory for these systems and must be completed before arriving onsite to install the system.

It is the responsibility of the customer, the sales team or business partner, and the IBM quality practitioners or TAM to ensure that the TDA is completed so that the SSR can use information in the TDA checklist to install and initially configure the solution.

Two types of TDA checklists are as follows:

Presale TDA checklist
    Provides a roadmap of questions that are associated with a configuration and proposal for IBM products and solutions. The pre-sale TDA is not relevant to SSRs or to system installations.

Preinstall TDA checklist
: Provides a roadmap of questions that assess site readiness and prepares the environment for the installation. Information in the preinstallation TDA checklist is used in the installation and initial configuration.

TDAs are online forms, but can be exported in an `.xlsx` or `.csv` format, saved, and printed.

TDA checklists are in the IMPACT website (this site requires an IBM ID). The IBM practitioner creates the preinstallation checklist. Customers, their business partner, IBM quality practitioner or TAM, and the assigned SSR can access this site to look at and to work on the TDA.

Package contents for an MTM 5149-S05 switch

The package includes the hardware that is shown in the next figure.

Important: If parts are missing or damaged, do not proceed with the installation. Contact your next level of support.

An MTM 5149-S05 switch package includes the following components.

- One switch (includes 2 PSUs and 6 fans)
- Rail kit
  - One air duct (bottom and top parts)
  - Two rack mounting rails
  - Two rack mounting blades
  - Eight M6 standard cage nuts
  - Eight M6 standard pan-head Phillips screws
  - Six flat-head Phillips screws
- (Optional) AC power cords

Installing the MTM 5149-S05 switch

Complete this procedure to install the MTM 5149-S05 switch.

Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be installed. Installation locations at 32U and higher require two people to lift the switch. For installation locations less than 32U, one person can safely lift the switch.

Remember: Do not start the installation procedure before all the components are unpacked and placed on an antistatic flat surface.

Warning: During the installation, for safety and reliability, use only the accessories and screws that are provided with your MTM 5149-S05 switch. By using other accessories and screws, the switch might be damaged. The warranty does not cover any damages that result of the use of unapproved accessories.

Two people are required to complete steps 5 and 6 of this procedure.

1. Attach the bottom part of the air duct to the switch enclosure.
   a) Place the bottom of the air duct near the PSUs side of the switch enclosure.

      Make sure that the open side of this part is up.

      And that the two holes for the securing screws are near the PSUs side of the switch enclosure.

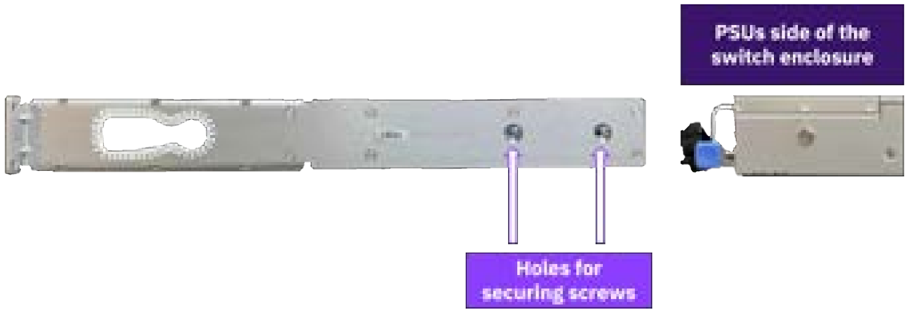


Figure 101. Aligning the bottom of the air duct with the PSUs side of the switch enclosure

b) By using the PSU handles, gently lift the enclosure about 2.5 inches and slide it into the bottom of the air duct until the four enclosure pins (two per side) are aligned with the four pin slots (two per side) of the bottom of the air duct.

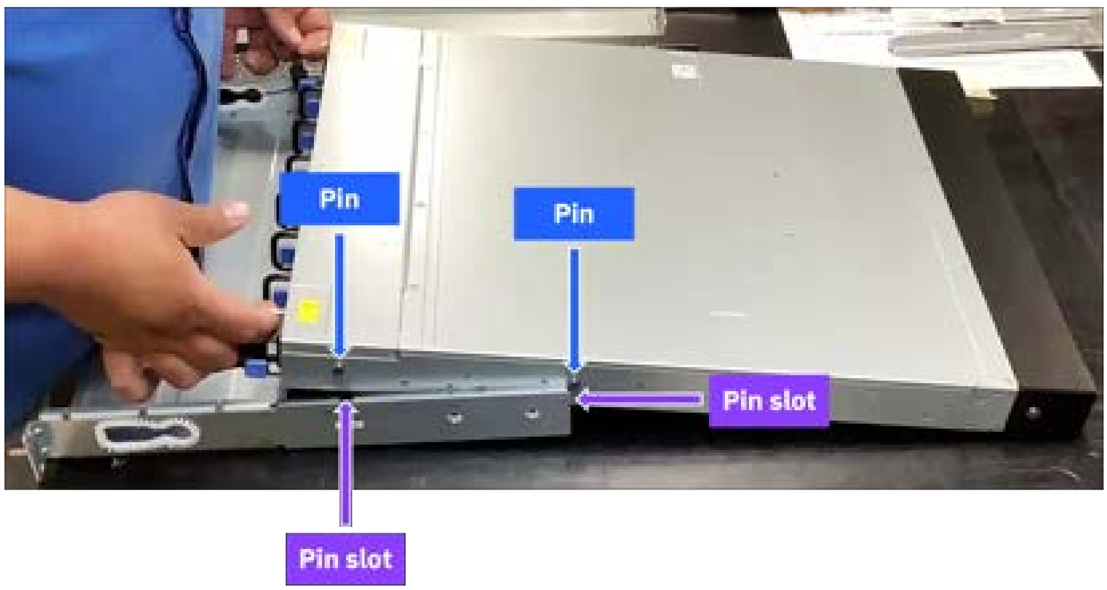


Figure 102. Sliding the switch enclosure into the bottom of the air duct

c) Gently slide the enclosure pins on their corresponding slots.

d) For each side, use two flat-head Phillips screws to secure the bottom of the air duct.

2. Attach the rack mounting rails to the enclosure.

   a) From the switches side of the enclosure, face the flat side of a rack mounting rail with one side of the enclosure.

   b) Align the enclosure pins with the key holes in the rack mounting rail.

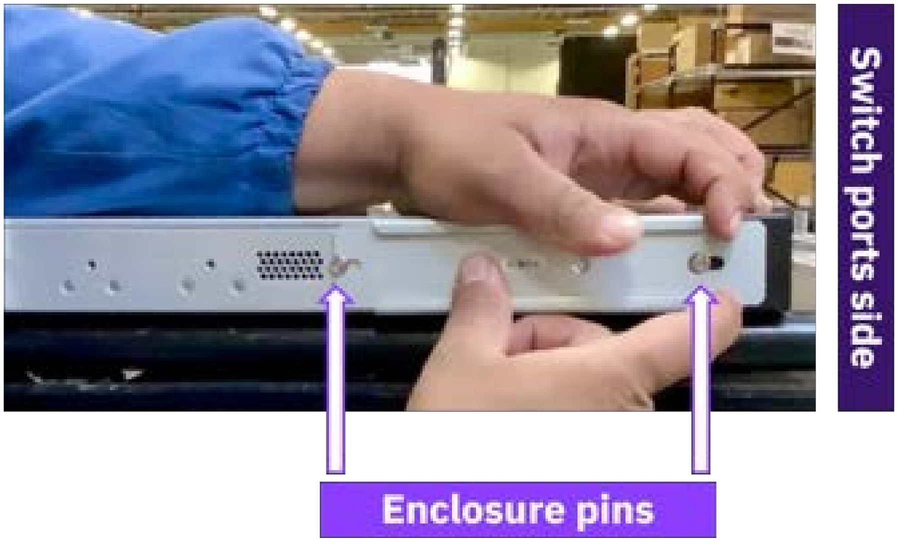


Figure 103. Aligning the enclosure pins with the key holes of the rack mounting rail

c) Gently push the rack mounting rail toward the PSUs side of the enclosure, passing the enclosure pins through the slider key holes until the enclosure pins are locked.

d) Use one flat-head Phillips screw to secure the rack mounting rail.

Figure 104. Securing the rack mounting rail

e) Repeat substeps a through d for the other side of the enclosure.

3. Insert the eight cage nuts in the rack posts, in the position where the switch enclosure is meant to be installed in the rack.

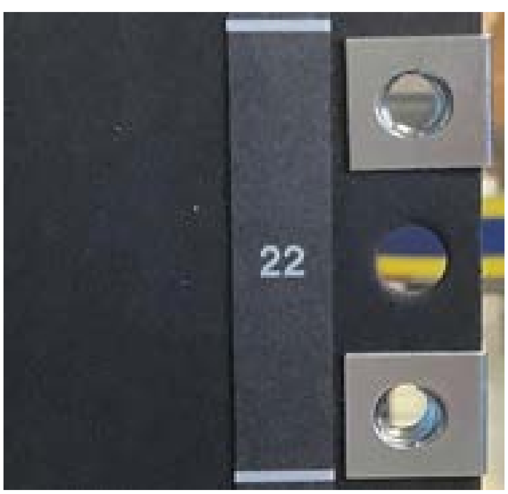


Figure 105. Inserting the cage nuts (example of two cage nuts that are inserted in position in a left rack post)

4. Loosely secure the rack mounting blades on one side of the rack.

   Important: Do not tighten the screws during this substep. The rack mounting blades must have some slack to be inserted in the rack mounting rails in a following step.

   a) Place one rack mounting blade on top of two of the previously inserted cage nuts and use two M6 standard pan-head Phillips screws to loosely keep rack mounting blade in position.
   b) On the same side of the rack but on the other rack post, repeat the previous substep for the other rack mounting blade.

Figure 106. Placing a rack mounting blade on a right rack post

5. Place the enclosure switch in its position in the rack.
   a) Take the enclosure switch to the contrary side of the rack where you installed the rack mounting blades.
   b) Face the switch ports side toward the rack mounting blades and slightly tilt the switch enclosure to make it fit in the rack.

Figure 107. Tilting the switch enclosure to install it

   c) With the help of a second person who is at the rack side where the rack mounting blades were previously placed in position, insert the rack mounting blades in the rack mounting rails.

Figure 108. Inserting the rack mounting blades in the rack mounting rails

6. Secure the switch enclosure to the rack.

   While the second person holds in place the enclosure switch, complete the next substeps.

   a) Tighten the four M6 standard pan-head Phillips screws that you previously used to loosely keep in position the rack mounting blades.
   b) Go to the contrary side of the rack and use four M6 standard pan-head Phillips screws to secure the bottom of the air duct.

Figure 109. Securing the rack mounting rail

7. Pass the PSUs cables through the cables grommet.

   Warning: Do not connect the PSUs cables to the external AC power source yet.

Figure 110. Displaying one of the cables grommets of an air duct

8. Install the top of the air duct.
   a) On bot sides, align the edge tabs of the air duct top with the edges of the air duct bottom.

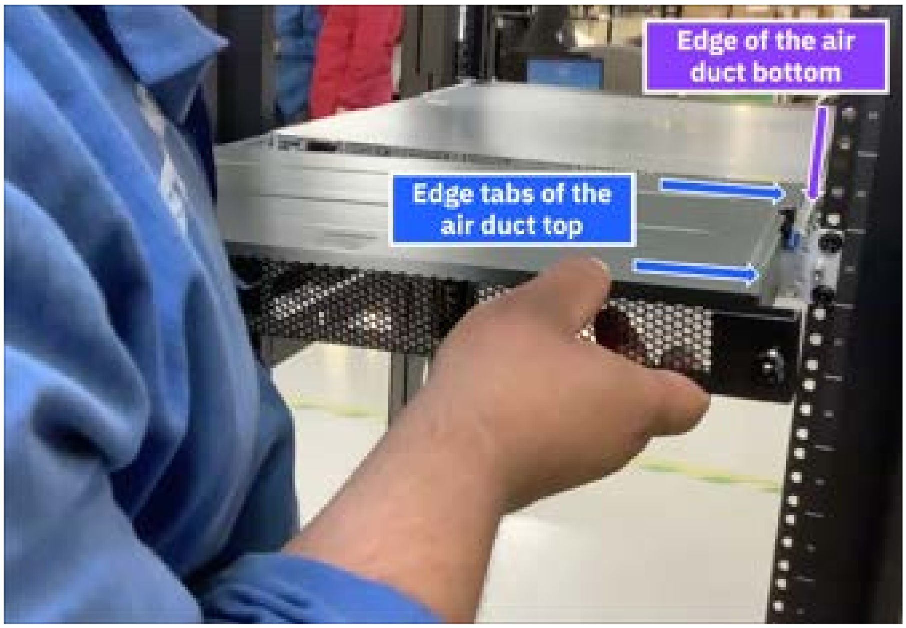


Figure 111. Aligning the right edge tabs of the air duct top with the right edge of the air duct bottom

   b) Insert the edges of the air duct bottom on the edge tabs of the air duct top.

Figure 112. Sliding the air duct top into the air duct bottom

   c) Secure the air duct top by tightening its two thumbscrews.

i) While you use one hand to evenly apply a light pressure on the air duct top and keep it in place, use your other hand to tighten one of the thumbscrews.

ii) Repeat for the other thumbscrew.

Figure 113. Securing the air duct top

9. Connect an external AC power source to the PSUs.
10. Make the network connections.
11. Make the management connections.

MTM 5149-S05 switch service

Service procedures and a list of replaceable parts are provided for the MTM 5149-S05 switch service.

Important: Before the service of any replaceable part, you must consult “Replaceable parts” on page 111 and “Replacing a customer replaceable unit” on page 111 to determine the replacement designation and avoid warranty infringement.

Key replaceable parts for MTM 5149-S05 switch

The following table lists key replaceable parts for the switch. Because part numbers can change over time, in a repair situation, verify that the correct CRU or FRU is obtained.

Part	FRU part number	CRU or FRU	Concurrent replace
MTM 5149-S05 switch	03GX235 03MT564	FRU	No
PSU	01PG798	FRU	Yes
Fan module	01PG799	FRU	Yes
Rail mounting kit (includes air duct)	01PG843	FRU	No
200 G QSFP56 Ethernet optical transceiver	03NK246	CRU Tier 1	No
Power cord 4.3 m (14 ft) (drawer to IBM PDU, 250 V / 10 A)	39M5510	CRU Tier 1	Yes

Part	FRU part number	CRU or FRU	Concurrent replace
Power cord 4.3 m (14 ft) (drawer to IBM PDU, 250 V / 10 A) For India only	01KV682	CRU Tier 1	Yes
Power cable 2.8 m (9.2 ft) (drawer to IBM PDU, 250 V / 10 A)	39M5392	CRU Tier 1	Yes
Power cable 2.8 m (9.2 ft) (drawer to IBM PDU, 250 V / 10 A) For India only	01PP689	CRU Tier 1	Yes

Replaceable parts

Each replaceable part is assigned one of three designations. Most of the parts for this switch are classified as Tier 1 CRU. None of the parts for this switch are Tier 2 CRU.

For more information, see the following table.

Replacement designation	Description
CRU -Tier 1	Replacing a Tier 1 CRU is a customer responsibility, but dependent on if they purchased a service upgrade: • Under the standard warranty – If an IBM representative installs a Tier 1 CRU at the customer's request, the customer is charged for the installation. – If an IBM representative installs a Tier 1 CRU at IBM Support's request, the customer is not charged for the installation. • Under an IBM Storage Expert Care service upgrade: The customer can request that IBM install the CRU at no additional cost.
CRU -Tier 2	Replacing a Tier 2 CRU can be completed by the customer or can be installed at the customer's request by an IBM representative. If a Tier 2 CRU is replaced by an IBM representative: • If the product is under warranty: There is no charge for the part or for the installation if the product. • If the product is no longer under warranty: The customer is charged for the part and the installation.
FRU	Replacing a FRU must be performed by an IBM service personnel or an IBM authorized warranty service provider.

Replacing a customer replaceable unit

Use this procedure for servicing a customer replaceable unit (CRU).

Before you start a replacement, the customer must complete the following tasks:

- If the replacement procedure might put data at risk, the customer must back up the system or logical partition (including operating systems, licensed programs, and data).
- Review the replacement procedure for the part.
- Obtain the required tools for the replacement procedure.

- Inspect the replacement part. If any of the parts are incorrect, missing, or visibly damaged, they should contact the provider of the part or IBM Support.

The replacement best practices are as follows:

- For all replacements performed on an operational system, remove only one module from the enclosure at a time, unless directed otherwise by a replacement procedure.
- To reduce the amount of time required for the replacement, remove the failed module only when the replacement module is available, unpackaged, and ready for installation.
- Follow all recommended procedures for handling devices that are sensitive to electrostatic discharge (ESD).

1. Identify the system that contains the part to replace.
2. Attach the ESD wrist strap. The ESD wrist strap must connect to an unpainted metal surface until the service procedure is completed.
3. Locate the CRU part to be serviced.
4. Remove and replace the CRU part.
5. Verify that the installed part is working as expected.

Replacing an MTM 5149-S05 switch

Refer to the service procedure to remove and replace an MTM 5149-S05 switch enclosure.

If possible, the customer should back up the switch configuration and its firmware before the replacement procedure.

For steps 6 and 7 of this procedure, two people are needed.

The high-level steps to replace an MTM 5149-S05 switch are as follows.

1. Locate the faulty switch.
2. Label all data and power cables that connect to the faulty MTM 5149-S05 switch.
3. Remove the top of the air duct.
4. Disconnect the power supply cables from their PDU connections.
5. Disconnect all data cables from the faulty MTM 5149-S05 switch.
6. While a second person holds the MTM 5149-S05 switch enclosure, remove the four securing cage or clip nuts.
7. Pull the faulty MTM 5149-S05 switch enclosure out of the rack and place it on a flat antistatic surface.
8. Install the new MTM 5149-S05 switch.

Important:

- Before you remove the switch from the rack, make sure that the switch is powered down, with all network and power cables disconnected.
- Two people are needed to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch is meant to be replaced. Replacement locations at 32U and higher require two people to lift the switch. For replacement locations lower than 32U, one person can safely lift the switch.

To perform this service, you need access to the PSUs side of the enclosure, where the top of the air duct is installed.

1. Locate the faulty switch.

   The service ticket must include location information. Confirm the switch location by looking for fault LEDs on the switch.
2. Label all data and power cables that connect to the faulty MTM 5149-S05 switch.

   Take note of which cables are plugged into the specific ports before you remove any cable. This record is needed for the installation of the new MTM 5149-S05 switch.

3. Remove the top of the air duct.
   a) While you use one hand to evenly apply a light pressure on the air duct top to prevent it from falling, use your other hand to loosen the thumbscrews of the air duct top.

   Figure 114. Loosening the thumbscrews of the air duct top
   b) Pull the air duct top out of the air duct bottom.
   c) Place the air duct top on an antistatic flat surface.
4. Disconnect the power supply cables from their PDU connections.
5. Disconnect all data cables from the faulty MTM 5149-S05 switch.
6. While a second person holds the MTM 5149-S05 switch enclosure, remove the eight screws and securing cage nuts.

   Figure 115. Removing the securing hardware
7. Pull the faulty MTM 5149-S05 switch enclosure out of the rack and place it on a flat antistatic surface.
8. Install the new MTM 5149-S05 switch.

   To install the new MTM 5149-S05 switch, follow the steps from "Installing the MTM 5149-S05 switch" on page 124.

Replacing the PSU of an MTM 5149-S05 switch

The MTM 5149-S05 switch has two PSUs in a redundant configuration. Only one healthy PSU is required for the switch to function.

Before you remove a PSU, ensure that the other PSU is connected and shows a solid green LED.

Figure 116. Locating the release latch and handle of a PSU

1. Remove the power cord from the faulty PSU.
2. Remove the faulty PSU.
   a) Press the release latch.
   b) While you still keep pressing the release latch, pull the handle toward you.

   As the fan unseats, its status LED turns off.

   Figure 117. Removing a PSU
3. Install the PSU.

   CAUTION: Do not insert a PSU whose power cables are connected. Disconnect power cords from outlets before you start the installation.

   a) Insert the FRU PSU by sliding it into the slot until you feel a slight resistance.

      When the latch snaps into place, the installation is correct.
   b) Insert the power cord into the FRU PSU connector.
   c) Insert the other end of the power cord into an outlet of the correct voltage.
   d) Wait for the FRU PSU LED to be solid green.

      If the LED does not turn solid green, restart the procedure, and make sure that the PSU is correctly seated.

Replacing a fan module of an MTM 5149-S05 switch

The MTM 5149-S05 switch can work with one failed fan module. Operating with more than one failed fan module is not supported.

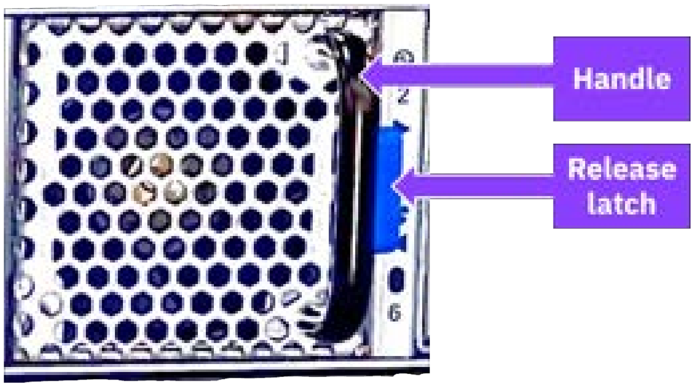


Figure 118. Locating the release latch and handle of a fan

1. Remove the faulty fan module.
   a) Press the release latch.
   b) While you still keep pressing the release latch, pull the handle toward you.

   As the fan unseats, its status LED turns off.

   Figure 119. Removing a fan module
2. Insert a fan module.
   a) Insert the fan module by sliding it into the opening until slight resistance is felt. Continue pressing the fan module until it seats completely.
   b) Wait for the fan module LED to illuminate solid green. If the fan module LED does not turn solid green, restart the procedure and ensure that the fan module is correctly seated.

The MTM 5149-S54 switch

This Edgecore switch, IBM MTM 5149-S54 switch, is a 1U, Gigabit Ethernet L2, 54-ports switch that offers 108 Gbps of switching capacity and 61.6 Mbps in forwarding rate. The MTM 5149-S54 switch can be installed in the same rack as the IBM Storage Scale System solution, in a similar way as other supported high-speed data switches.

Note: If Licensee purchases and incorporates IBM MTM 5149-S54 switch ("Switch") as part of your IBM Storage Scale System, the Switch may not comply with all of the requirements included in IBM's Security and Privacy by Design practices: https://www.ibm.com/support/pages/ibm-security-and-privacy-design.

Overview of the MTM 5149-S54 switch

Use the MTM 5149-S54 switch as a management switch in data center networking, enterprise access networking, or campus access networking.

This open network switch includes the following components:

- 48 10/100/1000 BASE-T ports
- 6 1G/10G SFP+ uplink ports
- 2 hot-swappable AC PSUs
- 2+1 redundant, fixed fans (front-to-back airflow or back-to-front airflow)
- Open Network Install Environment (ONIE) software that supports the installation of an open source or commercial compatible network operating system (NOS).

The following figures show the PSUs side and the ports side of an MTM 5149-S54 switch.

Figure 120. PSUs view of the MTM 5149-S54 switch

Figure 121. Ports views of the MTM 5149-S54 switch

For more information about the MTM 5149-S54 switch, see EPS121 in Edgecore documentation.

The following table lists the IBM MTM and the Edgecore system model.

Table 41. IBM IBM and Edgecore system model

IBM MTM	Edgecore system model
5149-S54	EPS121 AS4625-54T

Key features and components of the MTM 5149-S54 switch

The key benefits and components of this entry-level open networking switch include features and components like 108 Gbps of switching capacity 61.6 Gbps in forwarding rate, 54 switch ports and 3 management ports, options to enable UEFI Secure Boot.

In the next table, consult the performance characteristics of an MTM 5149-S54 switch.

Feature	Performance
Switching capacity	108 Gbps
Forwarding rate	61.6 Gbps
MAC addresses	64 000 minimum 112 000 maximum
VLAN IDs	4000

Feature	Performance
VLAN translation	4000 ingress 2000 egress
Jumbo frames	Up to 12 288 bytes
Packet buffer size	4 MB integrated, packet buffer memory
LAG (802.3ad)	128 LAG groups 256 members in total 8 members maximum per group
VRF	1000
L3 hosts	IPv4 32K or IPv6 16 000
L3 multicast groups	4000 IPMC groups
ECMP	256 groups 1024 members in total 64 members maximum per group

Hardware components:

- 1GbE connections to server node and storage node management ports in rack, with uplinks to management network aggregation switches.
- 54 switch ports:
  - 48 10/100/1000 BASE-T RJ-45 ports
  - 6 SFP + uplink ports supporting 1GE/10GE
- 3 management ports:
  - 1 RJ-45 serial console
  - 1 RJ-45 100/1000BASE-T management port
  - 1 micro-USB storage port
- Intel Atom COM Express module with C3508 4-Core 1.6 GHz x86 processor.
- 130 Gbps, Broadcom BCM56277 Trident III switch silicon
- Infineon OPTIGA TPM SLB 9670XQ2.0
- Full line-rate L2/L3 forwarding and switching
- Hot-swappable, load-sharing, redundant AC PSUs
- 2+1 redundant fixed fans
- 1 8GB SO-DIMM
- 32GB MLC M.2 SSD

Software features:

- Options to enable UEFI Secure Boot.
- Preloaded ONIE for automated loading of compatible open source or commercial NOS offerings.
- SPI flash with capacity of 16 Mb x 2.

Views of the MTM 5149-S54 switch

The switch ports are accessible from one of the sides of the enclosure; the fixed fans and the PSUs are placed on the contrary side of the enclosure.

Switch ports view

The following figure shows the 48 BASE-T ports, the 6 SFP +uplink ports, and the 3 management ports of an MTM 5149-S54 switch.

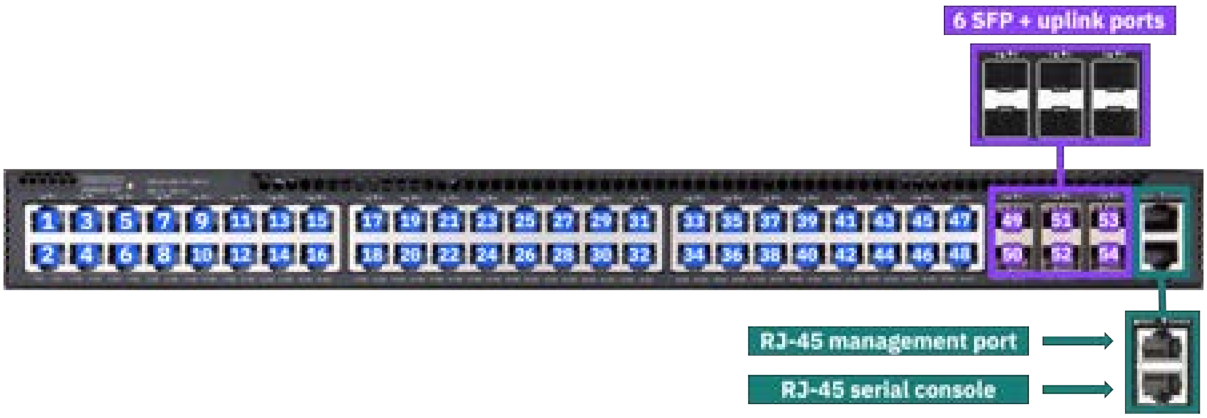


Figure 122. Ports of an MTM 5149-S54 switch on the rear of the enclosure

Port numbers are printed in pairs and placed above the top row of switches, as highlighted in the following figure.

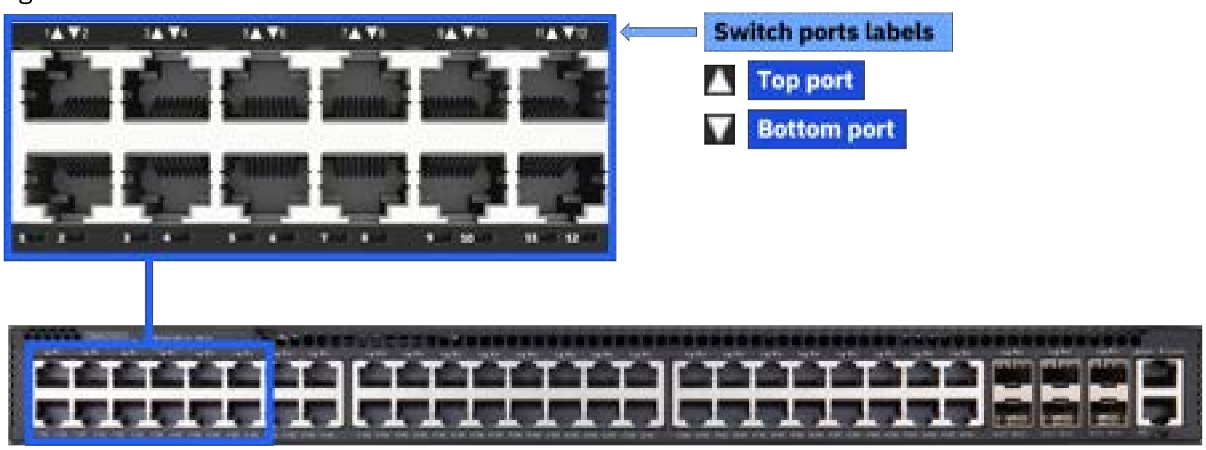


Figure 123. Switch port numbers of an MTM 5149-S54 switch

PSUs and fixed fans view

The following figure highlights the placement of the 3 fixed fans and the 2 PSUs of an MTM 5149-S54 switch.

Figure 124. PSUs and fixed fans of an MTM 5149-S54 switch

Labels of the MTM 5149-S54 switch

The details of the MTM 5149-S54 switch labels are provided in this section.

PSUs label

The PSUs label is at the bottom of the PSUs.

Figure 125. PSUs label

Switch labels

On the bottom of the switch, you can find the regulatory label, and the MTM label, as shown in the following figure. These labels contain the part number, serial number, and GUID of the MTM 5149-S54 switch.

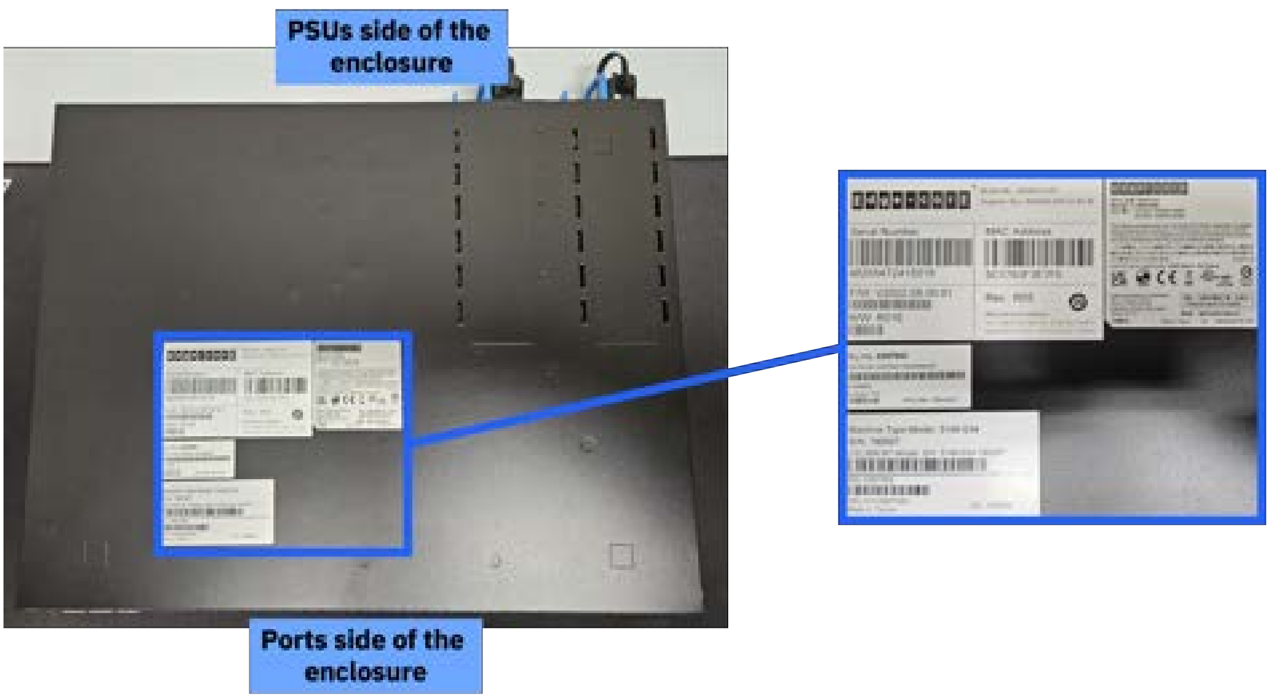


Figure 126. Labels on the bottom of the switch enclosure

Specifications of the MTM 5149-S54 switch

The following specifications provide brief information about the MTM 5149-S54 switch.

System dimensions and weight

System weight and dimensions are listed in the following table.

Height (1U rack standard)	Width (19" rack standard)	Depth	Weight
4.4 cm (1.73 in)	44.0 cm (17.32 in)	350.35 cm (13.79 in)	5.8 kg (12.77 lb)

Note: Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be installed. Installation locations at 32U and higher require two people to lift the switch. For installation locations less than 32U, one person can safely lift the switch.

Air temperature

Air flows from the power side of the switch to the port side (power-to-port air flow configuration). Temperature requirements for the switch are listed in the following table.

Air flow direction	Operating temperature and humidity	Storage
Power-to-connector (forward)	0°C to 45°C (32°F to 113°F) 5% to 95% non-condensing humidity	-40°C to 70°C (-40°F to 158°F)

Important: Regardless of the system specifications about absolute maximum temperature, the nonoperating environmental conditions of the system solid-state devices must adhere to the *JEDEC JESD218 Enterprise-class SSD* requirements for data retention endurance. These requirements must be met particularly for the system storage, transit, or shipping maximum temperatures. To best maintain the data retention of the system solid-state devices, keep an *optimal* condition for nonoperating temperatures by making sure that the system does not remain for longer than 3 months at temperatures that exceed 40°C.

Power

The switch has two redundant, load-sharing, hot-swappable 150W AC PSUs that comply with 80 PLUS Platinum. The maximum power consumption per PSU is 150W (at 25°C).

Power requirements for each PSU are listed in the following table.

Input	Frequency
100-120 VAC	50/60 Hz 4A maximum
200-240 VAC	50/60 Hz 2A maximum

Service indicators: LEDs of the MTM 5149-S54 switch

Service indicators for the switch are implemented by using LEDs to provide status information.

LEDs provide a visual method to identify, clarify, and resolve system issues. The system-level LEDs indicate the overall switch health and status. LEDs on individual modules, such as the PSUs, indicate the health and status of the specific module.

System LEDs (rear of the enclosure)

The following figure shows system state LEDs, which are visible on the port side of an MTM 5149-S54 switch.

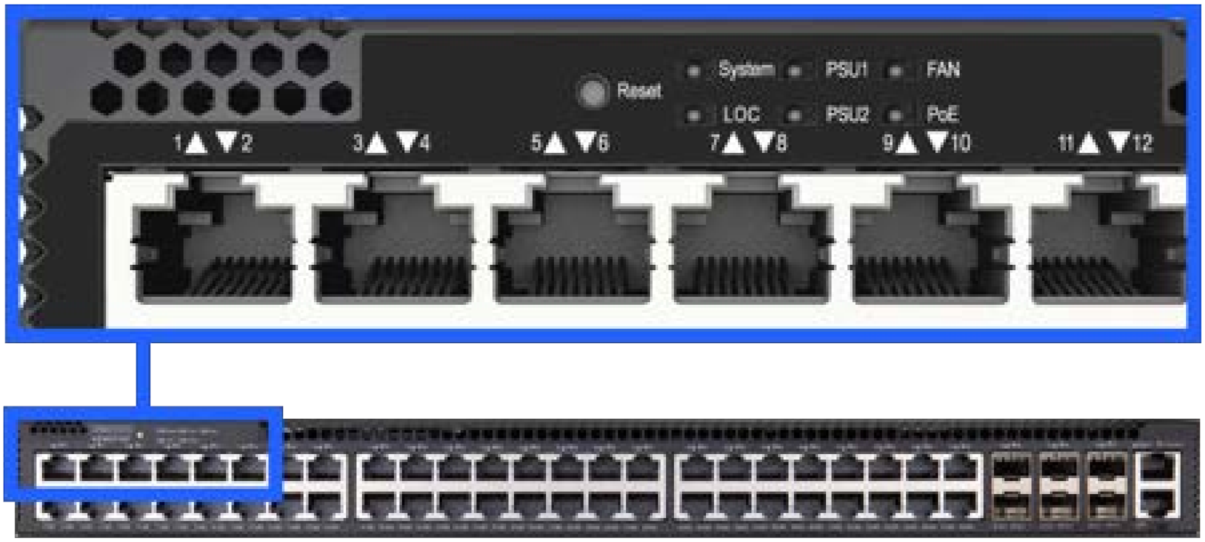
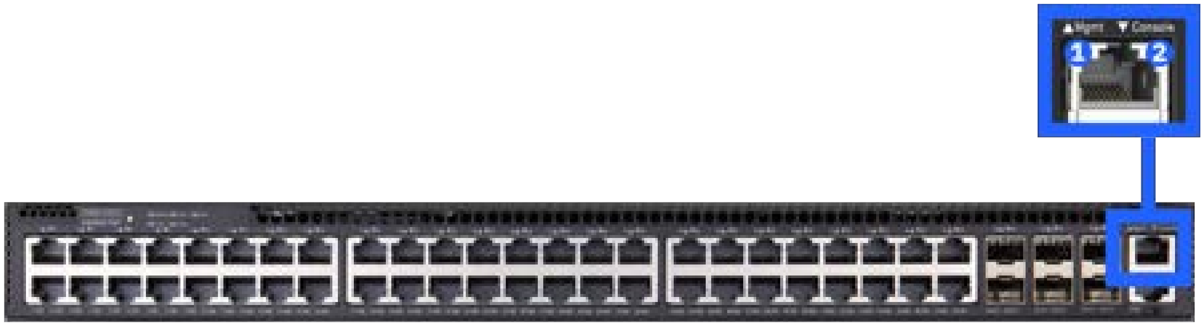


Figure 127. LED indicators for system, LOC, PSUs, fans, and PoE of an MTM 5149-S54 switch

The following figures and tables describes the LEDs on the port side (front of the enclosure).

State LED	Color	Description
System	Solid green	Okay
	Solid amber	Fault detected
Location (LOC)	Intermittent amber	Switch locator.
PSUs	Solid green	Okay
	Solid amber	Fault detected
Fans	Solid green	Okay
	Solid amber	Fault detected
Power over Ethernet (PoE)	Solid green	Working within PoE budget.
	Solid amber	Working near PoE budget limit.

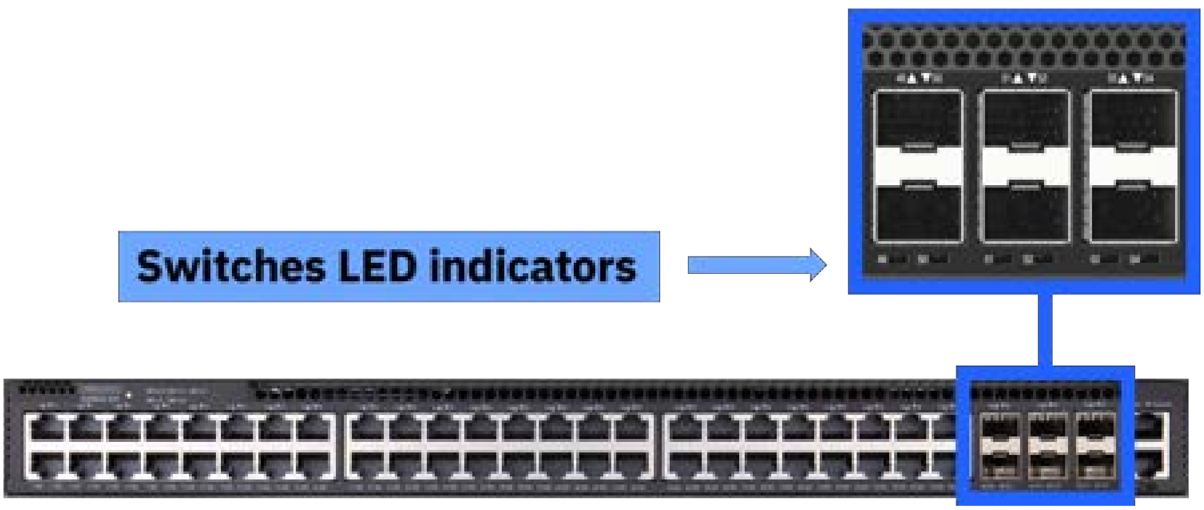
The management port has two LEDs, one on each side. The left LED (see 1 in the next figure) indicates that a link is established, and the right LED (see 2 in the next figure) indicates that the port is active.



1 (left)	Link
2 (right)	Activity

Figure 128. LED indicators for the management port of an MTM 5149-S54 switch

The LED of an SFP + uplink port uses green to indicate a 10G link and amber to indicate a 1G link.



Green	10G
Amber	1G

Figure 129. LED indicators for the 6 SFP + uplink ports of an MTM 5149-S54 switch

MTM 5149-S54 switch installation

This switch is installed by the customer or by IBM, depending on the deployment. Installing a system requires planning to facilitate a smooth installation, initialization, and configuration.

The MTM 5149-S54 switch is typically installed in an IBM Storage Scale System. IBM Storage Scale Systems can be purchased preracked or unracked. SSRs are responsible for installing the MTM 5149-S54 switch in an unracked IBM Storage Scale System.

Planning and preinstallation TDA checklist for an MTM 5149-S54 switch

For IBM Storage Scale System deployments, IBM is responsible for installing the switch, whether done by manufacturing for the racked solutions or by the SSR onsite for the unracked solutions.

Customers must complete a technical and delivery assessment (TDA) that the assigned SSR then reviews with them. In conjunction with the TDA, planning information is available in IBM Docs. The SSR and the customer must also review and complete all of the planning information in IBM Documentation. Before scheduling the trip onsite to perform the installation, the SSR should attend meetings with the customer to perform planning tasks.

TDA checklists ensure that the customer prepares the site and gathers information that is needed to deploy a system. The IBM sales team or business partner helps the customer complete the TDA checklists, with an assigned IBM quality practitioner or technical account manager (TAM) managing the communications.

Note: TDA completion is mandatory for these systems and must be completed before arriving onsite to install the system.

It is the responsibility of the customer, the sales team or business partner, and the IBM quality practitioners or TAM to ensure that the TDA is completed so that the SSR can use information in the TDA checklist to install and initially configure the solution.

Two types of TDA checklists are as follows:

Presale TDA checklist
Provides a roadmap of questions associated with a configuration and proposal for IBM products and solutions. The pre-sale TDA is not relevant to SSRs or to system installations.

Preinstall TDA checklist
Provides a roadmap of questions that assess site readiness and prepares the environment for the installation. Information in the pre-install TDA checklist is used the installation and initial configuration.

TDAs are online forms, but can be exported in an `.xlsx` or `.csv` format, saved, and printed.

TDA checklists are in the IMPACT website (this site requires an IBM ID). The IBM practitioner creates the pre-install checklist. Customers, their business partner, IBM quality practitioner or TAM, and the assigned SSR can access this site to look at and to work on the TDA.

Package contents for an MTM 5149-S54 switch

The package includes the hardware that is shown in the next figure.

Important: If parts are missing or damaged, do not proceed with the installation. Contact your next level of support.

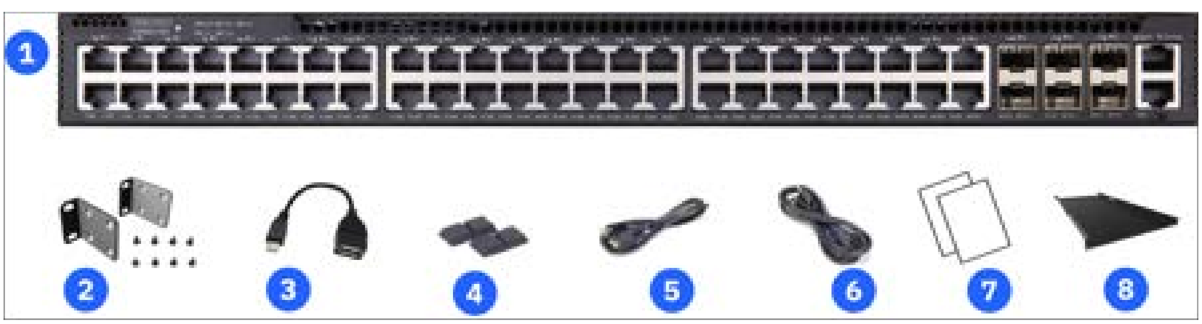


Figure 130. Contents of the package

An MTM 5149-S54 switch package includes the following components.

- One switch (includes 2 PSUs) –see 1 in the previous figure.
- Rack mounting kit (2 brackets, vented cover plate, and 8 M4x8 screws) –see 2 in the previous figure.
- USB micro-b to USB-A cable –see 3 in the previous figure.
- (Dispensable) Four adhesive rubber feet –see 4 in the previous figure.
- Console cable (RJ-45 to DE-9) –see 5 in the previous figure.

- (Optional) AC power cord –see 6 in the previous figure.
- Documentation (*Edgecore Quick Start Guide* and *Safety and Regulation Information*) –see 7 in the previous figure.
- Airflow control shelf (includes 2 solid cover plates) –see 8 in the previous figure.
- Mounting hardware:
  - 12J5289 - M6 hexagonal-head black screws
  - 12J5288 - M6 cage nuts
  - 46C6380 - M5 hexagonal-head black screws
  - 81Y2820 - M5 cage nuts
  - 00N8709 - C clip

Installing the MTM 5149-S54 switch

Complete this procedure to install the MTM 5149-S54 switch.

Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch is meant to be installed. Installation locations at 32U and higher require two people to lift the switch. For installation locations less than 32U, one person can safely lift the switch.

Important:

- For steps 1 and 3 of this procedure, two people are required.
- The airflow control shelf is meant to be installed beneath the switch.
- If your computer does not have a DE-9 serial port, you need a USB-to-serial adapter cable (part number 03KX086) to connect to the MTM 5149-S54 switch and complete step 6.

Clearance requirements: To handle the hardware for installation, you need a clearance of 6U rack space (see the next figure for reference). After the switch and its airflow control shelf are installed, they cover 2U of rack space in total.

Figure 131. 6U rack space of clearance that are required for installation

The next figures show a properly installed switch (rack position 41) and its airflow control shelf (rack position 40).

Figure 132. View from the switch ports side: Example of a properly installed MTM 5149-N64 switch and airflow control shelf

Figure 133. View from the switch PSUs side: Example of a properly installed MTM 5149-N64 switch and airflow control shelf

Warning: During the installation, for safety and reliability, use only the accessories and screws that are provided with your MTM 5149-S54 switch. By using other accessories and screws, the switch might be damaged. The warranty does not cover any damages that result of the use of unapproved accessories.

Note: The MTM 5149-N64 switch includes the software installer of Open Network Install Environment (ONIE) preinstalled, but no software image. For information about compatible software, see the Edgecore documentation.

1. Beneath the predetermined rack position for the switch, install the airflow control shelf.

   Remember: To complete this step, two people are required. One person holds and handles the hardware from the rack front and the other person does it from the rack rear.

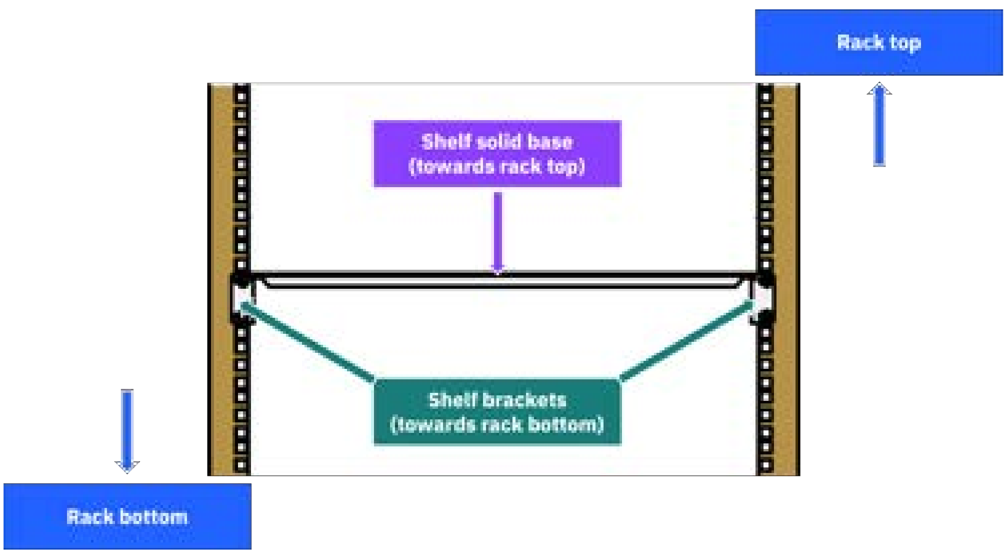


   Figure 134. Orienting the airflow control shelf

   a) Orient the airflow control shelf with its solid base toward the rack top.

   b) In the rack position that is beneath the predetermined 1U rack position for the switch, place four M6 cage nuts to mark the 1U rack position for the airflow control shelf.

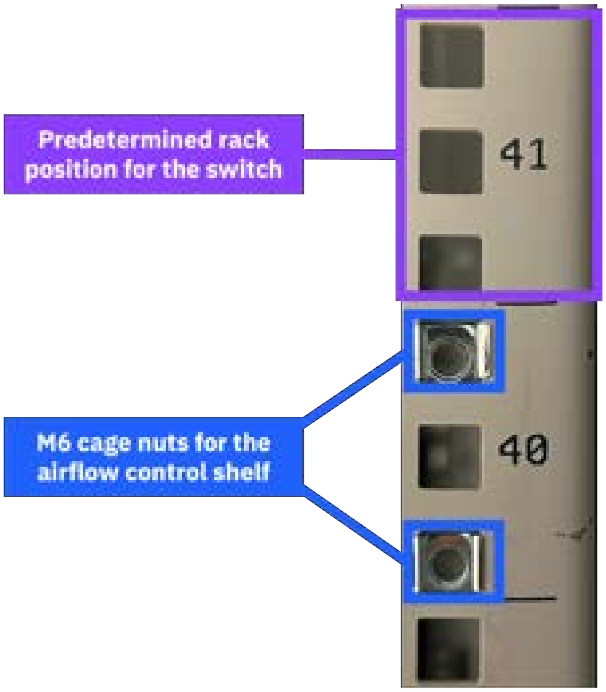


Figure 135. Marking the rack position for the airflow control shelf (example)

c) Tilt the airflow control shelf to insert it in the rack, and then hold it in its rack position.

d) Align the holes of the shelf brackets with the M6 cage nuts in the rack post holes.

e) On one side of the rack enclosure, provisionally secure the airflow control shelf by using two M6 screws.

f) On the opposite side of the rack enclosure, align the holes of the solid cover plate with the holes of the four M6 cage nuts.

g) While a person holds the airflow control shelf, use four M6 screws to secure the solid cover plate and the airflow control shelf to the rack posts.

h) Back on the opposite side of the rack enclosure, while a person holds the airflow control shelf, remove the two M6 screws that were provisionally securing that side of the airflow control shelf.

i) Repeat step 1g.

2. Attach the rack brackets to the switch.

   a) Take one of the front brackets and orient its front (rack securing side) toward the switch ports side of the enclosure, while aligning the enclosure-facing side of the bracket with one of the sides of the switch enclosure.

      The two holes on the front of the ports side of the front bracket are meant to secure the switch to the rack posts.

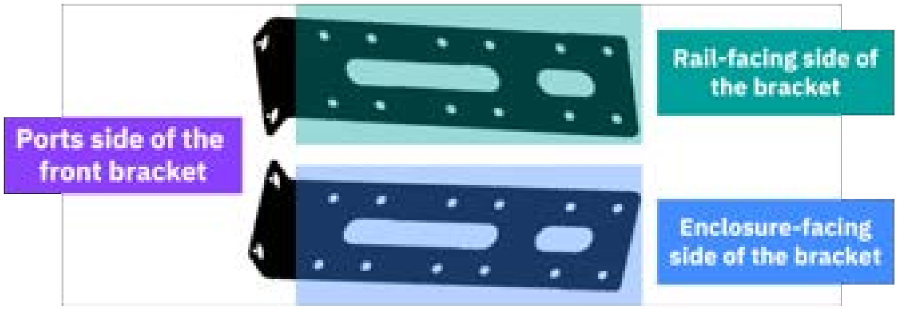


Figure 136. Aligning the bracket sides for installation

b) Align the first set of four holes (front to rear of the front bracket) of the front bracket with the first set of four holes (ports side to PSUs side of the enclosure) of the switch.

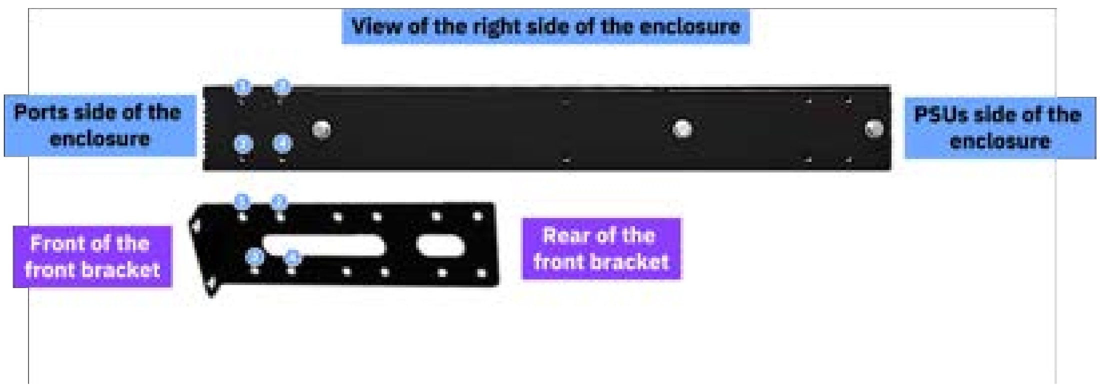


Figure 137. Aligning the front bracket and the enclosure for installation

c) Use four M4x8 screws to secure the front bracket to the switch.

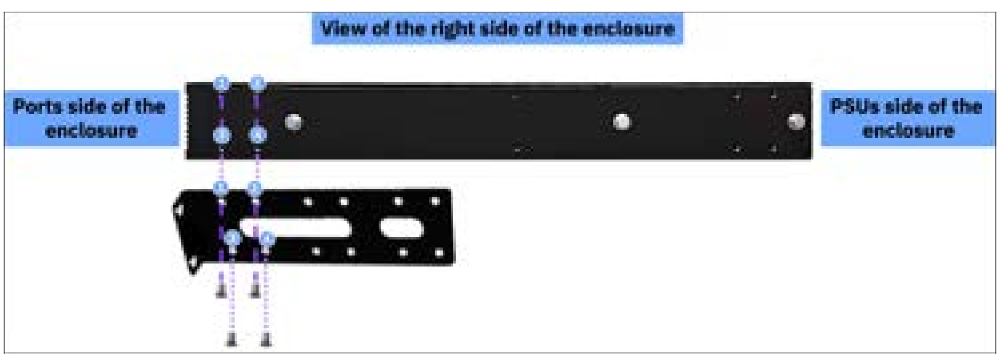


Figure 138. Attaching the rack brackets to the MTM 5149-S54 switch enclosure

d) Repeat steps 2a through 2c for the other bracket and the other side of the switch.

3. Secure the switch to the rack enclosure.

   Remember: To complete this step, two people are required. One person holds and handles the hardware from the rack front and the other person does it from the rack rear.

   a) In the predetermined rack position the 1U switch, place four M5 cage nuts to mark the position for the switch.

b) Insert the switch and hold it in that position.

c) Align the holes of the rack brackets with the M5 cage nuts in the rack post holes.

d) Use four M5 cage nuts and four M5 screws to secure the switch in its position.

e) In the opposite side of the rack enclosure but in the same rack position where you secured the switch, use four M5 cage nuts and four M5 screws to secure the vented cover plate of the switch to the rack posts.

Figure 139. Securing the MTM 5149-S54 switch to the rack

4. Connect to power.

   If the PSUs are not preinstalled, install one or two. Then, connect an external AC power source to the PSUs.

Figure 140. Connecting an external AC power source to the PSUs of an MTM 5149-S54 switch

5. Make the network connections.

   a) For the RJ-45 ports, connect Category 5, 5e, or better twisted-pair cables.

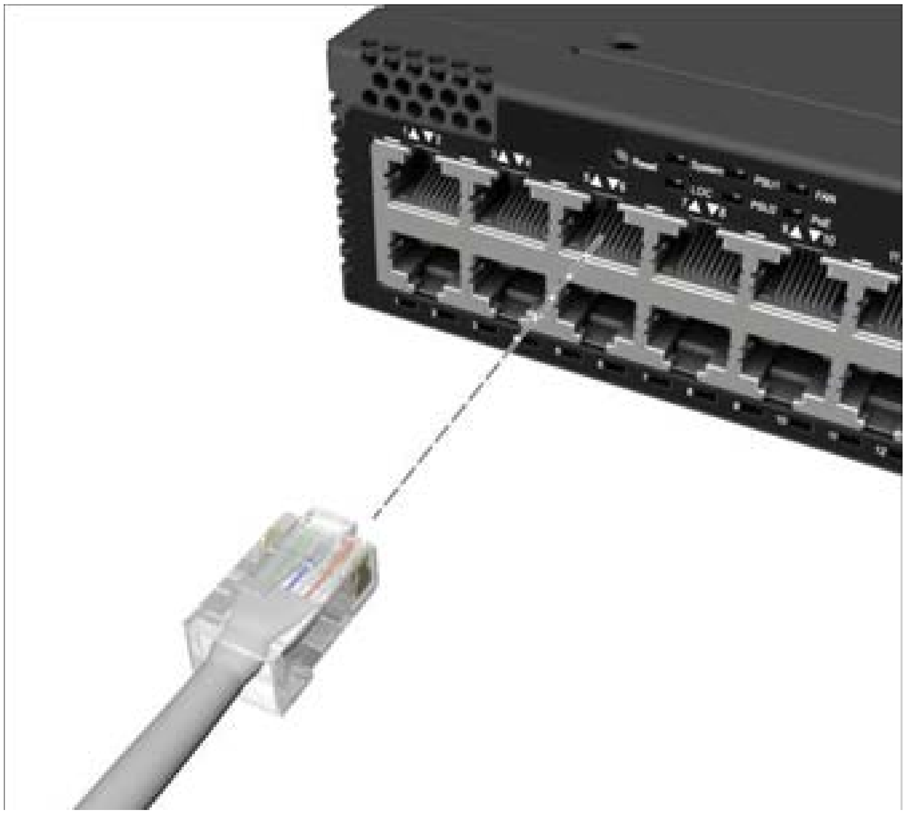


Figure 141. Connecting cables to the RJ-45 port of an MTM 5149-S54 switch

b) For the SFP+ ports, install transceivers and then connect fiber optic cabling to the transceiver ports.

Figure 142. Connecting cables the SFP+ port of an MTM 5149-S54 switch

Alternative: Connect AEC/AOC/DAC cables directly to the SFP+ slots.

The transceivers included in the next list are supported in the SFP+ ports.

- 10G BASE-SR, LR, CR, T
- 1000 BASE-SX, LX, T

6. Make the management connections.

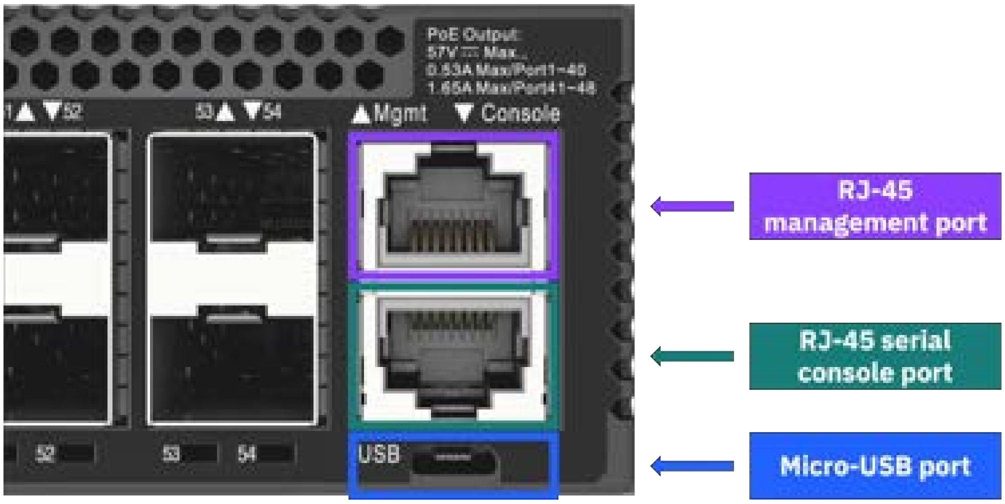


Figure 143. Making the management connections for an MTM 5149-S54 switch

a) Use the micro-USB cable that is included with your MTM 5149-S54 switch to connect it to an external storage device through the micro-USB port of the MTM 5149-S54 switch.

b) For the RJ-45 management port, use a Category 5, 5e, or better twisted-pair cable.

c) For the RJ-45 console port, use the RJ-45-to-DE-9, null-modem, console cable that is included with your MTM 5149-S54 switch to connect it to a computer that runs a terminal emulator software.

With computers that do not have a DE-9 serial port, use a USB-to-serial adapter cable (not included).

Configure the serial connection with the following characteristics.

- 11 5200 bps
- 8 characters
- 1 stop bit
- 8 data bits
- No parity
- No flow control

Adding additional management switches

If a single management switch is already using all its available ports (MTM 5149-S54 switch to MTM 5149-S54 switch), or if a new MTM 5149-S54 switch is meant to be added to an existing switch (MTM 5149-S54 switch to MTM 5149-S54 switch, or MTM 5149-S54 switch to MTM 8831-S52 switch), use the following guidance on how to extend the VLANs.

To add the new switch (*switch 2*, an MTM 5149-S54 switch) to the existing switch (*switch 1*, either MTM 5149-S54 switch or MTM 8831-S52), connect the next ports.

- Connect port 45 (`Vlan102`) of switch 1 to port 47 of switch 2.
- Connect port 46 (`Vlan101`) of switch 1 to port 48 of switch 2.

These connections extend both the management and the service VLANs ports to the new switch.

MTM 5149-S54 switch service

Service procedures and a list of replaceable parts are provided for the MTM 5149-S54 switch service.

Important: Before the service of any replaceable part, you must consult “Replaceable parts” on page 111 and “Replacing a customer replaceable unit” on page 111 to determine the replacement designation and avoid warranty infringement.

Key replaceable parts for MTM 5149-S54 switch

The following table lists key replaceable parts for the switch. Because part numbers can change over time, in a repair situation, verify that the correct CRU or FRU is obtained.

Part	FRU part number	CRU or FRU	Concurrent replace
MTM 5149-S54 switch	03MT898 03MU201	FRU	No
Rail mounting kit	03MT863	FRU	No
Power cord 4.3 m (14 ft) (drawer to IBM PDU, 250 V / 10 A)	39M5510	CRU Tier 1	Yes
Power cord 4.3 m (14 ft) (drawer to IBM PDU, 250 V / 10 A) For India	01KV682	CRU Tier 1	Yes
Power cable 2.8 m (9.2 ft) (drawer to IBM PDU, 250 V / 10 A)	39M5392	CRU Tier 1	Yes
Power cable 2.8 m (9.2 ft) (drawer to IBM PDU, 250 V / 10 A) For India	01PP689	CRU Tier 1	Ye

Additional cables

Table 42. Cable part numbers

Description	CRU part number
Ethernet cable (3 m)	00E5376
Ethernet cable (5 m)	00E5377
Fiber-optic cable with lucent connectors (25 m)	15R8848
Fiber-optic cable with lucent connectors (25 m)	88Y6857

Replaceable parts

Each replaceable part is assigned one of three designations. Most of the parts for this switch are classified as Tier 1 CRU. None of the parts for this switch are Tier 2 CRU.

For more information, see the following table.

Replacement designation	Description
CRU -Tier 1	Replacing a Tier 1 CRU is a customer responsibility, but dependent on if they purchased a service upgrade: • Under the standard warranty – If an IBM representative installs a Tier 1 CRU at the customer’s request, the customer is charged for the installation. – If an IBM representative installs a Tier 1 CRU at IBM Support's request, the customer is not charged for the installation. • Under an IBM Storage Expert Care service upgrade: The customer can request that IBM install the CRU at no additional cost.
CRU -Tier 2	Replacing a Tier 2 CRU can be completed by the customer or can be installed at the customer’s request by an IBM representative. If a Tier 2 CRU is replaced by an IBM representative: • If the product is under warranty: There is no charge for the part or for the installation if the product. • If the product is no longer under warranty: The customer is charged for the part and the installation.
FRU	Replacing a FRU must be performed by an IBM service personnel or an IBM authorized warranty service provider.

Replacing a customer replaceable unit

Use this procedure for servicing a customer replaceable unit (CRU).

Before you start a replacement, the customer must complete the following tasks:

- If the replacement procedure might put data at risk, the customer must back up the system or logical partition (including operating systems, licensed programs, and data).
- Review the replacement procedure for the part.
- Obtain the required tools for the replacement procedure.
- Inspect the replacement part. If any of the parts are incorrect, missing, or visibly damaged, they should contact the provider of the part or IBM Support.

The replacement best practices are as follows:

- For all replacements performed on an operational system, remove only one module from the enclosure at a time, unless directed otherwise by a replacement procedure.
- To reduce the amount of time required for the replacement, remove the failed module only when the replacement module is available, unpackaged, and ready for installation.
- Follow all recommended procedures for handling devices that are sensitive to electrostatic discharge (ESD).

1. Identify the system that contains the part to replace.
2. Attach the ESD wrist strap. The ESD wrist strap must connect to an unpainted metal surface until the service procedure is completed.
3. Locate the CRU part to be serviced.
4. Remove and replace the CRU part.
5. Verify that the installed part is working as expected.

Replacing an MTM 5149-S54 switch

Refer to the service procedure to remove and replace an MTM 5149-S54 switch enclosure.

If possible, the customer should back up the switch configuration and its firmware before the replacement procedure.

For step 6 of this procedure, two people are required.

The high-level steps to replace an MTM 5149-S54 switch are as follows.

1. Locate the faulty switch.
2. Label all data and power cables that connect to the faulty MTM 5149-S54 switch.
3. Disconnect the power supply cables from their PDU connections.
4. Disconnect all data cables from the faulty MTM 5149-S54 switch.
5. While holding the MTM 5149-S54 switch enclosure, remove the four securing cage nuts.
6. Pull the faulty MTM 5149-S54 switch enclosure out of the rack and place it on a flat anti-static surface.
7. Install the new MTM 5149-S54 switch.

Important:

- Before removing the switch from the rack, ensure that the switch is powered down, with all network and power cables disconnected.
- Two people are required to lift the switch above 1.4 m (57 in). Check the TDA document to identify where the switch will be replaced. Replacement locations at 32U and higher require two people to lift the switch. For replacement locations lower than 32U, one person can safely lift the switch.

1. Locate the faulty switch.

   The service ticket should include location information. Confirm the switch location by looking for fault LEDs on the switch.
2. Label all data and power cables that connect to the faulty MTM 5149-S54 switch.

   Take note of which cables are plugged into the specific ports before you remove any cable. You will need this record for the installation of the new MTM 5149-S54 switch.
3. Disconnect the power supply cables from their PDU connections.
4. Disconnect all data cables from the faulty MTM 5149-S54 switch.
5. While a second person holds the MTM 5149-S54 switch enclosure, remove the four securing cage nuts.
6. Pull the faulty MTM 5149-S54 switch enclosure out of the rack and place it on a flat anti-static surface.
7. Install the new MTM 5149-S54 switch.

   To install the new MTM 5149-S54 switch, follow the steps from "Installing the MTM 5149-S54 switch" on page 144.

Configuration and cabling

For more information about switch setup, configuration, and cabling, see the manufacturer's site.

- For MTM 5149-N64 switch, see QM9700/QM9790 1U NDR 400Gb/s InfiniBand Switch Systems User Manual in NVIDIA documentation.
- For MTM 5149-S05 switch, see NVIDIA Spectrum SN3700 in NVIDIA documentation.

For detailed configuration and cabling information, see the corresponding *Hardware Planning and Installation Guide* in IBM Storage Scale System documentation.

Cables for MTM 5149-S05 switch

Product number	Description	Feature code
39M5510	Power cable, drawer to IBM PDU, 200V/10A	6458
01KV682	4m jumper cord	END2
01KV679	1m jumper cord	END3
01KV680	2m jumper cord	END3
01KV681	3m jumper cord	END3
39M5392	2.8m power cable, drawer to IBM PDU, 250V/10A	6665
01PP689	2.8m power cable, drawer to IBM PDU, 250V/10A (for India)	END5
41V2458	10m InfiniBand cable	EB2J
45D6369	30m InfiniBand cable	EB2K
01FT706	100GbE, QSFP28 optical transceiver	EB59
01FT724	10m cable	EB5T
01FT726	30m cable	EB5W
03NK243	200G Ethernet cable	AJRF
01LL968	30m QSFP56	AJR9
01AF039	1.5m cable	ECCG
01AF045	3m cable	ECCJ
02CL470	5m cable	ECCW
01AF040	3m cable	1111
01AF041	10m cable	1112
01AF038	10m cable	1116
01AF045	3m cable	1118
01AF043	10m cable	1119

Cables for MTM 5149-S54 switch

Product number	Description	Feature code
39M5510	Power cable, drawer to IBM PDU, 200-240V/10A	6458
01KV682	4m jumper cord	END2

Product number	Description	Feature code
39M5392	2.8m power cable, drawer to IBM PDU, 250V/10A	6665
01PP689	2.8m power cable, drawer to IBM PDU, 250V/10A (for India)	END5
01AF040	3m cable	1111
01AF041	10m cable	1112
01AF038	10m cable	1116
01AF045	3m cable	1118
01AF043	10m cable	1119
45D4774	5m cable	ECBD
15R8848	25m LC cable	ECBE
00E5376	3m ENET cable	EN02
00E5377	5m ENET cable	EN03
02CL470	5m cable	ECCW

Warranty and service

The standard warranty period for these solutions is a 1-year, 9x5 warranty, with options available for the expanded IBM Expert Care.

IBM warranty information

Warranty type	Duration	Coverage	Service level
Standard	1 year	9x5 next business day	• IBM installation • Limited on-site FRU service • Software maintenance - for 90 days only, the customer can install an update
IBM Storage Expert Care Basic	1-5 years	9x5 next business day	Standard warranty plus: • Optional CRU service • Software maintenance (SWMA) - at anytime, the customer can install an update
IBM Storage Expert Care Advanced	1-5 years	24x7 same business day service	IBM Storage Expert Care Basic plus:

Warranty type	Duration	Coverage	Service level
			• Enhanced 24x7 coverage

The following warranty upgrade options are available:

- IBM onsite-service
- 24x7 same day service

The following post-warranty options are available:

- IBM onsite-service
- 9x5 next business day service
- 9x5 same day service
- 24x7 same day maintenance
- 2-hour or 4-hour service (US only)

Service strategy

IBM attempts to resolve problems over the telephone. Following problem determination, if IBM determines a customer replaceable unit (CRU) has failed, the CRU part will be provided as part of the machine's standard warranty service. If on-site service is required, a service call is scheduled.

Installation

These switches are specified as IBM set up; installation is performed by an IBM representative.

Updates

- Under the standard warranty, the customer can obtain and install a software update for the first 90 days only. After 90 days, the customer can not obtain an update. There is no software support.
- Under Expert Care Basic, the customer can update software at anytime.
- Under Expert Care Advanced, the customer can update software at anytime.

Parts replacement

Some of the replaceable parts for these switches are classified as CRUs. IBM specifies in the materials shipped with a replacement CRU whether the defective CRU must be returned to IBM. When a return is required, return instructions are shipped with the replacement CRU. The customer might be charged for the replacement CRU if IBM does not receive the defective CRU within 15 days of receipt of the replacement.

Other replaceable parts for these switches are classified as FRUs. For FRU parts, an IBM service representative or IBM Authorized Warranty Service Provider performs the replacement.

Troubleshooting

Information from multiple sources is used to maintain and service the switches. These sources include switch status LEDs, component status LEDs, events logs,and user documentation. Customers should troubleshoot issues on their own, including collecting and reviewing logs.

Reference documentation

For more information about the switches covered in this document, refer the corresponding manufacturer's documentation.

Documentation for MTM 5149-N64 switch

- NVIDIA QM79xx User Guide (https://docs.nvidia.com/networking/display/qm97x0pub).
- NVIDIA QM79xx Installation instructions (https://docs.nvidia.com/networking/display/qm97x0pub/installation).
- NVIDIA QM9700/QM9790 Datasheet (https://nvdam.widen.net/s/k8sqcr6gzb/infiniband-quantum-2-qm9700-series-datasheet-us-nvidia-1751454-r8-web).

Documentation for MTM 5149-S05 switch

- NVIDIA Spectrum SN3700
- NVIDIA Spectrum SN3000 Series Switches Datasheet
- NVIDIA Spectrum-2 SN3000 1U and 2U Switch Systems Hardware User Manual

Documentation for MTM 5149-S54 switch

- EPS121 AS4625-54T
- EPS121 (AS4625-54T) Datasheet
- EPS121 (AS4625-54T) Quick Start Guide
- EPS121 (AS4625-54T) Safety and Regulatory Information

Accessibility features for the system

Accessibility features help users who have a disability, such as restricted mobility or limited vision, to use information technology products successfully.

Accessibility features

The following list includes the major accessibility features in IBM Storage Scale RAID:

- Keyboard-only operation
- Interfaces that are commonly used by screen readers
- Keys that are discernible by touch but do not activate just by touching them
- Industry-standard devices for ports and connectors
- The attachment of alternative input and output devices

IBM Documentation, and its related publications, are accessibility-enabled.

Keyboard navigation

This product uses standard Microsoft Windows navigation keys.

IBM and accessibility

See the IBM Human Ability and Accessibility Center (www.ibm.com/able) for more information about the commitment that IBM has to accessibility.

Glossary

This glossary provides terms and definitions for the IBM Storage Scale System 6000 solution.

The following cross-references are used in this glossary:

- *See* refers you from a non-preferred term to the preferred term or from an abbreviation to the spelled-out form.
- *See also* refers you to a related or contrasting term.

For other terms and definitions, see the IBM Terminology website (opens in new window):

http://www.ibm.com/software/globalization/terminology

B

building block
A pair of servers with shared disk enclosures attached.

BOOTP
See *Bootstrap Protocol (BOOTP)*.

Bootstrap Protocol (BOOTP)
A computer networking protocol that is used in IP networks to automatically assign an IP address to network devices from a configuration server.

C

CEC
See *central processor complex (CPC)*.

central electronic complex (CEC)
See *central processor complex (CPC)*.

central processor complex (CPC)
A physical collection of hardware that consists of channels, timers, main storage, and one or more central processors.

cluster
A loosely-coupled collection of independent systems, or *nodes*, organized into a network for the purpose of sharing resources and communicating with each other. See also *GPFS cluster*.

cluster manager
The node that monitors node status using disk leases, detects failures, drives recovery, and selects file system managers. The cluster manager is the node with the lowest node number among the quorum nodes that are operating at a particular time.

compute node
A node with a mounted GPFS file system that is used specifically to run a customer job. ESS disks are not directly visible from and are not managed by this type of node.

CPC
See *central processor complex (CPC)*.

D

DA
See *declustered array (DA)*.

datagram
A basic transfer unit associated with a packet-switched network.

DCM
See *drawer control module (DCM)*.

declustered array (DA)
A disjoint subset of the pdisks in a recovery group.

dependent fileset
A fileset that shares the inode space of an existing independent fileset.

DFM
See *direct FSP management (DFM)*.

DHCP
See *Dynamic Host Configuration Protocol (DHCP)*.

drawer control module (DCM)
Essentially, a SAS expander on a storage enclosure drawer.

Dynamic Host Configuration Protocol (DHCP)
A standardized network protocol that is used on IP networks to dynamically distribute such network configuration parameters as IP addresses for interfaces and services.

E

Elastic Storage System (ESS)
A high-performance, GPFS NSD solution made up of one or more building blocks. The ESS software runs on ESS nodes - management server nodes and I/O server nodes.

encryption key
A mathematical value that allows components to verify that they are in communication with the expected server. Encryption keys are based on a public or private key pair that is created during the installation process. See also *file encryption key (FEK)*, *master encryption key (MEK)*.

ESS
See *Elastic Storage System (ESS)*.

environmental service module (ESM)
Essentially, a SAS expander that attaches to the storage enclosure drives. In the case of multiple drawers in a storage enclosure, the ESM attaches to drawer control modules.

ESM
See *environmental service module (ESM)*.

F

failback
Cluster recovery from failover following repair. See also *failover*.

failover
(1) The assumption of file system duties by another node when a node fails. (2) The process of transferring all control of the ESS to a single cluster in the ESS when the other clusters in the ESS fails. See also *cluster*. (3) The routing of all transactions to a second controller when the first controller fails. See also *cluster*.

failure group
A collection of disks that share common access paths or adapter connection, and could all become unavailable through a single hardware failure.

FEK
See *file encryption key (FEK)*.

file encryption key (FEK)
A key used to encrypt sectors of an individual file. See also *encryption key*.

file system
The methods and data structures used to control how data is stored and retrieved.

file system descriptor
A data structure containing key information about a file system. This information includes the disks assigned to the file system (*stripe group*), the current state of the file system, and pointers to key files such as quota files and log files.

file system descriptor quorum
The number of disks needed in order to write the file system descriptor correctly.

file system manager
The provider of services for all the nodes using a single file system. A file system manager processes changes to the state or description of the file system, controls the regions of disks that are allocated to each node, and controls token management and quota management.

fileset
A hierarchical grouping of files managed as a unit for balancing workload across a cluster. See also *dependent fileset*, *independent fileset*.

fileset snapshot
A snapshot of an independent fileset plus all dependent filesets.

flexible service processor (FSP)
Firmware that provides diagnosis, initialization, configuration, runtime error detection, and correction. Connects to the HMC.

FQDN
See *fully-qualified domain name (FQDN)*.

FSP
See *flexible service processor (FSP)*.

fully-qualified domain name (FQDN)
The complete domain name for a specific computer, or host, on the Internet. The FQDN consists of two parts: the hostname and the domain name.

G

GPFS cluster
A cluster of nodes defined as being available for use by GPFS file systems.

GPFS portability layer
The interface module that each installation must build for its specific hardware platform and Linux distribution.

GPFS Storage Server (GSS)
A high-performance, GPFS NSD solution made up of one or more building blocks that runs on System x servers.

GSS
See *GPFS Storage Server (GSS)*.

H

Hardware Management Console (HMC)
Standard interface for configuring and operating partitioned (LPAR) and SMP systems.

HMC
See *Hardware Management Console (HMC)*.

I

IBM Security Key Lifecycle Manager (ISKLM)
For GPFS encryption, the ISKLM is used as an RKM server to store MEKs.

independent fileset
A fileset that has its own inode space.

indirect block
A block that contains pointers to other blocks.

inode
The internal structure that describes the individual files in the file system. There is one inode for each file.

inode space
A collection of inode number ranges reserved for an independent fileset, which enables more efficient per-fileset functions.

Internet Protocol (IP)
The primary communication protocol for relaying datagrams across network boundaries. Its routing function enables internetworking and essentially establishes the Internet.

I/O server node
An ESS node that is attached to the ESS storage enclosures. It is the NSD server for the GPFS cluster.

IP
See *Internet Protocol (IP)*.

IP over InfiniBand (IPoIB)
Provides an IP network emulation layer on top of InfiniBand RDMA networks, which allows existing applications to run over InfiniBand networks unmodified.

IPoIB
See *IP over InfiniBand (IPoIB)*.

ISKLM
See *IBM Security Key Lifecycle Manager (ISKLM)*.

J

JBOD array
The total collection of disks and enclosures over which a recovery group pair is defined.

K

kernel
The part of an operating system that contains programs for such tasks as input/output, management and control of hardware, and the scheduling of user tasks.

L

LACP
See *Link Aggregation Control Protocol (LACP)*.

Link Aggregation Control Protocol (LACP)
Provides a way to control the bundling of several physical ports together to form a single logical channel.

logical partition (LPAR)
A subset of a server's hardware resources virtualized as a separate computer, each with its own operating system. See also *node*.

LPAR
See *logical partition (LPAR)*.

M

management network
A network that is primarily responsible for booting and installing the designated server and compute nodes from the management server.

management server (MS)
An ESS node that hosts the ESS GUI and is not connected to storage. It must be part of a GPFS cluster. From a system management perspective, it is the central coordinator of the cluster. It also serves as a client node in an ESS building block.

master encryption key (MEK)
A key that is used to encrypt other keys. See also *encryption key*.

maximum transmission unit (MTU)
The largest packet or frame, specified in octets (eight-bit bytes), that can be sent in a packet- or frame-based network, such as the Internet. The TCP uses the MTU to determine the maximum size of each packet in any transmission.

MEK
See *master encryption key (MEK)*.

metadata
A data structure that contains access information about file data. Such structures include inodes, indirect blocks, and directories. These data structures are not accessible to user applications.

MS
See *management server (MS)*.

MTU
See *maximum transmission unit (MTU)*.

N

Network File System (NFS)
A protocol (developed by Sun Microsystems, Incorporated) that allows any host in a network to gain access to another host or netgroup and their file directories.

Network Shared Disk (NSD)
A component for cluster-wide disk naming and access.

NSD volume ID
A unique 16-digit hexadecimal number that is used to identify and access all NSDs.

node
An individual operating-system image within a cluster. Depending on the way in which the computer system is partitioned, it can contain one or more nodes. In a Power Systems environment, synonymous with *logical partition*.

node descriptor
A definition that indicates how ESS uses a node. Possible functions include: manager node, client node, quorum node, and non-quorum node.

node number
A number that is generated and maintained by ESS as the cluster is created, and as nodes are added to or deleted from the cluster.

node quorum
The minimum number of nodes that must be running in order for the daemon to start.

node quorum with tiebreaker disks
A form of quorum that allows ESS to run with as little as one quorum node available, as long as there is access to a majority of the quorum disks.

non-quorum node
A node in a cluster that is not counted for the purposes of quorum determination.

O

OFED
See *OpenFabrics Enterprise Distribution (OFED)*.

OpenFabrics Enterprise Distribution (OFED)
An open-source software stack includes software drivers, core kernel code, middleware, and user-level interfaces.

P

pdisk
A physical disk.

PortFast
A Cisco network function that can be configured to resolve any problems that could be caused by the amount of time STP takes to transition ports to the Forwarding state.

R

RAID
See *redundant array of independent disks (RAID)*.

RDMA
See *remote direct memory access (RDMA)*.

redundant array of independent disks (RAID)
A collection of two or more disk physical drives that present to the host an image of one or more logical disk drives. In the event of a single physical device failure, the data can be read or regenerated from the other disk drives in the array due to data redundancy.

recovery
The process of restoring access to file system data when a failure has occurred. Recovery can involve reconstructing data or providing alternative routing through a different server.

recovery group (RG)
A collection of disks that is set up by ESS, in which each disk is connected physically to two servers: a primary server and a backup server.

remote direct memory access (RDMA)
A direct memory access from the memory of one computer into that of another without involving either one's operating system. This permits high-throughput, low-latency networking, which is especially useful in massively-parallel computer clusters.

RGD
See *recovery group data (RGD)*.

remote key management server (RKM server)
A server that is used to store master encryption keys.

RG
See *recovery group (RG)*.

recovery group data (RGD)
Data that is associated with a recovery group.

RKM server
See *remote key management server (RKM server)*.

S

SAS
See *Serial Attached SCSI (SAS)*.

secure shell (SSH)
A cryptographic (encrypted) network protocol for initiating text-based shell sessions securely on remote computers.

Serial Attached SCSI (SAS)
A point-to-point serial protocol that moves data to and from such computer storage devices as hard drives and tape drives.

service network
A private network that is dedicated to managing POWER8 servers. Provides Ethernet-based connectivity among the FSP, CPC, HMC, and management server.

SMP
See *symmetric multiprocessing (SMP)*.

Spanning Tree Protocol (STP)
A network protocol that ensures a loop-free topology for any bridged Ethernet local-area network. The basic function of STP is to prevent bridge loops and the broadcast radiation that results from them.

SSH
See *secure shell (SSH)*.

STP
See *Spanning Tree Protocol (STP)*.

symmetric multiprocessing (SMP)
A computer architecture that provides fast performance by making multiple processors available to complete individual processes simultaneously.

T

TCP
See *Transmission Control Protocol (TCP)*.

Transmission Control Protocol (TCP)
A core protocol of the Internet Protocol Suite that provides reliable, ordered, and error-checked delivery of a stream of octets between applications running on hosts communicating over an IP network.

V

VCD
See *vdisk configuration data (VCD)*.

vdisk
A virtual disk.

vdisk configuration data (VCD)
Configuration data that is associated with a virtual disk.

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